

Source Diff for the Substantive Revision

Baseline commit 6b8d85e to the final revision working tree

September 2026

This change-marked artifact is a unified source diff. It covers the main manuscript, online appendix, generated table sources, and revision documents. Binary artifacts are not embedded in the patch. The canonical editable files and generated PDFs accompany this document.

```
diff --git a/paper/appendix/appendix.tex b/paper/appendix/appendix.tex
index 30bcb19..41b2831 100644
--- a/paper/appendix/appendix.tex
+++ b/paper/appendix/appendix.tex
@@ -3,12 +3,12 @@
 \usepackage{styles/working-paper}
 \usepackage{styles/yax-common}
 \graphicspath{{figures/}}
-\hypersetup{pdftitle={Online Appendix to AI Exposure or Occupational Composition?},pdfauthor={Lily Luo}}
+\hypersetup{pdftitle={Online Appendix to Occupational AI Exposure and Young-Worker Employment},pdfauthor={Lily Luo}}
\begin{document}
\begin{center}
{\Large\bfseries Online Appendix to\par}
\vspace{0.6em}
- {\Large ‘‘AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment’’\par}
+ {\Large ‘‘Occupational AI Exposure and Young-Worker Employment: Support and Comparisons in the CPS’’\par}
\vspace{0.8em}
Lily Luo\[\[0.25em]
September 2026
@@ -17,16 +17,13 @@
\tableofcontents
\clearpage
\appendix
-\input{appendix/sections/appendix_A_measurement.tex}
-\input{appendix/sections/appendix_B_mapping.tex}
-\input{appendix/sections/appendix_C_support.tex}
-\input{appendix/sections/appendix_D_inference.tex}
-\input{appendix/sections/appendix_E_stock.tex}
-\input{appendix/sections/appendix_F_dynamics.tex}
-\input{appendix/sections/appendix_G_flows.tex}
-\input{appendix/sections/appendix_H_reallocation.tex}
-\input{appendix/sections/appendix_I_FG_AE.tex}
-\input{appendix/sections/appendix_J_LOCO.tex}
-\input{appendix/sections/appendix_K_power.tex}
-\input{appendix/sections/appendix_L_reproducibility.tex}
+\input{appendix/sections/r3_A_data.tex}
+\input{appendix/sections/r3_B_mapping.tex}
+\input{appendix/sections/r3_C_occupational_support.tex}
+\input{appendix/sections/r3_D_characteristics.tex}
+\input{appendix/sections/r3_E_inference.tex}
+\input{appendix/sections/r3_F_dynamics.tex}
+\input{appendix/sections/r3_G_flows.tex}
+\input{appendix/sections/r3_H_bcc_architecture.tex}
+\input{appendix/sections/r3_I_reproducibility.tex}
\end{document}
diff --git a/paper/appendix/sections/appendix_A_measurement.tex b/paper/appendix/sections/appendix_A_measurement.tex
index 9d569c0..59f76f5 100644
--- a/paper/appendix/sections/appendix_A_measurement.tex
+++ b/paper/appendix/sections/appendix_A_measurement.tex
@@ -70,7 +70,7 @@
For the task family, the raw primitives satisfy  $\mathbb{E}[\alpha] = D$ ,  $\mathbb{E}[\beta] = D + S/2$ , and  $\mathbb{E}[\gamma] = D + S$ . On strict common support, the maximum absolute gap between raw beta and  $\mathbb{E}[\alpha + \gamma]/2$  is  $\$7.0 \times 10^{-13}$ . Mapping and aggregation preserve the identity. Separate employment-weighted standardization changes the relative location and scale and creates a maximum numerical midpoint gap of 0.5112; it does not create a third primitive.
```

-The external-architecture admission rule was written before fitting external outcomes. Webb AI and the reversed OECD capability gap meet the declared construct, public-source, non-title mapping, full-component, distinct-cut, and 80-percent coverage criteria. Webb software remains a computerization control; its source correlation with Webb AI is 0.7021. Frey--Osborne remains an automation-risk comparator, not an AI architecture.

+The external-architecture admission rule was written before fitting external outcomes. Webb AI and the reversed OECD capability gap meet the declared construct, public-source, non-title mapping, full-component, distinct-cut, and 80-percent coverage criteria. Webb software remains a computerization control; its source values are occupation-level patent--task overlap percentages on a 0--100 scale, while every fitted Webb interaction uses a preperiod employment-weighted standardized score. Its source correlation with Webb AI is 0.7021. Frey--Osborne remains an automation-risk comparator, not an AI architecture.

\subsection{Dependence and candidate census}

```
diff --git a/paper/appendix/sections/r3_A_data.tex b/paper/appendix/sections/r3_A_data.tex
new file mode 100644
index 0000000..426dc65
--- /dev/null
+++ b/paper/appendix/sections/r3_A_data.tex
@@ -0,0 +1,53 @@
+\section{Data Reconstruction and Treatment Contract}\label{app:data}
+
+\subsection{Source files and sample flow}
+
+
+The stock analysis uses two restricted IPUMS CPS extracts. The wide file supplies January 2017--July 2026 records; a second file supplies the five March Basic Monthly samples that replace incorrectly selected March ASCE samples in 2017--2021. The replication manifest records extract identifiers, creation metadata where available, byte sizes, and SHA-256 hashes without redistributing microdata. The March replacement occurs before eligibility filters or occupation routing. The superseded rows contribute no positive-final-weight analysis records; the replacement adds 252,862 eligible records.
+
+
+The wide file is IPUMS extract 9: its request was submitted on August 26, 2026, and the completed-file receipt was recorded four minutes later. Separate hashes authenticate its submitted specification, data, DDI, and basic codebook. The March repair is extract 11; its actual request, returned API response, data, DDI, and basic-codebook hashes are verified, but its surviving API response contains no submission or completion timestamp, so none is inferred from filesystem metadata. Flow models additionally use extract 10 for \texttt{LNKFWMWT}; it was recorded complete when checked on August 31, 2026, but the check time is not called an API completion time. Exact timestamps and hashes appear in the input manifest and provenance receipt. Restricted files, account metadata, download links, and direct identifiers are not redistributed.
+
+
+\begin{table}[\htbp]
+\centering
+\caption{Calendar and cell reconstruction audit}\label{tab:app_calendar}
+\begin{tabular}{lrr}
+\toprule
+Object & Count & Audit rule \\
+\midrule
+Expected calendar months & 115 & January 2017--July 2026 \\
+Observed months & 114 & October 2025 not collected \\
+Static months & 113 & December 2022 transition excluded \\
+Primary occupations & 468 & named fixed support \\
+Complete occupation--month grid & 52,884 &  $468 \times 113$  \\
+Positive-total fitted cells & 51,891 & includes one-sided zeros \\
+Both-age-zero cells & 993 & no likelihood term \\
+One-sided zero cells retained & 12,965 & finite boundary contributions \\
+\bottomrule
+\end{tabular}
+\tabnote{October 2025 is absent, not coded zero or interpolated. December 2022 is observed but excluded from the static pre/post fit.}
```

+\\end{table}

+\\

+Eligibility requires age 18--65, employment, a valid occupation, and positive \\texttt{WTFIML}. The analysis scans 9,843,021 source rows, retains 5,322,047 eligible source records, and creates 6,188,956 route-expanded descendants. The last count is fractional before 2020 and is not a count of distinct respondents. The principal age groups are formed after routing. Of the complete grid, 13,883 young cells and 1,068 older cells have zero respondent-equivalent records. These sparse-cell facts motivate the finite-sample audit; they do not justify selecting cells on realized young counts.

+\\

+\\subsection{Reconstructed preperiod treatment}

+\\

+The substantive treatment is rebuilt using only January 2017--November 2022 stock. On the fixed 468-occupation support, the employment-weighted beta mean is 0.37520 and standard deviation is 0.21041. The tie-preserving cut values are 0.153846, 0.324615, 0.456522, and 0.537037. Webb software exposure is standardized over the same 71-month construction window. Table-\\ref{tab:app_treatment} compares the rebuilt contract with the historical production contract.

+\\

+\\begin{table}[\\htbp]

+\\centering

+\\caption{Historical and reconstructed treatment contracts}\\label{tab:app_treatment}

+\\begin{tabular}{\\lrrr}

+\\toprule

+Contract & Construction window & Post stock used & Q5--Q1 coefficient \\\\

+\\midrule

+Historical production & full static panel & Yes & $\\$-0.134554\\$$ \\\\

+Reconstructed & Jan. 2017--Nov. 2022 & No & $\\$-0.132109\\$$ \\\\

+\\bottomrule

+\\end{tabular}

+\\tabnote{Both coefficients use the corrected 113-month outcome calendar and identical occupation support. Nine occupations change quintile. The reconstructed-minus-historical paired difference is 0.002445 with 95-percent interval $\\$[-0.001269,0.006158]\\$$.}

+\\end{table}

+\\

+The reconstructed contract contains 129, 99, 82, 67, and 91 occupations in Q1--Q5. Corresponding preperiod stock shares are 20.06, 19.96, 20.16, 20.31, and 19.51 percent. The complete membership file identifies all nine occupations whose historical and rebuilt labels differ. All revised substantive models use the reconstructed labels unless explicitly described as a historical audit.

+\\

+\\subsection{Survey-production dates}

+\\

+Official BLS documentation establishes population-control discontinuities in January 2025 and January 2026. BLS did not revise the preceding official months onto the new bases and does not publish counterfactual age-by-occupation weights. October 2025 was not collected; November collection began late, used reduced overlap and a modified weighting procedure, and had a lower response rate. IPUMS later processed the reissued January 2026 file and corrections to longitudinal identifiers. The stock and flow manifests preserve their respective extract dates and hashes. Endpoint and exclusion checks diagnose reliance on these dates, but no aggregate adjustment factor is imposed on detailed cells.

diff --git a/paper/appendix/sections/r3_B_mapping.tex b/paper/appendix/sections/r3_B_mapping.tex

new file mode 100644

index 0000000..e49f603

--- /dev/null

+++ b/paper/appendix/sections/r3_B_mapping.tex

@@ -0,0 +1,57 @@

+\\section{Occupational Mapping and Analysis Support}\\label{app:mapping}

+\\

+\\subsection{Bridge construction}

+\\

+The early panel reports Census-2010 occupations and the later panel Census-2018 occupations. Exposure inputs additionally span SOC vintages. The versioned bridge first routes 503 early source codes into 568 target codes. Fifty-six sources have more than one target, with maximum multiplicity seven. Official shares partition each source record's final weight; separate young and older routing would require validation data that the public adjacent-vintage files do not provide. The routed early and current stocks reconcile to their sources with relative error below $\\$2\\times 10^{-16}\\$$.

+\\

+In the frozen mapping-only lookup, the equal-administrative AIOE variant has full-route coverage for 480 of the 503 pre-2020 source codes. This 480-code mapping count is not the regression sample: intersecting the rebuilt beta and Webb inputs with positive young preperiod stock yields the 468-occupation primary analysis support below.

+\\

+The primary joint beta/Webb universe contains 468 occupations and 87.19 percent of stock in the audited candidate universe. Table-\\ref{tab:app_exclusions} gives mutually exclusive exclusion counts. A machine-readable file lists each included and excluded occupation, its preperiod stocks, scores, and reason. This file, rather than a hard-coded count in the paper, defines support.

+\\

+\\begin{table}[\\htbp]

+\\centering

+\\caption{Primary-support exclusions}\\label{tab:app_exclusions}

+\\begin{tabular}{\\lrr}

+\\toprule

+Reason & Occupations \\\\

+\\midrule

+Missing Rule-A beta only & 41 \\\\

+Missing Webb software only & 22 \\\\

+Missing both exposure inputs & 8 \\\\

+Nonpositive young preperiod stock & 2 \\\\

+\\midrule

+Total excluded & 73 \\\\

+Primary retained & 468 \\\\

+\\bottomrule

+\\end{tabular}

+\\end{table}

+\\

+\\subsection{Separating score correction from population expansion}

+\\

+A literal exact-code merge between SOC-2010 AIOE and post-2018 employment data leaves only 3.33 percent of computer-and-mathematical employment. The version-aware route reaches 97.7 percent. The four-row continuous comparison separates the two operations that an aggregate ‘‘crosswalk effect’’ would conflate:

+\\

+The source workbook standardizes AIOE without employment weights across 774 source occupations: its mean is approximately zero and its sample standard deviation is approximately one. A raw AIOE point is therefore one unit of that source index, not a probability or percentage. As a common descriptive increment, the source-workbook interquartile difference is 1.879 raw units. The historical four-row audit below uses a different, explicitly fixed unit: one full-window employment-stock-weighted standard deviation of repaired equal-administrative AIOE on the expanded 495-occupation AIOE-plus-Webb support. The retained artifact does not store the numerical weighted standard deviation, so I do not manufacture a raw-point conversion.

+\\

+\\begin{table}[\\htbp]

+\\centering

+\\caption{AIOE taxonomy repair and support expansion}\\label{tab:app_crosswalk}

+\\begin{tabular}{\\llrr}

+\\toprule

+Score & Support & Occupations & Coefficient per fixed SD \\\\

+\\midrule

+Literal original & original & 410 & $\\$-0.01885\\$$ \\\\

+Repaired & same original & 410 & $\\$-0.01920\\$$ \\\\

+Repaired & expanded & 495 & $\\$-0.03156\\$$ \\\\

+Repaired & expanded, excluding SOC15 & --- & $\\$-0.02940\\$$ \\\\

+\\bottomrule

+\\end{tabular}

+\\tabnote{The first row is an invalid vintage merge retained as an implementation audit, not a defensible specification. Row 1 to 2 isolates score repair on fixed support; row 2 to 3 changes the population; the last row tests whether readmitted computer-and-mathematical occupations alone account for that expansion movement. Every coefficient uses the fixed weighted-SD unit defined in the text; none is a percentage or a coefficient per raw AIOE point.}

+\\end{table}

+\\

+The invalid merge is documented as a failure encountered in this project. I do not claim that a published paper made the same error without direct verification of its code and occupational universe.

+\\

+\\subsection{Split-route and age-allocation sensitivity}

+\\

+Within the primary support, split sources contribute 14.93 percent of young and 17.75 percent of older early stock. Ninety-nine target occupations can receive a split route. A one-to-one-target restriction and an aggregation to stable Census-2010 categories both retain negative historical-contract estimates, but each changes the target population. They are mapping sensitivities, not repairs of the primary bridge.

+\\

+Using the same route proportions by age is transparent but unvalidated. Accounting ranges that allocate every ambiguous source in all feasible ways place the 2017--2019 young-to-older stock ratio between 0.1031 and 0.1480 in Q1 and between 0.0790 and 0.1175 in Q5. These are stock-accounting bounds, not bounds on the nonlinear regression coefficient. A limited odds-tilt sensitivity changes the historical-contract coefficient only modestly because 6.88 percent of early stock is eligible, but it cannot validate age-specific routing. No adjacent coding vintage is treated as a same-person validation sample.

diff --git a/paper/appendix/sections/r3_C_occupational_support.tex b/paper/appendix/sections/r3_C_occupational_support.tex

new file mode 100644

```

index 0000000..62cb61c
--- /dev/null
+++ b/paper/appendix/sections/r3_C_occupational_support.tex
@@ -0,0 +1,40 @@
+ \section{Broad-Family Support, Profiles, and Influence} \label{app:families}
+
+ \subsection{Family-by-quintile support}
+
+
+ \Table{\ref{tab:app_family_support} reports, for every SOC two-digit family, the number of retained occupations, preperiod stock share, exposure range, quintiles present, direct-tail overlap, and nuisance-adjusted Q5 information share. Only arts/media (SOC27), health practitioners (SOC29), health support (SOC31), and sales (SOC41) span Q1 and Q5 under the reconstructed comparison. Other families connect the Q5--Q1 regression coefficient through intermediate quintiles and the imposed common Q2--Q5 slopes.
+
+
+ \input{tables/r3_appendix_family_support.tex}
+
+
+ \Table{\ref{tab:app_family_information} makes the loss and concentration of identifying information explicit. Family-by-month conditioning retains 29.7 percent of the pooled Q5 target's nuisance-adjusted information, while the direct-tail comparison has only 6.2 effective occupations and its five largest occupation contributions supply 77.8 percent of information. Its normal-theory MDE{80} is 0.4576 log point. The continuous within-family companion is more diffuse and has MDE{80}=0.0313 per one within-family standard deviation, but it imposes a common slope across families.
+
+
+ \input{tables/r3_appendix_family_information_precision.tex}
+
+
+ \The direct-tail population retains only occupations in the four spanning families that belong to Q1 or Q5. It contains 29 occupations and 5.03 percent of primary preperiod stock. In the SOC2-by-calendar-month model its coefficient is 0.1494, with interval  $[-0.1694, 0.4682]$ . This benchmark is valuable because it exposes direct overlap; it is not a preferred estimate for the original population, and the interval includes large effects of either sign.
+
+
+ \subsection{Full exposure profile}
+
+
+ \The pooled Q2--Q5 coefficients are jointly different from zero, but pairwise movements do not establish a monotone dose response. After SOC2-by-calendar-month conditioning, the four coefficients are jointly indistinguishable from zero. Table\ref{tab:app_profile} reports pointwise and simultaneous intervals and the common-draw movement for every group.
+
+
+ \input{tables/r3_appendix_profile.tex}
+
+
+ \The continuous companion deems beta within family using preperiod employment weights and imposes a common linear slope. With SOC2-by-calendar-month conditioning, the rebuilt coefficient is  $-\$0.0025$  per 0.1084 raw beta units. Its interval is  $[-\$0.0240, 0.0191]$  in that standardized within-family unit. The unconditioned continuous coefficient is essentially zero as well. This is a different functional form from the tail contrast.
+
+
+ \subsection{Leave-one-family and occupation influence}
+
+
+ \Leave-one-family-out estimates are computed from the same family-conditioning program and reported with the omitted family's stock and target-information contribution. These rows answer whether a broad comparison is concentrated; they are not used to choose which families belong in the main estimate. Table\ref{tab:app_lofo} displays the eight largest absolute movements. Omitting office/administrative support or food preparation moves the coefficient in opposite directions by about 0.04 log point, and every paired deletion interval includes zero.
+
+
+ \input{tables/r3_appendix_lofo.tex}
+
+
+ \Figure{\ref{fig:family_paths} shows the four direct-tail families, selected by descending nuisance-adjusted Q5 information rather than coefficient sign. Within each exposure tail, the young and older employment stocks are shown separately and indexed to their own 2019 means. The figure therefore exposes which age series drives a tail's relative path without treating either age group as untreated.
+
+
+ \begin{figure} [htbp]
+ \centering
+ \includegraphics[width=0.96\textwidth]{r3_figure3_family_paths.pdf}
+ \caption{Young and older employment stocks in direct-tail families} \label{fig:family_paths}
+ \tabnote{Families are selected by direct-tail, SOC2-by-calendar-month Q5 nuisance-adjusted information, with ties broken by SOC2 code; selection does not use coefficient sign. Gray and blue distinguish Q1 and Q5; solid and dashed series distinguish ages 22--25 and 26--65. Each series equals 100 at its own 2019 mean, and the red dashed line marks January 2023. These are descriptive weighted-stock paths without design-based bands.}
+ \end{figure}
+
+
+ \Occupation leave-one-out calculations are interpreted in the same way. Fast-food and counter workers have high signed influence because low-exposure service employment recovered differently after the pandemic. Other occupations push in the opposite direction. Historical-contract diagnostics show that deleting the five largest absolute prior influences moves the coefficient toward zero, whereas deleting ten or twenty can make it more negative. Symmetric influence trimming stays near the pooled estimate. Because every such deletion is outcome adaptive, none replaces the broad-family regression.
+
+
+ \Econometrically defined service exclusions are cleaner diagnostics but still change the population. Removing Q1 food-service occupations attenuates the historical-contract absolute coefficient by less than one tenth; removing the broader SOC35/37/39 set of food, building-and-grounds, and personal-care service occupations does not attenuate it. Protective-service occupations (SOC33) are not part of that exclusion. These results reject a one-occupation explanation, not hypotheses about reopening, immigration, wages, or labor supply that the CPS design does not separately identify.
diff -git a/paper/appendix/sections/r3_D_characteristics.tex b/paper/appendix/sections/r3_D_characteristics.tex
new file mode 100644
index 0000000..b060e1d
--- /dev/null
+++ b/paper/appendix/sections/r3_D_characteristics.tex
@@ -0,0 +1,37 @@
+ \section{Occupational Characteristics and Heterogeneity} \label{app:characteristics}
+
+ \subsection{Inputs, timing, and common-support comparisons}
+
+
+ \The characteristic exercise uses one variable at a time before considering a parsimonious joint model. Every row first re-estimates the unaugmented exposure model on exactly the support available for that characteristic. It then adds the characteristic interacted with young status and the post indicator, reuses the same occupation multipliers, and forms the augmented-minus-baseline difference within each draw. Sample loss and coefficient movement are therefore distinct objects. The corresponding paired minimum detectable effect is the two-sided, five-percent normal-theory quantity  $\$(1.96+0.84)SE(\Delta)$ ; it describes resolution and is never used as a second rejection rule.
+
+
+ \The leading computer-use variable is O*NET item 4.A.3.b.1, "Interacting With Computers," from release 24.3 (May 2020). Remotability is the Dingel--Neiman occupation classification. Wage and observed education composition are calculated from the CPS preperiod, while externally coded education requirements are kept conceptually separate. Routine and manual indices use preperiod O*NET inputs. All scales, source files, checksums, weighted correlations, and support counts are stored in the characteristic result directory.
+
+
+ \The focused rebuilt-treatment comparison uses the 408 occupations with both computer use and remotability, so every row has identical support. The baseline coefficient is  $-\$0.1070$  and the computer-use-conditioned coefficient is  $-\$0.2124$ . Their paired difference is  $-\$0.1054$ , with 95-percent interval  $[-\$0.1835, -\$0.0273]$  and paired  $\$MDE_{80}=0.1131$ . Thus computer use is a suppressor in this particular regression rather than an attenuating control. This result neither validates the residual as AI nor makes computer use exogenous.
+
+
+ \Adding remotability on the same support gives  $-\$0.1155$ ; its paired difference from  $-\$0.1070$  is  $-\$0.0085$ , with interval  $[-\$0.0437, 0.0267]$  and paired  $\$MDE_{80}=0.0510$ . Adding both computer use and remotability gives  $-\$0.2120$  and a paired movement of  $-\$0.1049$ , with interval  $[-\$0.1814, -\$0.0284]$ . The other one-at-a-time characteristics likewise do not produce individually detected movements. The expanded module's joint characteristic support falls to 341 occupations and 76.77 percent of rebuilt treatment-construction stock; its wider interval precludes a claim that the combined residual is a purified AI contrast.
+
+
+ \subsection{Expanded support-specific and cumulative audit}
+
+
+ \The separate characteristic module in Tables\ref{tab:appendix_characteristic_estimates} and \ref{tab:appendix_characteristic_information} reports the support-specific comparisons and the uncontaminated portion of the cumulative sequence. Its one-at-a-time rows use each characteristic's maximal finite support, whereas its cumulative rows use the literal 341-occupation common support. An audit found that an earlier version of this module had retained historical full-static-panel exposure groups. The reported module now authenticates and uses the canonical corrected-preperiod membership and Webb normalization throughout; its 468-occupation baseline exactly reproduces  $-\$0.1321$ . The support-specific computer comparison retains 455 occupations and holds those rebuilt primary-universe groups fixed, while the focused exercise requires both computer-use and remotability data and rebuilds groups on its 408-occupation support. Their coefficients therefore differ by design, not by transcription. Rows at and after the superseded unbounded shortfall control are removed rather than mixed with the corrected shortfall population. Every addition and this corrective rerun are labeled post-outcome exploratory.
+
+
+ \input{tables/r3_appendix_characteristic_estimates.tex}
+
+
+ \input{tables/r3_appendix_characteristic_information.tex}
+
+
+ \subsection{Pandemic shortfall}
+
+
+ \The corrected exercise uses ages 22--65 throughout and admits only one-to-one occupation mappings with sufficient 2017--2019 histories. It retains 363 occupations, rather than the central 468. Linear trend predictions for total employment are bounded below at zero; predicted young employment shares are bounded to  $[\$0, 1]$ . This is consequential: before bounding there are 174 negative total predictions and 1,258 out-of-range share predictions. The characteristics are the average bounded trend-prediction-minus-realized gaps from January 2020 through November 2022; December 2022 remains the transition month.
+
+
+ \On this narrower common support, the pooled and family-month baseline coefficients are  $-\$0.16417$  and  $-\$0.08708$ . Total-stock conditioning moves them by 0.00040 and  $-\$0.00050$ ; young-relative conditioning moves them by 0.00114 and 0.00613. Table\ref{tab:appendix_shortfall} reports linked-household sampling sensitivities that regenerate the shortfall inside each draw. All four conditioning-movement intervals include zero. Jointly regenerating exposure labels changes an average of 1.99 occupations per draw and leaves the young-relative family-month movement at 0.00613 with sampling-sensitivity standard error 0.00695.

```

+
+The controls are generated from related CPS outcomes and may share sampling error and regression-to-the-mean with the dependent stocks. A deterministic household split gives a similar average but heterogeneous directions; it does not eliminate PSU-level common error. The bounded functional form and changed population also remain consequential. The exercise therefore does not establish that pandemic dynamics are absent, and the linked-household intervals are not complete CPS design-based inference.

+
+\input{tables/r3_appendix_shortfall.tex}
+
+\subsection{Industry, education, and exact age}
+
+The industry analysis is rebuilt at the occupation-by-broad-industry-by-age-by-month level rather than assigning each occupation a dominant industry. The risk set covers all 468 occupations and 3,127 preperiod-connected occupation-industry cells. Relative to the stratified baseline coefficient of $\$-0.1376\$$, adding 13 broad-industry-specific young-by-post shifts gives $\$-0.0987\$$. The paired movement is $\$0.0389\$$, with interval $[\$0.0082, \$0.0697\$]$ and paired $\mathbf{\text{\textit{MDE}}}_{\{80\}} = 0.0444\$$. The conditioned coefficient remains below zero. These are statements about the microcell objective, not firm-level adoption.

+
+Education-specific models are estimated on a common occupational support. The BA-or-higher estimate is $\$-0.1380\$$ and the non-BA estimate is $\$-0.1226\$$; their paired difference is $\$-0.0154\$$, with interval $[\$-0.1557, \$0.1250\$]$ and paired $\mathbf{\text{\textit{MDE}}}_{\{80\}} = 0.2014\$$. The comparison is too imprecise to distinguish moderate education differences. Exact-age estimates for ages 22, 23, 24, and 25 are reported as one jointly estimated exploratory collection with simultaneous intervals. No single age is selected after inspection.

+
+\input{tables/r3_appendix_heterogeneity.tex}
diff --git a/paper/appendix/sections/r3_E_inference.tex b/paper/appendix/sections/r3_E_inference.tex
new file mode 100644
index 00000000..e50d0bc
--- /dev/null
+++ b/paper/appendix/sections/r3_E_inference.tex
@@ -0,0 +1,31 @@
+\section{Inference and Finite-Sample Audit}\label{app:inference}
+
+\subsection{Stochastic targets}
+
+The principal wild-score interval treats occupation as the shock cluster and conditions on the constructed stocks, exposure labels, occupational bridge, and retained support. It accommodates arbitrary serial dependence within an occupation but not unrestricted dependence across occupations. A 22-family Webb-weight exercise instead allows common shocks within broad SOC families. A separate household/sample-unit multiplier rebuilds cell stocks from microdata, preserves all observed records of the sampled unit across months, and assigns the same multiplier to every fractional descendant of a routed source record. That exercise describes repeated-sample sensitivity conditional on released CPS final weights; the public Basic Monthly extracts do not supply the design variables needed to call it complete CPS design-based inference.

+
+These variances are not mechanically added. Occupation-shock and household-resampling procedures can contain overlapping components, so summing them without a derived orthogonal decomposition could double count uncertainty. They are reported as separate sensitivity targets.

+
+\subsection{Score and bootstrap implementation}
+
+The conditional likelihood retains one-sided zero cells and drops only occupation-month cells with zero total stock. Scores use the fitted grouped-binomial objective, and nuisance-adjusted target scores use the Schur-complement projection implied by the fitted information matrix. The same 9,999 Rademacher occupation multipliers are reused for paired comparisons. Each draw first forms the two refitted or score-updated targets and then their difference; intervals for movement therefore retain covariance. The finite-cluster correction, random seed, multiplier matrix hash, convergence checks, and score-conservation checks are written to the execution receipts.

+
+For the reconstructed pooled comparison, the analytic standard error is 0.04517 and the 9,999-draw 95-percent interval is $[\$-0.2206, -0.0437\$]$. After SOC2-by-calendar-month conditioning, the coefficient is $\$-0.0217\$$, with standard error 0.07132 and interval $[\$-0.1607, 0.1173\$]$. The paired movement is 0.11043, with standard error 0.05189 and interval $[\$0.0107, 0.2102\$]$. The corresponding two-sided normal-theory paired $\mathbf{\text{\textit{MDE}}}_{\{80\}}$ is 0.14539. The detected movement remains smaller than this planning quantity; that is not a contradiction because an MDE is a power summary, not an acceptance threshold for an observed draw.

+
+Table-\ref{tab:appendix_precision_summary} consolidates the principal precision diagnostics. It states the scale and whether the quantity is a coefficient or a paired difference; the same numerical formula is not reinterpreted as an equivalence margin.

+
+\input{tables/r3_appendix_precision_summary.tex}
+
+\subsection{Broad-family and household sensitivity}
+
+The few-family exercise uses 22 broad-family weights and reports baseline, conditioned, and paired targets from common draws. Its role is sensitivity to cross-occupation dependence, not a declaration that SOC2 is the uniquely correct cluster. The household/sample-unit exercise performs 399 positive-weight full refits. With labels fixed, its basic interval is $[\$-0.1865, -0.0794\$]$ for the pooled coefficient, $[\$-0.1212, 0.0587\$]$ for the family-conditioned coefficient, and $[\$0.0362, 0.1753\$]$ for their movement. Regenerating preperiod exposure labels within the same draws gives $[\$-0.1864, -0.0762\$]$, $[\$-0.1236, 0.0665\$]$, and $[\$0.0355, 0.1771\$]$. A median of two occupations changes quintile per regenerated-label draw. Webb standardization by itself is an affine transformation absorbed by the specified fixed effects, so this difference is a quintile-membership sensitivity rather than a separate normalization channel.

+
+The source does not expose official Basic Monthly replicate weights or all confidential design identifiers. Household identifiers are therefore a dependence-preserving public-data approximation, not an ASEC replication-weight procedure. Link-specific flow analyses additionally report person- and household-cluster sensitivities using their own risk sets.

+
+\subsection{Calendar-time HAC and stress validation}
+
+The time-HAC audit operates on nuisance-adjusted score sums and implements occupation clustering plus an aggregate elapsed-calendar HAC term minus within-occupation elapsed-calendar HAC terms. Calendar lags are measured in months, so the missing October 2025 observation does not make September and November adjacent. Under rebuilt exposure labels, lag-16 standard errors are 0.04449 for the pooled target, 0.05522 for the family-conditioned target, and 0.04501 for their paired movement. None of the 15 rebuilt five-parameter covariance matrices across the three targets and five lags is indefinite, and no projection or eigenvalue clipping is applied. Indefinite positive-lag matrices from the superseded historical-treatment/SOC2-post audit are retained only as labeled implementation history.

+
+The completed finite-sample audit uses 11 declared designs, 799--1,599 retained outer replications per design, 9,999 common multiplier draws per joint refit, and independently computed pseudo-targets. Absolute coefficient bias is at most 0.00102 log point in the empirical observed-effect design. Interval performance is target-specific: occupation-Rademacher coverage is 0.865 for the pooled projection, 0.921 for the family-month projection, and 0.889 for their paired movement. Family-Rademacher coverage is respectively 0.990, 0.895, and 0.992. Rademacher and Webb weights give nearly identical coverage at a fixed cluster level, so changing the multiplier law does not repair the cluster-level mismatch.

+
+The earlier 26.7-percent and 11.3-percent rejection figures are withdrawn rather than reinterpreted. In the clean family-variance-zero design, the family-month pseudo-target is zero to numerical precision; its five-percent rejection rates are 6.1 percent under occupation Rademacher weights, 8.9 percent under family Rademacher weights, and 5.6 percent for the cross-fit full-refit oracle. In the empirically calibrated structural-null design, removing the structural exposure coefficient still leaves a pooled pseudo-target of $\$-0.11035\$$ and a paired pseudo-target of 0.11035 because the data-generating process retains the observed fitted family-by-month surface. The near-nominal oracle then rejects zero for the pooled projection 96.5 percent of the time. This is a restatement of the broad-family decomposition built into the calibration, not independent evidence that composition caused the empirical gap. The simulations therefore limit both the inferential and structural interpretation: none of the feasible cluster procedures is adequate for every target and significance of the pooled projection does not by itself test a structural exposure effect.

diff --git a/paper/appendix/sections/r3_F_dynamics.tex b/paper/appendix/sections/r3_F_dynamics.tex
new file mode 100644
index 00000000..4c60b0d
--- /dev/null
+++ b/paper/appendix/sections/r3_F_dynamics.tex
@@ -0,0 +1,29 @@
+\section{Dynamics, Endpoints, and Trend Sensitivity}\label{app:dynamics}
+
+\subsection{Quarterly system}
+
+The corrected dynamic model interacts every nonreference exposure quintile with calendar quarter. The omitted period is 2022Q4, which contains October and November because December 2022 is the transition month and is excluded. The resulting Q5--Q1 path has 23 preperiod and 15 postperiod coefficients. The pooled and SOC2-by-calendar-month versions are estimated on identical reconstructed support, and the archive stores the complete Q2--Q5 coefficient vector, full covariance matrix, influence matrix, and simultaneous critical values.

+
+The joint null that all 23 Q5 preperiod coefficients equal zero is rejected in both the pooled model ($\$p=.0149\$$) and the family-conditioned model ($\$p=.0053\$$). As a precision diagnostic, an OLS linear functional through those coefficients is $\$-0.0078\$$ log point per year in the pooled model and $\$0.0041\$$ after family conditioning. Its occupation-cluster standard errors are 0.0119 and 0.0292, implying normal-theory $\mathbf{\text{\textit{MDE}}}_{\{80\}}$ values of 0.0332 and 0.0819 log point per year. Thus an insignificant linear slope cannot rule out economically relevant smooth drift and is not a substitute for the rejected unrestricted joint test. Simultaneous bands exclude zero only in late pooled quarters and never in the family-conditioned path. These results do not establish which preperiod force produces the differences; they make a parallel-evolution interpretation of the January 2023 split untenable.

+
+The postperiod dynamic functional averages the 15 post-quarter coefficients with predeclared calendar weights. Its pooled value is $\$-0.1199\$$, with interval $[\$-0.2636, 0.0239\$]$. Its family-conditioned value is $\$-0.2074\$$, with interval $[\$-0.4252, 0.0103\$]$. This linear functional is not the nonlinear static grouped-binomial coefficient. The mapping file reports their paired relationship rather than silently treating them as identical.

+
+\subsection{Official HonestDiD implementation}
+
+The trend-sensitivity program passes the joint event-time vector and covariance matrix to the official `\texttt{HonestDiD}` implementation. The environment receipt fixes HonestDiD 0.2.8 and its source commit, and records the compatible official CVXR and solver versions. The program does not feed a single static coefficient and standard error into an event-study procedure. It reports both the complete 2017Q1--2022Q3 calibration and the prespecified consecutive 2021Q1--2022Q3 post-reopening calibration. The latter does not delete an internal pandemic quarter or compress a calendar gap. For the family-conditioned vector its $\$M=0\$$ smoothness interval excludes zero, but the interval includes zero at $\$M=0.005\$$; the pooled intervals include zero even at $\$M=0\$$. The conventional event-functional interval already includes zero in both structures. The correct positive zero-exclusion breakdown is therefore undefined: there is no positive robustness threshold to lose. Earlier installation and API failures remain in the execution log; the successful official run is not replaced with a home-built approximation.

```

+
+\input{tables/r3_appendix_honestdid.tex}
+
+\subsection{Onset, endpoint, seasonality, and survey dates}
+
+Onset sensitivity reports every monthly split from November 2022 through June 2023. January 2023 is retained because December 2022 was designated as the transition month after the November 30, 2022
public release of ChatGPT; robustness across nearby dates does not identify that event as the cause. The endpoint grid includes December 2024, September and December 2025, and July 2026
wherever the calendar permits. October 2025 is never imputed. Extending from the 2025 endpoints to July 2026 makes the pooled coefficient about 0.02 log point more negative, and its paired
interval excludes zero; the analogous conditioned movements remain near zero.
+
+\input{tables/r3_appendix_endpoint_sensitivity.tex}
+
+Exposure-group-by-month-of-year interactions supply the lower-dimensional seasonality check and move the target little. More saturated occupation-specific seasonality and a shortened post-2020 fit
fail to converge under the declared algorithm; both attempts and diagnostics are retained. Excluding September and November 2025 around the collection gap does not detectably change the full-
window estimate. These rows address sensitivity to calendar construction and survey production, not counterfactual reweighting for the January 2025 or January 2026 population controls.
+
+\subsection{Official-weight and respondent-equivalent era comparison}
+
+Table-\ref{tab:appendix_population_eras} reports a post-outcome exploratory joint-model comparison of 2023--2024 with 2025--2026. The official-weight row retains the paper's employment-stock estimand
; the respondent-equivalent row is a deliberately different routed-count outcome used only to diagnose whether the later pattern mechanically appears only in final-weight stocks. It cannot
recover unpublished age-by-occupation counterfactual weights.
+
+\input{tables/r3_appendix_population_eras.tex}
diff --git a/paper/appendix/sections/r3_G_flows.tex b/paper/appendix/sections/r3_G_flows.tex
new file mode 100644
index 0000000..8b20797
--- /dev/null
+++ b/paper/appendix/sections/r3_G_flows.tex
@@ -0,0 +1,19 @@
+\section{Linked Flows and Worker Outcomes}\label{app:flows}
+
+\subsection{Link construction and risk sets}
+
+Person links follow official IPUMS/Census matching guidance and use the available longitudinal weight for each eligible link. Adjacent-month and twelve-month endpoints are distinct samples. A twelve
-month link is not the sum of twelve observed monthly transitions and may conceal intervening changes. The annual young link rate is 50.98 percent, compared with 70.55 percent for the older
group, so the longer horizon entails appreciable differential selection.
+
+Employment exit begins with an employed person in an occupation on the analysis support. Reported occupation change requires employment at both endpoints and a valid occupation at each. Entry-
destination allocation begins with a nonemployed origin, ends with observed employment, and asks whether the destination falls in Q5 rather than Q1. It therefore conditions on entry and is
neither an employment-finding probability nor an employer hiring rate. Every crosswalk descendant of one source record shares its link and resampling weight.
+
+\subsection{Results and uncertainty}
+
+At the adjacent-month horizon, the young-relative Q5--Q1 estimate is 0.132 for employment exit, 0.003 for reported occupation change, and  $-\$0.070\$$  for entry-destination allocation. Their respective
95-percent intervals are  $[-\$0.042, \$0.307]\$,  $[-\$0.118, \$0.123]\$, and  $[-\$0.250, \$0.110]\$$ . At twelve months, the three estimates are 0.123,  $-\$0.018\$$ , and  $-\$0.052\$$ . Their intervals are  $[-\$0.111, \$0.357]\$$ ,
 $[-\$0.097, \$0.061]\$, and  $[-\$0.251, \$0.147]\$$ . All include zero. Person- and household-cluster sensitivities, descriptive transition rates, effective risk counts, and named occupation influence
appear in the machine-readable flow directory.
+
+Usual-hours changes are estimated within linked employed persons. The adjacent estimate is  $-\$0.119\$$  hours per week with interval  $[-\$0.295, \$0.057]\$$ . Weekly earnings use the eligible outgoing-rotation
sample and its earnings weight; the conditional log-earnings coefficient is  $-\$0.0245\$$  with interval  $[-\$0.1024, \$0.0534]\$$ . The earnings extract covers 74 months through March 2023 and therefore
contains only three postperiod months. Both outcomes condition on employment. Unemployment duration is described only among persons observed unemployed, and no occupation is imputed to never-
employed or out-of-labor-force respondents. These restrictions prevent the outcomes from being misreported as population employment effects.
+
+\input{tables/r3_appendix_exit_components.tex}
+
+\subsection{Why no stock--flow calibration is reported}
+
+A compatible accounting identity would require entries from nonemployment, exits, switches among all occupations, aging into and out of ages 22--25, corresponding movements for the older denominator,
job-to-job moves, and population/sample-composition residuals. The CPS endpoint links do not measure all of these components on a common employer-match risk set, and the BCC hiring and
separation objects are not identical to CPS employment transitions. A single implied stock response would therefore be assumption driven. The revision reports the measurable margins and the
precise comparability gap instead of manufacturing a calibration.
diff --git a/paper/appendix/sections/r3_H_bcc_architecture.tex b/paper/appendix/sections/r3_H_bcc_architecture.tex
new file mode 100644
index 0000000..d53aa80
--- /dev/null
+++ b/paper/appendix/sections/r3_H_bcc_architecture.tex
@@ -0,0 +1,33 @@
+\section{Payroll Bridge and Exposure-Construction Diagnostics}\label{app:bcc_architecture}
+
+\subsection{BCC version and grouping bridge}
+
+The comparison uses the August 12, 2026 version of Brynjolfsson--Chandar--Chen and records the accessed paper, dashboard, and workbook in the evidence ledger. Their reported 19-percent young-worker
"kept pace" shortfall is a stock-growth counterfactual in ADP data, not a regression coefficient, employment probability, or hiring rate. Their public dashboard forms equal-occupation
exposure groups; later regressions employ other weights and employer controls. Exact dashboard memberships and the treatment of ties are not released.
+
+The CPS bridge therefore applies an equal-occupation, tie-preserving top-two-versus-bottom-three approximation on the present support and labels it as such. The pooled estimate is  $-\$0.0720\$$ , with
interval  $[-\$0.1218, -\$0.0222]\$$ . The SOC2-by-calendar-month estimate is  $-\$0.0168\$$ , with interval  $[-\$0.0810, \$0.0474]\$$ . Their paired movement is 0.0553, with interval  $[-\$0.0024, \$0.1129]\$$ . The
archive provides group summaries and a membership file; it cannot provide concordance with undisclosed assignments. Differences in occupation universe, age comparison, denominator, endpoint,
firm controls, and employer-matched flows remain.
+
+\input{tables/r3_appendix_bcc_bridge.tex}
+
+\subsection{Task-capability weighting}
+
+Write the two Eloundou primitives as  $\$D=E1\$$  and  $\$S=E2\$$ . The score family  $\$D+\lambda \$S$  is evaluated on a fixed grid from zero to one under common preperiod weights and common bootstrap draws. The
implementation verifies numerically that  $\lambda=0.5$  reproduces raw beta before grouping. Any earlier mismatch between that row and the headline arose from a different support or treatment
contract and is preserved only as implementation history.
+
+Two versions are kept separate. The native-rank analysis recomputes the score normalization and tie-preserving groups at each  $\lambda$ , so both scale and membership can change. The fixed-membership
analysis changes only the continuous score representation while holding the beta groups fixed. For every grid point the result files report coefficient, paired difference from beta, interval,
occupation coverage, tail turnover, and correlations with computer use and remotability. A focused 408-occupation run adds both computer use and remotability on common support. None of these
paths measures realized deployment.
+
+\subsection{Primitive and architecture comparisons}
+
+The joint primitive model reports  $\$D\$$  and  $\$S\$$  in raw share units, in preperiod-standard-deviation units, their covariance, and an illustrative common change. This is a two-regressor parameterization,
not a causal decomposition into technologies. Architecture comparisons likewise report both component coefficients and their paired difference on each declared support. Webb-AI comparisons
are shown first on identical support and then on broader native support so that score changes are not confused with population changes. Coordinate rotations that do not change the fitted
predictor are not interpreted as additional scientific results.
+
+Raw AIOE is a standardized source-occupation index: one point is approximately one unweighted standard deviation across the 774 occupations in the source workbook, and the source interquartile
difference is 1.879 points. It is not a percentage. Equal-administrative and source-employment-weighted mappings preserve that source scale through averaging, although their target-support
distributions need not retain unit variance. The direct-ability reconstruction has its own raw scale and is not numerically interchangeable. Categorical AIOE estimates use ranks and tie-
preserving groups; continuous comparisons must state whether their increment is a raw point or a current-support weighted standard deviation.
+
+\input{tables/r3_appendix_primitive_ds.tex}
+
+\input{tables/r3_appendix_architecture_pairs.tex}
+
+\input{tables/r3_appendix_webb_support.tex}
+
+\subsection{What validation would require}
+
+None of the sampling intervals above treats a score as a noisy observation of a validated true exposure. A useful validation sample would begin at the versioned occupation--task level and oversample
both broad families and tasks on which architectures disagree. It would record separate outcomes for technical capability, realized time savings without complementary software, realized time
savings with software, observed workplace adoption, and task importance. Conflating these labels would reproduce the construct problem the validation is meant to solve.
+
+Labels should come from repeated, blinded, independent expert assessments and, where feasible, audited product-use or task-performance records. Calibration and evaluation occupations should be$$$ 
```

separated so that crosswalk and occupation aggregation choices are tested out of sample. The task description, model snapshot, prompt, complementary tools, rater instructions, occupational bridge, and employment weights all require version records. Sample size cannot be stated as one universal number: it depends on within-occupation task correlation, rater reliability, unequal task weights, the latent quantity being corrected, and whether the downstream target is a continuous slope or tail assignment. In the absence of these inputs, the revision specifies the validation design but does not apply a generated-regressor correction.

```
diff --git a/paper/appendix/sections/r3_I_reproducibility.tex b/paper/appendix/sections/r3_I_reproducibility.tex
new file mode 100644
index 00000000..d956ae2
--- /dev/null
+++ b/paper/appendix/sections/r3_I_reproducibility.tex
@@ -0,0 +1,19 @@
+ \section{Reproducibility, Numerical Consistency, and Unresolved Items}\label{app:reproducibility}
+
+ \subsection{Execution contract}
+
+
+ The revision archive separates restricted inputs from distributable code and derived aggregate outputs. Two commands serve different purposes. The restricted-data orchestrator authenticates the licensed inputs, refuses to overwrite an existing output root, runs the rebuilt baseline before dependent modules, passes named treatment objects downstream, invokes each self-check, and runs the pinned official HonestDiD implementation. A separate aggregate-only command regenerates exhibits from committed results, checks numerical and prose consistency, runs repository tests, and builds the LaTeX documents. The latter neither reads CPS microdata nor independently re-estimates the empirical models. Environment records report Python and R package versions, seeds, scheduler resources, source hashes, and git state where captured. Every module declares its treatment contract, calendar, support, objective, inference target, and output paths before interpreting the result.
+
+
+ The Git repository is not itself labeled a sanitized public-release artifact. A machine-readable positive allowlist defines the files eligible for staging; unlisted files and suffixes are excluded. The release scan rejects private absolute roots, credential-token patterns, restricted-data formats, symlinks, caches, and tabular columns containing direct CPS link identifiers, followed by manual disclosure and metadata review. Licensed CPS records, the historical private access receipt, account metadata, download links, longitudinal identifier values, credentials, and operational path history remain outside the staged package. The manifest records IPUMS extract numbers 9, 10, and 11 and the available request, data, DDI, and codebook hashes; unavailable API timestamps are left blank rather than inferred.
+
+
+ The canonical results ledger is the numerical source of truth. Manuscript tables are generated from or checked against it, and repeated estimates use one declared draw set. If an interval comes from a different stochastic target---occupation shocks, broad-family shocks, or household resampling---the table labels that procedure rather than presenting multiple intervals as interchangeable versions of one result. The test suite verifies calendar membership, one-sided zeros, route conservation, fixed-support pairing, covariance symmetry and rank, draw hashes, and table/prose consistency.
+
+
+ \subsection{Implemented failures}
+
+
+ The failure registry is part of the replication evidence. It includes the literal vintage merge, the original March sample selection, nonconvergence of saturated seasonality variants, and indefinite positive-lag joint HAC matrices from the superseded historical-treatment audit. A failure is never converted into a different estimator without an explicit amendment. Successful reruns preserve the earlier log and record the replacement's package and solver versions; operational installation history remains outside the scientific exposition.
+
+
+ \subsection{Unresolved limitations}
+
+
+ The analysis cannot observe firm adoption, employer identifiers, worker--firm hiring, confidential CPS design variables, exact BCC dashboard memberships, age-specific occupational route probabilities, or a validated latent "true" AI exposure. The public data therefore do not identify an AI causal effect or a complete stock--flow mechanism. The direct within-family tail population is small, unrestricted preperiod differences are present, several linked-flow intervals are wide, and January 2025--2026 survey-production changes cannot be undone with published aggregate factors. These are limits on interpretation, not missing robustness rows that a specification search can solve.
+
+
+ Author affiliation, email, acknowledgments, funding, conflicts, data-access wording, and journal-required disclosures remain marked \texttt{[AUTHOR TO COMPLETE]} wherever author-supplied information is unavailable. No identifying or disclosure information is inferred from directory names, credentials, or institutional access.
diff --git a/paper/main/abstract_restat.tex b/paper/main/abstract_restat.tex
index 5e4b5a00..3e0879d 100644
--- a/paper/main/abstract_restat.tex
+++ b/paper/main/abstract_restat.tex
@@ -1 +1,1 @@
-I ask whether detailed occupational AI exposure explains a public-CPS young-worker employment pattern beyond broad occupational composition. The corrected highest-versus-lowest exposure contrast is $-0.1346$ log points. Conditioning on two-digit occupation-family post changes reduces it to $-0.0315$ (95-percent interval $[-0.1676, 0.1046]$), while family-specific monthly paths give $-0.0317$ $-0.1346$ log points. The reduction is statistically detectable in a paired comparison, but the conditional models retain only 30 percent of baseline target information and exposure alternatives include zero. The CPS association is principally a broad-occupation comparison, not an identified effect of detailed AI susceptibility.
+Recent evidence associates generative-AI exposure with weaker employment among workers in their early twenties. I reconstruct this in public monthly CPS data. The estimand is the post-December-2022 change in the log conditional-mean employment-stock ratio of ages 22--25 to 26--65 in the highest relative to lowest GPT-4-exposure quintile, conditional on intermediate quintiles and preperiod software exposure. Its estimate is $-0.132$ log points; broad occupation-family paths reduce it to $-0.022$, while direct tail support spans 29 occupations. Preperiod dynamics reject parallel evolution. The pattern is a descriptive comparison across broad occupational structure, not an identified causal AI effect.
diff --git a/paper/main/abstract_working.tex b/paper/main/abstract_working.tex
index a77901b..0006f97 100644
--- a/paper/main/abstract_working.tex
+++ b/paper/main/abstract_working.tex
@@ -1 +1,1 @@
-I rebuild a widely discussed young-worker employment pattern in public Current Population Survey data and ask whether detailed occupational AI exposure carries information beyond broad occupational composition. In the corrected January 2017--July 2026 panel, the post-January-2023 young-to-older employment-stock contrast between the highest and lowest Eloundou exposure quintiles is $-0.1346$ log points. Conditioning on two-digit occupation-family-specific post changes reduces it to $-0.0315$ (95-percent interval $[-0.1676, 0.1046]$); allowing each family its own monthly young-relative path gives $-0.0317$. The reduction is statistically detectable in a paired comparison, but the conditional models retain only 30 percent of baseline target information and change the estimand. Food-service exclusions explain little of the reduction. Architecture comparisons are less informative than their point estimates suggest: the baseline design has an 80-percent minimum detectable effect of $0.1244$ log points, and paired differences across seven alternative constructions all include zero. A public implementation of Brynjolfsson--Chandar--Chen's top-two-versus-bottom-three exposure grouping yields a smaller $-0.0728$ CPS contrast. Taxonomy repair and construction choices remain consequential, but the defensible conclusion is narrower: the CPS association is principally a broad-occupation comparison, not an identified effect of detailed AI susceptibility.
+Recent evidence associates generative-AI exposure with weaker employment among workers in their early twenties. I reconstruct the public monthly Current Population Survey comparison and ask which occupational contrasts support it. The estimand is the post-December-2022 change in the log conditional-mean employment-stock ratio of ages 22--25 to ages 26--65 in the highest relative to the lowest GPT-4-exposure quintile, conditional on intermediate quintiles and preperiod software exposure. Its estimate is $-0.132$ log points. Broad occupation-family paths reduce it to $-0.022$; the paired movement excludes zero under the occupation-clustered procedure, but direct tail support covers only 29 occupations and cannot resolve a within-family gradient. Preperiod computer-use conditioning makes the association more negative, industry conditioning moves it toward zero, and preperiod dynamics reject parallel evolution. Linked CPS estimates do not distinguish exit, reported occupation change, or entry destinations. The pattern is therefore a descriptive comparison across broad occupational structure, not an identified causal AI effect or a replication of employer hiring results.
diff --git a/paper/main/preamble.tex b/paper/main/preamble.tex
index 12483c3..2255d43 100644
--- a/paper/main/preamble.tex
+++ b/paper/main/preamble.tex
@@ -8,5 +8,5 @@
\fi
\usepackage{styles/yax-common}
\graphicspath{{figures/}}
-\hypersetup{pdftitle={AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment},pdfauthor={Lily Luo}}
-\newcommand{\papertitle}{AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment}
+\hypersetup{pdftitle={Occupational AI Exposure and Young-Worker Employment: Support and Comparisons in the CPS},pdfauthor={Lily Luo}}
+\newcommand{\papertitle}{Occupational AI Exposure and Young-Worker Employment: Support and Comparisons in the CPS}
diff --git a/paper/main/sections/01_introduction.tex b/paper/main/sections/01_introduction.tex
index 097fdfa..45e1620 100644
--- a/paper/main/sections/01_introduction.tex
+++ b/paper/main/sections/01_introduction.tex
@@ -1,28 +1,28 @@
-Recent evidence has linked occupational exposure to generative artificial intelligence with weaker employment growth among workers in their early twenties. The pattern matters because it bears on whether new technology is already changing entry-level work. It is also unusually hard to interpret. Occupational exposure is fixed before the labor outcome is observed, is highly correlated with broad occupational structure, and is available in several constructed forms that do not necessarily measure the same object. A negative exposed-versus-unexposed contrast can therefore reflect detailed AI susceptibility, broad occupational composition, or both.
+Recent evidence associates generative-AI exposure with weaker employment growth among workers in their early twenties. Interpretation is difficult because exposure is constructed rather than observed, high- and low-exposure jobs differ broadly, and the period around ChatGPT's release includes several aggregate disruptions.
-This paper asks a deliberately narrower question: \emph{does a detailed occupational AI score explain the public-CPS early-career employment pattern beyond the evolution of broad occupation families?} I aggregate Basic Monthly Current Population Survey records into occupation-by-age-by-month employment stocks from January 2017 through July 2026. The design compares ages 22--25 with ages 26--65 within occupations and asks whether that relative stock changes after January 2023 across exposure groups. The outcome is a stock, not hiring, and the design is descriptive. It does not observe employer adoption and does not turn the release of ChatGPT into an instrument.
+This paper asks which occupational comparisons support the public Current Population Survey (CPS) pattern. I rebuild monthly employment stocks from January 2017 through July 2026 and classify occupations with the GPT-4 beta score of \cite{eloundou2024gpts}. The target is the post-December-2022 change in the log conditional-mean employment-stock ratio of ages 22--25 to ages 26--65 in the highest relative to the lowest exposure quintile, conditional on intermediate quintiles and preperiod Webb software exposure. I examine broad-family paths, direct within-family support, preperiod computer use, dynamics, and linked-CPS flows. Exposure measures susceptibility rather than adoption; January 2023 is chronology, not an instrument.
-The corrected-calendar baseline reproduces a sizeable pattern. The post-January-2023 young-relative coefficient for the highest versus lowest employment-weighted quintile of Eloundou et al.'s GPT-4 beta exposure is $-0.1346$ log points, with a canonical 9,999-draw 95-percent wild-score interval of $[-0.2223, -0.0468]$. The frozen 108-month chronology gives $-0.1311$ and its canonical interval $[-0.2171, -0.0451]$. These numbers describe the change in a young-to-older employment-stock ratio, not a percentage decline in individual employment probability.

```

+The reconstructed coefficient is $-\$0.132\%$ log points, with 95-percent occupation wild-score interval $[-\$0.221, -\$0.044]\%$; it is neither an individual probability nor an employer hiring rate. With two-digit occupation-family young-relative monthly paths, it is $-\$0.022\%$ ($[-\$0.161, 0.117]\%$). Their paired difference is $\$0.110\%$ ($[\$0.011, 0.210]\%$). This distinguishes conditional estimands without making a causal decomposition: family paths may absorb AI-related and non-AI-related evolution.

-The central result changes the interpretation. Adding two-digit-occupation-family by young by post interactions reduces the corrected coefficient to $-\$0.0315\%$ with interval $[-\$0.1676, 0.1046]\%$. Allowing every broad family its own young-relative monthly path gives $-\$0.0317\%$. The paired baseline-to-conditioned movement is about $\$0.103\%$ log points and its interval excludes zero. Yet this is not an additive decomposition: the conditioning model asks whether detailed quintile status predicts outcomes among the comparisons left after broad-family evolution is absorbed. Only four of 22 broad families contain both Q1 and Q5, and conditional target information falls to 30 percent of the baseline. The evidence says that the baseline is predominantly a comparison across broad occupation families and that the remaining within-family association is weakly measured. It does not assign a causal share to broad composition.

+Direct support is thin. Four families span both exposure tails; their 29 occupations represent 5.03 percent of preperiod stock, and the interval includes large effects of either sign. The conditioned comparison otherwise relies on intermediate quintiles and common-coefficient restrictions, so its near-zero estimate is not economic equivalence or evidence of no AI effect.

-Nor does a simple food-service story account for the result. Excluding low-exposure food-service occupations attenuates the coefficient by 8.8 percent, to $-\$0.1196\%$. Excluding food, personal-care, and protective-service occupations makes it slightly more negative. Jointly removing the five occupations with the largest prior leave-one-out movements yields $-\$0.1011\%$, while symmetric trimming and continuous down-weighting return estimates close to baseline. Influence is consequential and signed in both directions; ‘food-service recovery explains the pattern’ is not supported.

+Adding a pre-2022 O*NET computer-use measure makes the common-support coefficient more negative; one control therefore does not resolve computerization. Unrestricted quarterly coefficients reject preperiod equality, while linked-flow intervals include zero. The data neither isolate an AI counterfactual nor attribute the stock pattern to exit, switching, or entry allocation.

-The second result concerns precision. The realized occupation-clustered standard error is 0.0444, compared with 0.0122 in the planning simulation. The frozen design’s normal-theory minimum detectable effect at 80-percent power is 0.1244 log points, approximately the size of the headline coefficient. Paired minimum detectable differences across alternative exposure constructions range from 0.0609 to 0.1689. Every paired Q5–Q1 interval comparing beta with an alternative contains zero. The data therefore do not detect architecture differences at conventional levels; they also do not establish that the architectures are economically equivalent. This distinction replaces earlier language that treated smaller or insignificant alternative point estimates as evidence that the conclusion did not generalize.

-

-The third result connects the public-data exercise directly to \cite{brynjolfsson2026canaries}. Implementing their published GPT-4 beta classification---the top two exposure quintiles against the bottom three---inside the same CPS model yields $-\$0.0728\%$ (interval $[-\$0.1240, -\$0.0216]\%$). This is a reproducible component of their design, not a replication of proprietary ADP outcomes, job-title mapping, firm controls, hiring, or separation. Reapplying the grouping across exposure architectures produces coefficients between approximately $-\$0.088\%$ and $-\$0.008\%$ on common support; only beta’s difference from the OECD capability measure is detected under that grouping. The contrast itself is part of the empirical object.

-

-Measurement still matters, but in more specific ways than the earlier manuscript claimed. A literal SOC-2010 exact-code merge into post-2018 data covers only 3.33 percent of computer-and-mathematical employment; the documented taxonomy repair covers 97.7 percent. Most of the associated coefficient change comes from readmitting occupations rather than changing exposure values on fixed support. The original six implementations also span little independent variation: two weighted principal components explain 96.11 percent of their common-support variance. These facts motivate a taxonomy that separates implementation errors, same-construct measurement changes, construct changes, support changes, estimand changes, representation changes, and pure reparameterizations.

-

-The paper contributes a public-data diagnosis of a prominent descriptive pattern and a disciplined way to audit constructed treatments. Its substantive conclusion is negative but informative: in the CPS, detailed AI quintiles do not carry a precisely estimated association after broad occupational evolution is absorbed. Its measurement conclusion is equally bounded: alternative constructed scores can differ in coverage and classification while available outcomes remain too noisy to distinguish many downstream coefficients. The paper does not estimate a causal AI effect, infer economic equivalence from a confidence interval containing zero, or claim that broad-family conditioning identifies the non-AI counterfactual.

-

-The remainder of the paper defines the construction layers, documents the corrected data and inferential target, presents the broad-family result, and then evaluates architecture precision, reference groups, influence, population controls, and timing. Mobility and algebraic reparameterizations are demoted to the online appendix because they do not resolve the employment-stock question.

-

-\begin{figure}[htbp]

-\centering

-\includegraphics[width=\textwidth]{figure1_architecture_genealogy_revised.pdf}

-\caption{From exposure construction to an economic statement}\label{fig:genealogy}

-\tabnote{The six initial implementations belong to two selected families. Webb AI and the OECD capability gap change the construct as well as the score. Mapping, support, representation, comparison, and conditioning determine the downstream estimand. Potential exposure is not observed adoption.}

-\end{figure}

+The paper shows that a sizeable public-CPS association is identified predominantly by comparisons spanning broad occupation families, quantifies the resolution of direct and flow evidence, and records calendar and taxonomy failures that silently alter support. This localization tells future adoption or quasi-experimental studies which comparisons they must replace rather than inherit. It narrows rather than overturns the motivating evidence: it neither replicates proprietary payroll outcomes nor identifies AI, computerization, or pandemic recovery as the cause.

diff --git a/paper/main/sections/02_literature.tex b/paper/main/sections/02_literature.tex

index 07c75b4..4ec7bb3 100644

--- a/paper/main/sections/02_literature.tex

+++ b/paper/main/sections/02_literature.tex

@@ -1,38 +1,15 @@

-\section{Constructed Treatments and Related Evidence}\label{sec:literature}

+\section{Related Evidence and the Empirical Object}\label{sec:literature}

-

-\subsection{A layered empirical object}

+The closest motivating study is the August 2026 revision of \cite{brynjolfsson2026canaries}. Using ADP payroll records through June 2026, it reports weaker employment growth among ages 22--25 in occupations with high GPT-4 beta exposure and studies employer-match hiring and separation. The CPS cannot reproduce its employer panel, standardized-title mapping, firm controls, or worker--firm flows. Because that study also reports public CPS and ACS evidence, novelty here lies in auditing a fixed national-CPS estimand, not discovering the age pattern.

-

-Write a constructed-treatment analysis as a sequence from an economic construct $\mathcal{C}\$$, through a measurement rule $\mathcal{M}\$$, a harmonization rule $\mathcal{H}\$$, a retained support $\mathcal{S}(\Omega)\$$, and a representation $\mathcal{R}\$$, to an estimand $\mathcal{Q}\$$:

-\[

-\mathcal{C}\xrightarrow{\mathcal{M}}\mathcal{N}\xrightarrow{\mathcal{H}}\mathcal{X}\xrightarrow{\mathcal{S}(\Omega),\mathcal{R}}\mathcal{Q}

-\mathcal{X}\xrightarrow{\mathcal{R}}\mathcal{P}\]

-\]

-Calling every perturbation a different ‘measure’ conceals economically important distinctions. An invalid taxonomy merge is an implementation error. Reweighting the same application--ability links is a same-construct measurement change. Moving from LLM task acceleration to patent overlap or a broad capability gap changes the construct. Expanding mapped occupations changes the target population. Converting a level to quintiles changes the treatment representation. Replacing Q5–Q1 by Q5–Q2 changes the comparison. Rotating a fixed two-regressor column space changes only coordinates.

-

-Table~\ref{tab:architectures} gives the reporting taxonomy used throughout. A row can carry more than one label: repairing a mapping both corrects an implementation and may readmit occupations. The analysis separates the fixed-support score change from the support expansion rather than calling their sum a ‘crosswalk effect.’

-

-\input{tables/table1_architectures.tex}

-

-Two simple invariance results discipline the exercise. First, for fixed positive weights and a fixed tie rule, a positive affine transformation preserves ranks and weighted-quantile membership. Second, if two regressor matrices are related by a nonsingular coordinate transformation and sample, objective, nuisance terms, and constraints remain fixed, fitted values and transformed contrasts are identical. The $\mathcal{F}/\mathcal{G}\$$ to $\mathcal{A}/\mathcal{E}\$$ exercise in the prior draft is therefore a basis change, not additional empirical corroboration; it is retained only in the online appendix.

-

-The task-family construction has an equally useful structure. If $\mathcal{D}_o\mathcal{S}$ is the share of tasks directly accelerated by an LLM and $\mathcal{S}_o\mathcal{S}$ the share accelerated only with complementary software, then

-\begin{equation}

-\mathcal{X}_{(\alpha\ o)}=\mathcal{D}_o\mathcal{S}\mathcal{X}_{(\beta\ o)}=\mathcal{D}_o\mathcal{S}+\frac{1}{\alpha}\mathcal{D}_o\mathcal{S}\mathcal{X}_{(\gamma\ o)}=\mathcal{D}_o\mathcal{S}+\mathcal{S}_o\mathcal{S}\mathcal{X}_{(\delta\ o)}\end{equation}

-\end{equation}

-For any pair of occupations, rankings along $\mathcal{D}+\lambda\mathcal{S}$ can change only where their affine score paths cross. Weighted-bin assignments are consequently piecewise constant in $\lambda\mathcal{S}$ on fixed support. There is no reason, however, for estimated categorical coefficients to be monotone in $\lambda\mathcal{S}$.

-

-\subsection{Exposure architectures}

+Administrative and workplace studies observe adoption or tool access in particular settings \cite{humlum2025still,brynjolfsson2025generative}; occupation scores instead describe tasks a technology might affect. Remote-work evidence supplies another competing interpretation \cite{emanuel2026proximity}. None turns a capability score into observed deployment.

-

-The core audit contains three AIOE implementations and three Eloundou implementations. AIOE links progress in ten AI applications to 52 O*NET abilities and then to occupational ability requirements \cite{felten2018method,felten2021occupational}. Administrative, direct-ability, and source-employment-weighted versions vary reconstruction or aggregation within that system. The Eloundou family changes the technology boundary through equation~\eqref{eq:ds} \cite{eloundou2024gpts}. These are six implementations from two families, not six independent measurements of a known latent exposure.

+The primary score is Eloundou et al.’s GPT-4 beta: tasks are labeled by whether an LLM, alone or with complementary software, could reduce completion time at constant quality \cite{eloundou2024gpts}. AIOE maps AI applications to abilities \cite{felten2018method,felten2021occupational}; Webb measures patent--task overlap \cite{webb2020impact}; Dingel--Weiman classifies teleworkability \cite{dingel2020home}. Computer use, routine content, education, and remotability are therefore overlapping occupational characteristics, not clean alternative treatments. The task literature likewise allows technology to substitute for some tasks, complement others, and reallocate work within occupations \cite{autor2003skill,autor2013growth,acemoglu2019automation}.

-

-The treatment-side dependence is strong. On the exact 444-occupation common support, the first weighted principal component explains 80.04 percent of standardized-score variance and the first two explain 96.11 percent; three explain 99.58 percent. The intersection--union sign calculation for the six coefficients is therefore retained only as a predeclared statement about a dependent set. It is not presented as six independent confirmations, and the failed simultaneous-band criterion remains in the design record.

+This distinction changes the burden of interpretation. Agreement across correlated scores can show that a direction is not unique to one coding rule, but cannot validate a latent exposure construct. Conversely, coefficient differences can arise from task definitions, occupational support, or treatment-group membership. The empirical work below therefore matches supports where possible and names population changes where it is not.

-

-Two externally admitted constructions use different primitives. Webb maps patent text to occupational tasks, while the OECD measure compares nine capability domains with occupational requirements \cite{webb2020impact,oecd2026exposure}. They are different economic constructs, not reliability draws for the same LLM variable. Frey--Osborne remains an automation comparator rather than being relabeled as AI. A repository-bound census records six core implementations from two families, two admitted external implementations from two additional families, one scope failure, and several published candidates whose score artifacts or mappings are not yet locally verified. This is not claimed to be a timeless census of the literature.

-

-\subsection{Relation to labor-market evidence}

-

- \cite{brynjolfsson2026canaries} use proprietary ADP payroll records and emphasize employment declines among ages 22--25 in highly exposed jobs, operating largely through hiring. Their public rule places GPT-4 beta occupations in employment-weighted quintiles and contrasts the top two with the bottom three. Section-\ref{sec:results} implements that public classification in CPS. It cannot recover their employer panel, title-to-occupation mapping, firm-time controls, hiring, or separations.

-

- Other studies answer different questions. \cite{humlum2025still} link reported adoption to Danish administrative outcomes; workplace deployments observe actual tool access rather than occupational potential \cite{brynjolfsson2025generative}. These designs distinguish exposure from use but do not validate one universal occupation score. The broader task literature likewise emphasizes substitution, complementarity, and occupational adjustment rather than a one-dimensional technology dose \cite{autor2003skill,autor2013growth,acemoglu2019automation}.

-

- This paper is closest in spirit to sensitivity analysis that asks which moments or trends support an empirical result \cite{andrews2017sensitivity,rambachan2023credible} and to work on generated covariates \cite{battaglia2024inference,christensen2026valid,duan2026inference,ludwig2026large}. Conventional occupation-clustered inference here conditions on labels, mapping, and support. The online appendix specifies what a validation sample would need to contain before generated-covariate corrections could address occupational AI exposure itself.

+ I distinguish six layers:

+ \[

+ \(\longrightarrow M \longrightarrow H \longrightarrow \Omega \longrightarrow R \longrightarrow Q\),

+ \]

+ the intended construct, measurement rule, occupational harmonization, retained support, representation, and regression estimand. A vintage mismatch is an implementation error; score repair on fixed support changes measurement; occupation readmission changes population; and a different exposure contrast changes the estimand. Common-draw paired inference tests coefficient movement, while each component interval measures its own resolution. Neither a detected movement nor a nondetection assigns a causal share or establishes equivalence. This focus connects the exercise to specification-sensitivity and generated-covariate inference \cite{andrews2017sensitivity,rambachan2023credible,battaglia2024inference,christensen2026valid,duan2026inference,ludwig2026large}.

diff --git a/paper/main/sections/03_measurement.tex b/paper/main/sections/03_measurement.tex

index 17767a..6b927fa 100644

--- a/paper/main/sections/03_measurement.tex

+++ b/paper/main/sections/03_measurement.tex

@@ -1,33 +1,29 @@

-\section{Data, Calendar, and Occupational Harmonization}\label{sec:data}

+\section{Data and Exposure Construction}\label{sec:data}

-

-\subsection{CPS stocks and the corrected calendar}

+\subsection{CPS stocks and calendar}

-

- The public-data application uses IPUMS Basic Monthly CPS records from January 2017 through July 2026. Employed respondents with positive final weights are aggregated into occupation-by-age-group-by-month stocks. Ages 22--25 form the young group and ages 26--65 the comparison group. Age is not labor-market experience, and the outcome can change through employment entry, exit, occupational switching, or the older denominator.

+ I use IPUMS Basic Monthly CPS data from January 2017 through July 2026. Employed respondents ages 18--65 with positive final weights are aggregated to occupation--month stocks; the regression compares ages 22--25 with ages 26--65. Age is not experience, and older workers need not be untreated.

-

- The original wide extract accidentally selected the March ASEC sample ('03s') rather than the Basic Monthly sample ('03b') in 2017--2021. ASEC records carry zero Basic Monthly final weights for this purpose. A separately authenticated extract restores those five March samples. October 2025 remains absent because no CPS was collected during the federal shutdown. The combined files contain 9,843,021 records and 114 observed survey months. Omitting December 2022 as a transition leaves 113 months in the substantive static model.

+ The corrected source calendar contains 114 of 115 possible months. Five March files in the wide extract were ASEC samples and are replaced by authenticated Basic Monthly files for 2017--2021. October 2025 is absent because no CPS was collected during the funding lapse; it is not interpolated. December 2022 is observed but excluded as the transition month, leaving 113 static months.

-

- The corrected calendar is now the substantive baseline. Restoring March changes the coefficient from \$-0.1311\$ to \$-0.1346\$, with a 9,999-draw interval \$[-0.2223, -0.0468]\$. The frozen result remains visible as the confirmatory chronology record but no longer defines the default exposition.

+ The reconstruction scans 9,843,021 source rows and retains 5,322,047 eligible records before routing. Fractional pre-2020 crosswalk descendants are not distinct respondents. The 468-occupation grid contains 52,884 cells; 51,891 enter the likelihood, including 12,965 valid one-sided zeros, while 993 both-age-zero cells contribute nothing. Young cells are sparse: 26.25 percent have zero respondent-equivalent records and 69.83 percent have fewer than five.

-

-\subsection{Taxonomy bridge and universe reconciliation}

+\input{tables/r3_table1_data.tex}

-

- CPS occupations use Census-2010 codes through 2019 and Census-2018 codes thereafter, while source exposures use SOC vintages. Pre-2020 records are expanded along official Census-2010-to-2018 routes and aggregated to the common target taxonomy. In a one-to-many route, official conversion proportions are applied equally to young and older source records. This preserves the source occupation's age ratio across target components by construction; official files do not supply age-specific probabilities.

+\subsection{Harmonization and support}

-

- The full route-expanded preperiod universe contains 539 target occupations with positive young and older stock. Two counts that appeared to be a five-occupation discrepancy are instead nonnested, architecture-specific supports. The stored 490-occupation universe requires full beta coverage. The 495-occupation continuous-mapping universe requires repaired AIOE and Webb coverage. They overlap in 466 occupations: 24 appear only in beta support and 29 only in AIOE-plus-Webb support. Adding Webb to beta leaves the 468-occupation primary model. The online appendix names every member and reason.

+ Pre-2020 Census-2010 occupations are routed to Census-2018 using versioned official shares. Applying the same shares to young and older records preserves source stock and age ratios by construction, but does not validate age-specific routing. The route reaches 99.546 percent of eligible early stock and conserves routed weight to machine precision.

-

- Within primary support, split sources contribute 14.93 percent of young and 17.75 percent of older weighted stock during 2017--2019. Ninety-nine target occupations can receive a split route and carry 23.38 percent of fitted target information; proportional allocation of information by split-stock share assigns 4.44 percent to split stock. The former is a conservative touched-target diagnostic and the latter is not an identified decomposition.

+ Primary support requires finite GPT-4 beta and Webb software scores and positive preperiod young and older stock. It contains 468 occupations and 87.19 percent of candidate-universe stock; a named file records all 73 exclusions. Raw AIOE is a standardized source-occupation index, not a percentage; the source workbook has mean approximately zero, unit sample standard deviation, and an interquartile difference of 1.879 raw points. A literal SOC-2010 AIOE merge into post-2018 employment data covers only 3.33 percent of computer-and-mathematical employment, versus 97.7 percent after version-aware routing. Fixed-support score repair changes little; occupation readmission changes the population. I treat the literal merge as invalid and do not attribute it to another paper without implementation evidence.

-

- The negative association does not depend on split routes. Restricting the corrected model to 369 targets with no structurally possible split inbound route gives \$-0.1674\$ under fixed labels and \$-0.1471\$ after recomputing the comparison on that support. Aggregating to a stable Census-2010 taxonomy gives \$-0.1522\$. Each changes the population and should not be called a correction of the primary estimate.

+ The support file separates missing exposure, missing Webb, and absent young preperiod stock. This matters because a common-support regression estimates a population defined by all required scores, whereas native-support rows can change both the exposure definition and who remains. The appendix's four-step crosswalk table holds support fixed before adding newly reachable occupations, so measurement correction is not conflated with sample composition.

-

- Age-specific route allocation remains uncertain. Under unrestricted official-route accounting, the 2017--2019 young/older stock ratio can range from .1031 to .1480 in Q1 and from .0790 to .1175 in Q5. These are sharp accounting ranges for stocks, not bounds on the nonlinear regression coefficient. A predeclared tilt that changes young-versus-older allocation odds toward the highest-beta eligible target by factors from .5 to 2 moves the coefficient from \$-0.1292\$ to \$-0.1391\$. Only 6.88 percent of early stock is eligible for that scenario, so it is not a global robustness claim.

+\subsection{Exposure groups and production breaks}

-

-\subsection{A detectable workflow failure}

+ The primary exposure is Eloundou et al.'s GPT-4 beta share. CPS stock from January 2017 through November 2022 sets normalization and employment-weighted quintile cuts, with ties retained. The quintiles contain 129, 99, 82, 67, and 91 occupations and about one fifth of preperiod stock each. No postperiod stock enters the rebuilt labels.

-

- Literal exact-code matching of SOC-2010 AIOE into a post-2018 outcome taxonomy covers just 3.33 percent of computer-and-mathematical employment. The documented bridge reaches 97.7 percent. A machine-readable compatibility gate now rejects a merge when declared source and target vintages are not identical and no versioned route is supplied.

+ Employment weighting balances stock rather than occupation counts. Ties are assigned to the lower group at a cut and never split to force exact fifths. Consequently, group membership is a versioned treatment object rather than a percentile transformation that can be recreated from rounded published values.

-

- The corresponding continuous sequence separates layers: original score/original 410-occupation support gives \$-0.0188\$; repaired score/same support gives \$-0.0192\$; repaired score/expanded 495-occupation support gives \$-0.0315\$; excluding computer/math gives \$-0.0294\$. The invalid exact-code result is not an economically defensible alternative. The small first movement concerns score correction; the larger second movement is population re-admission.

+ A historical production treatment instead used the full static panel for weights. On the corrected calendar its coefficient is \$-0.1346\$. Rebuilding from the 71-month preperiod changes nine assignments and moves the coefficient by 0.0024 log points (\$[-0.0013, 0.0062]\$). The revised analysis uses only the rebuilt treatment; the historical row is a checkpoint.

-

-\subsection{Treatment formation}

-

- The primary Eloundou beta score and Webb software control are standardized using preperiod employment weights. Beta is divided into employment-weighted quintiles with ties retained. On the 468-occupation primary support, Q1 contains 133 occupations and 20.02 percent of employment; Q5 contains 91 and 19.53 percent. All support and sensitivity rows state whether cutoffs and scales are frozen or recomputed. Respondent-equivalent counts after route expansion are constructed records, not unique people.

-

-\input{tables/table_sample_flow_revised.tex}

+ Official documentation also identifies population-control changes in January 2025 and January 2026, modified November 2025 collection and weighting, and later longitudinal-identifier corrections. These establish survey-production breaks, not their causal effect on the coefficient. I report endpoint and shutdown-month sensitivities, preserve elapsed calendar time in lags, and do not apply aggregate population adjustments to age-by-occupation cells without the required counterfactual controls.

diff --git a/paper/main/sections/04_data_design.tex b/paper/main/sections/04_data_design.tex

index c61e12f..2f2d0ef 100644

--- a/paper/main/sections/04_data_design.tex

+++ b/paper/main/sections/04_data_design.tex

@@ -1,41 +1,39 @@

-\section{Estimand and Inference}\label{sec:design}

-\\subsection{Employment-stock contrast}

-

-Let $\$N_{\{oat\}}$ be the CPS-weighted employment stock in occupation $\$o$, age group $\$a$, and month $\$t$. Let $\$Y_{\{a\}}$ identify ages 22--25, $\$P_{\{t\}}$ identify January 2023 onward, $\$Q_{\{oq\}}$ denote exposure quintile $\$q$, and $\$W_{\{os\}}$ denote standardized Webb software exposure. The maintained mean is

+Let $\$N_{\{oat\}}$ be CPS-weighted employment stock in occupation $\$o$, age group $\$a$, and month $\$t$; $\$Y_{\{a\}}$ indicate ages 22--25; $\$P_{\{t\}}$ indicate January 2023 onward; $\$Q_{\{oq\}}$ indicate rebuilt exposure quintile $\$q$; and $\$W_{\{os\}}$ be standardized Webb software exposure. The pooled mean is

+\\begin{equation}

+E[N_{\{oat\}}|X]=\exp\{\left[\gamma_{\{oa\}}+\delta_{\{ot\}}+\lambda_{\{at\}}\right.}

+\\sum_{\{q=2\}}^{\{5\}}\beta_{\{q\}}Y_{\{aP\}}t+\theta_{\{W_{\{os\}}\}}\left.\right]\}.\label{eq:headline}

+E[N_{\{oat\}}|\text{mid }X]=\exp\{\left[\gamma_{\{oa\}}+\delta_{\{ot\}}+\lambda_{\{at\}}\right.}

+\\sum_{\{q=2\}}^{\{5\}}\beta_{\{q\}}Y_{\{aP\}}t+\theta_{\{W_{\{os\}}\}}\left.\right]\}.

+\\label{eq:headline}

+\\end{equation}

-Occupation-by-age, occupation-by-month, and age-by-month fixed effects absorb time-invariant age composition within occupation, common occupation shocks, and aggregate age-specific shocks. The target $\$beta_{\{a\}}$ is the change in the young-to-older employment-stock ratio for Q5 relative to Q1 after January 2023, conditional on Q2--Q4 and Webb software exposure. It is not an individual employment probability, a hiring rate, or a continuous exposure slope.

+The target $\$beta_{\{a\}}$ is the postperiod change in the young-to-older stock ratio for Q5 relative to Q1, conditional on Q2--Q4 and Webb. It is not an individual probability, continuous slope, or hiring rate.

-

-With two age groups, conditioning independent Poisson pseudo-likelihood contributions on $\$T_{\{ot\}}=N_{\{oyt\}}+N_{\{oot\}}$ gives

+With two age groups, conditioning on $\$T_{\{ot\}}=N_{\{oyt\}}+N_{\{oot\}}$ removes the occupation--month effect and gives

+\\begin{equation}

+\\ell_{\{ot\}}=N_{\{oyt\}}\log p_{\{ot\}}+N_{\{oot\}}\log(1-p_{\{ot\}}).\quad

+\\operatornamename{logit}(p_{\{ot\}})=\log(\mu_{\{oyt\}}/\mu_{\{oot\}}).\label{eq:conditional}

+\\operatornamename{logit}(p_{\{ot\}})=\log(\mu_{\{oyt\}}/\mu_{\{oot\}}).

+\\label{eq:conditional}

+\\end{equation}

-The occupation-month effect cancels. CPS final weights construct the two continuous cell stocks and are not applied a second time to the objective. The grouped-binomial expression is an estimating equation for weighted stocks, not a literal model of independent survey-weighted trials.

-

-The central conditioning exercise augments equation-\\ref{eq:headline} with SOC2-family-specific $\$Y_{\{aP\}}$ terms. A saturated version instead absorbs SOC2-family-specific young-relative effects in every month. These models ask whether detailed exposure groups predict outcomes after broad-family evolution is conditioned out. They do not decompose the baseline coefficient into causal AI and non-AI components, because the conditioning set and therefore the estimand change.

+Final weights construct continuous cell stocks once; equation-\\ref{eq:conditional} is an estimating criterion, not a literal independent-binomial sampling model.

-

-\\subsection{Maintained source of randomness}

+The central alternative gives each SOC two-digit family its own young-relative calendar-month path; a less saturated companion uses family-specific post shifts. Both change the conditional estimand. Neither the residual coefficient nor its movement from equation-\\ref{eq:headline} is an AI remainder or composition share.

-

-The principal confidence interval treats occupation-level economic score contributions as independent across 468 occupation clusters and permits arbitrary serial dependence within occupation. It conditions on the observed CPS-weighted cells, constructed exposure labels, taxonomy mapping, support, quintile cuts, and fitted specification. Thus it is model-based uncertainty for a stochastic occupational-shock parameter. It is not a complete design-based CPS variance estimator and does not cover exposure-label generation, taxonomy-allocation uncertainty, or every dependence induced by respondents moving across occupations.

+The occupational-conditioning, influence, dynamic, and flow analyses added in this revision were designed after the initial outcome pattern was known. They are post-outcome exploratory analyses, not preregistered confirmatory tests; simultaneous inference does not undo that selection.

-

-For transparency, let $\$x_{\{ot\}}$ be the target and nuisance regressors after the grouped-binomial fit, $\$p_{\{ot\}}$ the fitted young share, and $\$h_{\{ot\}}=T_{\{ot\}}p_{\{ot\}}(1-p_{\{ot\}})$. The implementation absorbs the fixed effects from $\$x_{\{ot\}}$ under weights $\$h_{\{ot\}}$, producing $\$\\widetilde{x}_{\{ot\}}$. It forms the information matrix

-\\[

-H=\\sum_{\{o,t\}}h_{\{ot\}}\\widetilde{x}_{\{ot\}}\\widetilde{x}_{\{ot\}}',

-\\]

-the occupation score $\$s_o=\\sum_t\\widetilde{x}_{\{ot\}}(N_{\{oyt\}}-T_{\{ot\}}p_{\{ot\}})$, and the one-step influence $\$H^{-1}s_o$, with the finite-cluster factor $\$\\sqrt{G/(G-1)}$. Each wild-score draw multiplies the entire occupation influence vector by one Rademacher sign. The analytic occupation-cluster standard error studentizes the centered draw; the 95-percent critical value is the empirical .95 quantile of the absolute studentized shifts. Paired comparisons reuse the same signs for both fits, thereby preserving estimated covariance. The procedure does not create pseudo-outcomes or refit each draw.

+\\subsection{What the design establishes}

-

-All central revised intervals use 9,999 multipliers. The sole canonical interval for the frozen $\$-0.1311$ estimate is $\$[-0.2171,-0.0451]$; previously displayed variants came from different finite sets of 999 multipliers, except a minimum-count row that changed the sample. Pair-specific member intervals remain labeled as such and never replace the canonical interval.

+The design measures the magnitude, support, and specification sensitivity of a public-CPS association. It distinguishes paired coefficients and locates the occupational comparisons carrying the contrast, but cannot allocate movement among AI, computerization, pandemic recovery, or other forces. Attenuation is not a causal share, survival is not an AI remainder, and an interval containing zero is nondetection rather than economic equivalence. The limitations section returns to preperiod evolution, sparse direct support, exposure without adoption, and incomplete flow and survey-design information.

-

-\\subsection{Precision, sparse cells, and rotation sensitivity}

+For shared observations I reuse bootstrap multipliers and form each draw of $\$\\Delta=\\widehat{\\beta}_2-\\widehat{\\beta}_1$. A paired interval excluding zero shows statistical distinguishability under that procedure; it does not identify the reason. An interval including zero means nondetection, not economic equivalence. I report $\$\\mathrm{MDE}_{\{80\\}}\\simeq(1.96+0.84)SE$ only as a precision diagnostic.

-

-The realized primary SE is 0.0444, versus 0.0122 in the planning simulation. Its normal-theory 80-percent minimum detectable effect is therefore 0.1244 log points, or 13.25 percent on the relative-stock-ratio scale. The paper uses MDEs to characterize what the design can resolve; it does not substitute an arbitrary smallest effect of interest or interpret a confidence interval containing zero as equivalence.

-

-Young cells are sparse: the 10th, 50th, and 90th percentiles of respondent-equivalent counts are 0, 2, and 16; 26.3 percent are zero and 69.9 percent are below five. Older-cell percentiles are 4, 25, and 199. One-sided cells contribute valid finite terms to equation-\\ref{eq:conditional}; only 965 occupation-months with both stocks zero are omitted. Conditioning on at least five observed young respondents attenuates the coefficient to about $\$-0.109$, but this selects on realized outcomes, retains fewer than half of occupation-months, and changes the estimand.

+To describe where a target is learned, let $\$h_{\{ot\}}=T_{\{ot\}}p_{\{ot\}}(1-p_{\{ot\}})$ and let $\$r_{\{ot\}}$ be the target regressor after projection on every nuisance regressor in the fitted-information metric. I report

+\\[

+I=\\sum_{\{o,t\}}h_{\{ot\}}r_{\{ot\}}^2,\quad

+s_o=\\frac{\\sum_t h_{\{ot\}}r_{\{ot\}}^2}{I},\quad

+G_{\\mathrm{eff}}=\\left(\\sum_o s_o^{-2}\\right)^{-1}.

+\\]

+These are fitted target information and concentration diagnostics, not an effective number of CPS sampling clusters. Family and occupation shares, leave-one-out estimates, and the top-five information share are reported without using influence to select a preferred sample.

-

-Quarterly aggregation gives $\$-0.1310$ under the frozen chronology and $\$-0.1345$ under the repaired chronology, essentially matching monthly estimates and SEs. To probe CPS rotation and common monthly shocks, a post-outcome score-covariance sensitivity adds contemporaneous and lagged covariance in aggregate month scores while subtracting the occupation-month intersection. At a 12-month Newey--West lag the SE rises to 0.0491 and the normal interval is $\$[-0.2273,-0.0349]$. Occupation clustering already permits arbitrary within-occupation serial dependence; this extra calculation addresses declared cross-occupation time covariance but remains model-based rather than a CPS replicate-weight variance.

+The principal 9,999-draw wild-score interval treats occupations as shock clusters and allows arbitrary serial dependence within occupation. It conditions on weighted cells, exposure labels, mapping, support, and specification. Separate sensitivities use 22 broad-family shocks, household/sample-unit full refits under released weights, and elapsed-calendar time-HAC scores. A sparse-cell and family-shock simulation audits finite-sample behavior. These procedures have different stochastic targets; they are not mechanically combined, and none supplies complete CPS design inference or exposure-label uncertainty.

-

-Finally, a historical simulation that preserves each donor month's entire contemporaneous cross-occupation residual vector under a sharp global-sign design closes none of the prospective-to-realized precision gap. It rules out that specific implementation as the explanation; it does not validate the planning calculation or identify which unmodeled dependence generated the gap.

+The household exercise keeps every month and fractional route descendant of a positive CPS household identifier together. It still omits unavailable multistage-selection and calibration uncertainty. The time-HAC construction uses elapsed calendar lags, so the missing October 2025 is not treated as adjacency between September and November.

diff --git a/paper/main/sections/05_support.tex b/paper/main/sections/05_support.tex

index 0983e58..aa82f9c 100644

--- a/paper/main/sections/05_support.tex

+++ b/paper/main/sections/05_support.tex

@@ -1,41 +1,30 @@

-\\section{Where the Employment Association Comes From}\\label{sec:results}

+\\section{Which Occupational Comparisons Support the Pattern?}\\label{sec:results}

-

-\\subsection{Baseline and broad occupation families}

+Table-\\ref{tab:occupation} begins with the rebuilt pooled result. The Q5--Q1 coefficient is $\$-0.1321$, with standard error 0.0452 clustered by occupation. Its 95-percent interval is $\$[-0.2206,-0.0437]$. Giving every SOC two-digit family its own young-relative calendar-month path yields $\$-0.0217$ ($\$[-0.1607,0.1173]$). On common draws, the conditioned-minus-pooled movement is 0.1104 ($\$[0.0107,0.2102]$), with paired $\$\\mathrm{MDE}_{\{80\\}}=0.1454$. Thus the conditional estimands are distinguishable while the residual gradient is imprecise.

-

-The substantive baseline uses the repaired 113-month calendar and frozen beta treatment definition. The Q5--Q1 coefficient is $\$-0.1346$ with occupation-clustered SE 0.0450 and canonical 95-percent interval $\$[-0.2223,-0.0468]$. The frozen design record remains $\$-0.1311$ with the canonical interval $\$[-0.2171,-0.0451]$. Calendar repair therefore does not explain the association.

+\\input{tables/r3_table2_occupation.tex}

-

-The direct regression analogue of the earlier within-SOC2 permutation does. Adding broad-family-specific young-relative post terms reduces the coefficient to $\$-0.0315$; allowing each family its own young-relative monthly path gives $\$-0.0317$. Both intervals include zero. On common draws, the SOC2-by-post estimate minus baseline is $\$+0.1031$, with paired interval $\$[0.0035,0.2026]$; the

monthly version gives $\$0.1028$, with interval $[\$0.0035, \$0.2021]$. Thus the movement is detected even though the residual coefficient is not. The ratio $\$0.1104/0.1321$ is not a causal ‘‘share explained.’’ Broad families differ in AI-relevant tasks, computerization, education, work location, and pandemic recovery; family paths absorb all such common evolution. The result says that the pooled estimate relies mainly on comparisons across families, not what would have happened without AI.

-This is strong evidence about the comparison and weak evidence about a structural mechanism. In point-estimate terms, conditioning absorbs about 76 percent of the baseline magnitude. But the target changes from a cross-economy Q5–Q1 contrast to a coefficient assembled from comparisons remaining within broad families after their differential evolution is removed. Only four of 22 SOC2 families contain both Q1 and Q5, and conditional target information falls from 26.4 million to 8.0 million, about 30 percent of baseline. The conditional coefficient is connected through intermediate quintiles in most families. It should not be described as ‘‘the remaining AI effect,’’ and the difference should not be described as the causal contribution of broad composition.

+The full Q2–Q5 profile reaches the same conclusion. Pooled coefficients are jointly different from zero but do not establish a monotone dose response. After family-month conditioning, the four coefficients are jointly indistinguishable from zero. Simultaneous intervals and paired movements appear in the online appendix.

-\input{tables/table2_support.tex}
+ \subsection{Direct support and restrictions}

-\begin{figure}[htbp]
-\centering
-\includegraphics[width=0.92\textwidth]{figure2_composition_major_revision.pdf}
-\caption{Broad occupational composition and alternative comparisons}\label{fig:composition}
-\tabnote{All rows are post-outcome exploratory and use 9,999 occupation wild-score draws. SOC2 rows change the conditioning estimand. Food and influence rows retain frozen groups but change support. The BOC row changes the exposure-group contrast. Intervals are therefore informative within rows but the coefficients are not estimates of one invariant parameter.}
-\end{figure}

+Only SOC27, 29, 31, and 41 contain Q1 and Q5 occupations. Elsewhere the regression links partial segments---for example Q1–Q3 in one family and Q4–Q5 in another---through common quintile coefficients. That is valid regression variation but not repeated side-by-side tail comparisons.

+A direct-tail benchmark uses only Q1 and Q5 occupations in the four spanning families. Its 29 occupations account for 5.03 percent of preperiod stock. The family-month estimate is 0.1494 ($[-0.1694, 0.4682]$), so the CPS cannot precisely resolve a direct tail contrast within broad families. This row changes population and support and cannot replace the primary estimand.

-The prior within-SOC2 permutation is consistent with this result but is no longer asked to carry it. Randomly reassigning detailed beta scores within broad families produces a mean coefficient of $\$ -0.1097$ against the frozen $\$ -0.1311$. Its exchangeability assumption is not established, so the permutation percentile is descriptive. The conditioned regressions state the relevant comparison directly and show the support and information cost.

+A continuous companion instead subtracts each family’s preperiod-weighted beta mean and imposes a common within-family slope. Its family-month coefficient is $\$ -0.0025$ per 0.1084 raw beta units ($[-0.0240, 0.0191]$). It uses more residual variation but trades the tail contrast for linearity and a common slope. Together, the specifications expose the support--functional-form tradeoff.

-\subsection{Profile, reference tail, and stable classifications}
+ \subsection{Influence and observed paths}

-The fitted post coefficients relative to Q1 are $\$0$, $\$ -0.0855$, $\$ -0.0478$, $\$ -0.0970$, and $\$ -0.1311$. A three-restriction test does not reject a common coefficient for Q2–Q5 ($\$ = 119\%$), but cannot establish their equality. The visible Q3–Q2 increase also does not yield a rejection of monotone decline under a least-favorable max- t test ($\$ = 393\%$). The data therefore neither establish nor reject monotonicity.

+The online appendix reports nuisance-adjusted information, effective information counts, and family and occupation deletions. Fast-food and counter workers have substantial signed influence because low-exposure service employment recovered differently after the pandemic, but other occupations push oppositely. Outcome-selected deletion is not a design correction; symmetric trimming and economically defined service exclusions remain diagnostics.

-Changing the requested contrast clarifies Q1’s role. Q5–Q2 is $\$ -0.0456$ with interval $[-0.1248, 0.0335]$; Q5–Q4 is $\$ -0.0340$ with interval $[-0.1110, 0.0429]$; Q5–Q3 is $\$ -0.0833$ with interval $[-0.1581, -0.0084]$. These are different economic contrasts, not alternative omitted-category parameterizations. Figure-\ref{fig:paths} shows why: both the low-exposure trajectory and the older-worker denominator matter. The fitted contrast is not simply a collapse of young employment in Q5.

+Figure-\ref{fig:paths} separates young and older stocks for both tails. Because numerator and denominator move, the coefficient is not an individual young worker’s loss. The low-exposure reference trajectory is part of the result.

-\begin{figure}[htbp]
-\centering
-\includegraphics[width=\textwidth]{figure3_q1_q5_paths.pdf}
-\caption{Q1 and Q5 employment stocks and ratios}\label{fig:paths}
-\tabnote{Monthly CPS-weighted stocks under frozen beta membership. The upper row indexes to observed 2019 means and the lower row to observed 2022 means. The dashed line marks January 2023. October 2025 is absent because no CPS was collected. The paths are descriptive and do not additively decompose the nonlinear coefficient.}
+\includegraphics[width=0.94\textwidth]{r3_figure1_q1_q5_paths.pdf}
+\caption{Young and older employment stocks in the exposure tails}\label{fig:paths}
+\tabnote{Monthly CPS-weighted stocks use rebuilt beta labels and are indexed to each series’ 2019 mean. The ratio panel divides young by older stock before indexing. The vertical line marks January 2023; December 2022 is the transition month, and October 2025 is absent rather than interpolated. Paths are descriptive.}
-\end{figure}

-Only 46 occupations are always Q1 and 18 always Q5 across the six original implementations. These stable tails account for 9.74 percent of common-support employment. Their contrast is $\$ -0.2120$ with interval $[-0.4131, -0.0109]$, but has only 15.0 effective information occupations and a 46.7-percent top-five information share. The negative estimate survives in this narrow population; it does not show that unstable classification is innocuous in the broader one.

-\subsection{Influence and the food-service hypothesis}

-Influence is economically consequential but not one-directional. Jointly deleting the five occupations with the largest absolute prior leave-one-out movements reduces the estimate to $\$ -0.1011$. Deleting the top ten or twenty instead makes it more negative, $\$ -0.1522$ and $\$ -0.1555$, because later occupations have opposing signed influence. A symmetric 2.5-percent trim of each signed-influence tail gives $\$ -0.1281$, and Huber-style continuous down-weighting gives $\$ -0.1314$. These are outcome-adaptive diagnostics rather than preferred specifications.

-Cleaner occupation-side exclusions do not support the strong claim that food-service recovery explains the headline. Dropping the five Q1 food-service occupations yields $\$ -0.1196$, an 8.8-percent reduction in absolute magnitude. Dropping all food-service occupations gives $\$ -0.1201$. Dropping all food, building-and-grounds, personal-care, and protective-service occupations yields $\$ -0.1370$; dropping only their Q1 members gives $\$ -0.1387$. These exercises cannot isolate reopening, immigration, minimum wages, or labor supply, but they reject the simpler assertion that removing in-person service occupations makes the pattern disappear.

diff -git a/paper/main/sections/06_stock_results.tex b/paper/main/sections/06_stock_results.tex
index 427faae..649a9b7 100644
--- a/paper/main/sections/06_stock_results.tex
+++ b/paper/main/sections/06_stock_results.tex
@@ -1,36 +1,19 @@
-\section{What Exposure Construction Changes---and What the Data Resolve}\label{sec:family}
+\section{Characteristics and Inferential Resolution}\label{sec:characteristics}

-\subsection{Paired architecture precision}

+On the focused 408-occupation support, beta’s preperiod-weighted correlation is 0.794 with O*NET computer use and 0.589 with remotability. A static horse race is therefore descriptive: residual variation may be narrow, and a control may also be an AI channel.

-Alternative exposure constructions change occupation ranks, tail membership, and point estimates. The key inferential question is whether the downstream coefficient differences are resolved once both estimates use common support and common occupation multipliers. Table-\ref{tab:stock} answers no for the original Q5–Q1 comparison. Every beta-minus-alternative paired interval contains zero. The closest case is beta minus the OECD capability gap, $\$ -0.1115$ with interval $[-0.2257, 0.0027]$; its paired MDEs_{80} is 0.1689. Beta minus Webb AI is $\$ -0.0646$ with interval $[-0.1682, 0.0389]$ and MDEs_{80} $\$ 0.1499$.

+The leading computerization measure is the May 2020 O*NET rating ‘‘Interacting With Computers.’’ On common support the beta-plus-Webb coefficient is $\$ -0.107$. Adding computer use makes it $\$ -0.212$; the paired movement is $\$ -0.105$ ($[-0.183, -0.027]$). Computer use acts as a suppressor, rejecting the simple claim that this control necessarily attenuates the CPS contrast but not isolating AI. Remotability alone gives $\$ -0.116$ and an undetected $\$ -0.009$ movement. Rebuilt-treatment checks of wages, education requirements, and routine/manual content likewise do not yield individually detected movements on their declared maximal supports. A separately repaired pandemic-shortfall exercise uses bounded predictions and a stricter 363-occupation support; total-stock and young-relative conditioning move the pooled coefficient by only 0.0004 and 0.0011 log point. Regenerating these controls within linked-household draws widens uncertainty but does not resolve a conditioning movement. These are exploratory diagnostics, not equivalence results.

-\input{tables/table3_stock_results.tex}
+\input{tables/r3_table3_characteristics.tex}

-The correct conclusion is not that all architectures yield the same economic relationship, nor that a smaller alternative estimate shows the pattern ‘‘does not extend.’’ The design does not detect these paired differences and is too imprecise to rule out economically important movements in either direction. This is itself a result: the employment application reveals much more about how treatments are constructed than about which construction has a different population coefficient.

+Broad-family paths and a computer-use control need not move the coefficient alike: the former remove all family-common evolution, while the latter partials one correlated dimension. Neither is a causal technology decomposition.

-\begin{figure}[htbp]
-\centering
-\includegraphics[width=0.90\textwidth]{figure4_architecture_precision.pdf}
-\caption{Paired differences and the resolution of the CPS design}\label{fig:architectureprecision}
-\tabnote{Points are beta-minus-alternative Q5–Q1 coefficient differences on pair-specific common support. Thin lines are 95-percent common-draw paired intervals. Gray bars span plus or minus the normal-theory MDE at 80-percent power. An interval containing zero is a failure to detect a difference, not evidence of equivalence.}
-\end{figure}

+Industry conditioning is built from occupation--industry--age--month cells rather than dominant-industry labels. Allowing 13 broad industries their own young-relative post shifts moves the stratified coefficient from $\$ -0.138$ to $\$ -0.099$; the paired 0.039 interval $[\$ 0.008, \$ 0.070]$ excludes zero. It does not measure firm AI adoption. BA and non-BA estimates are $\$ -0.138$ and $\$ -0.123$; their paired difference is $\$ -0.015$ ($[-0.156, 0.125]$), with $\text{MDE}_{80} = 0.201$. Exact-age estimates vary but simultaneous paired intervals do not isolate a particular age mechanism.

-The original six-score intersection-union calculation is correspondingly demoted. All six marginal point estimates are negative on their literal intersection, and the maximum one-sided marginal tail area is .045, while the predeclared simultaneous-band criterion fails. With two principal components explaining 96.11 percent of score variance, this is a dependent-set sign statement, not six independent validations. Nothing in it overturns the paired-precision finding.

+The industry row changes the cell objective as well as the nuisance set, so it is reported separately from occupation-level characteristic controls. Education and exact-age estimates use literal common supports and common draws; no subgroup is selected by its point estimate. The age coefficients are jointly unequal, but each simultaneous age-minus-pooled interval contains zero.

-\subsection{A direct bridge to the motivating result}

+Table-\ref{tab:inference} separates uncertainty targets. Occupation wild scores allow serial dependence within occupation. A 22-family sensitivity permits common broad-family shocks. Household full-refit multipliers preserve repeated observations and co-residence under released weights but are not full CPS design inference. None covers exposure labels or route allocation.

-The published component of `\cite{brynjolfsson2026canaries}` that can be reproduced is a different representation: GPT-4 beta exposure divided into employment-weighted quintiles, with Q4--Q5 compared against Q1--Q3. Implementing that classification in the CPS model gives $\beta = -0.0728$ with interval $[-0.1240, -0.0216]$, roughly half the corrected Q5--Q1 log-point magnitude. The unconditioned result is $\beta = -0.0728$ as well. Young CPS stock rises 5.78 percent in the bottom-three group and falls 5.03 percent in the top-two group between November 2022 and June 2026, while older stocks rise by about 0.6 percent in each. These aggregate growth calculations do not inherit a hiring interpretation.

+`\input{tables/r3_table4_inference.tex}`

-Reapplying the BCC grouping rule to eight exposure constructions produces native-support coefficients from $\beta = -0.0817$ for Eloundou broad to $\beta = -0.0096$ for the OECD capability gap. On literal 426-occupation common support, β is $\beta = -0.0810$ and OECD is $\beta = -0.0076$; their paired difference is $\beta = -0.0733$ with interval $[-0.1390, -0.0076]$. The other six paired differences are not detected. This does not contradict Table-\ref{tab:stock}: grouping changes the estimand and its precision. It illustrates that the economic contrast is part of the constructed empirical object.

-

-`\input{tables/table4_reallocation.tex}`

-\subsection{Taxonomy repair and construction layers}

-

-Taxonomy repair is the clearest measurement finding because it separates an invalid workflow from defensible alternatives. Literal matching of SOC-2010 AIOE codes into a post-2018 occupation universe covers only 3.33 percent of computer-and-mathematical employment. The documented route reaches 97.7 percent. In a continuous comparison, the original score on original support gives $\beta = -0.01885$; the repaired score on the same 410 occupations gives $\beta = -0.01920$; expanding to 495 repaired-score occupations gives $\beta = -0.03156$; excluding computer/math from the expanded support gives $\beta = -0.02940$. Most movement comes from changing who enters, not correcting scores among already matched occupations.

-

-This sequence is not a multiverse of equally defensible estimates. The exact-code merge is an implementation failure. The fixed-support row measures score correction. The expansion changes the target population. The computer/math exclusion asks whether one family drives the expanded-support coefficient. Reporting these layers separately is the portable contribution.

-

-\subsection{Primitive models rather than rotated labels}

-

-The prior draft emphasized ‘‘consensus’’ and ‘‘disagreement’’ coordinates $\beta_F = (A+E)/2$ and $\beta_G = (A-E)/2$. Those coordinates are retained in the online appendix only as a complete record. A representative two-score model is more transparent: administrative AIOE enters at $\beta = 0.0449$ and β at $\beta = -0.0788$ when included jointly. In direct task units, β_D and β_S enter at $\beta = -0.0323$ and $\beta = -0.0303$. These are conditional associations among correlated constructed regressors, not causal technology channels. Transforming β_F/β_G back to β_A/β_E reproduces the same fitted predictor exactly and supplies no additional evidence.

+With 22 family clusters, the paired movement remains detected under Rademacher and Webb weights; the Webb interval is $[0.0059, 0.2150]$. The completed finite-sample audit finds small coefficient bias but target-specific interval performance. In the empirical observed-effect design, occupation-wild coverage is 0.865 for the pooled coefficient, 0.921 for the family-conditioned coefficient, and 0.889 for their paired movement; family clustering is conservative for the pooled and paired targets but covers the family-conditioned target only 0.895 of the time. In the clean zero-target design, rejection rates for the family-conditioned coefficient are 6.1 percent under occupation weights, 8.9 percent under family weights, and 5.6 percent for the cross-fit oracle. No one procedure is therefore validated for all targets and declared designs. In the rebuilt-treatment audit, elapsed-calendar HAC leaves scalar uncertainty similar and none of the 15 five-parameter covariance matrices across three targets and five lags is indefinite. Earlier indefinite matrices belong to the superseded historical-treatment audit and remain labeled in the archive.

diff --git a/paper/main/sections/07_reallocation.tex b/paper/main/sections/07_reallocation.tex
index f0b742d..2f7dc2f 100644
--- a/paper/main/sections/07_reallocation.tex
+++ b/paper/main/sections/07_reallocation.tex
@@ -1,23 +1,38 @@
-\section{Timing, Weights, and Remaining Interpretations}\label{sec:reallocation}
+\section{Dynamics, Flows, and External Comparison}\label{sec:dynamics}

-\subsection{Pre-AI pseudo-breaks and post-period dynamics}
+\subsection{Dynamics}

-The authenticated extract begins in January 2017, so the literal request for monthly pseudo-breaks beginning in 2015 is infeasible. I estimate every feasible break in the fully pre-AI January 2017--December 2019 window and emphasize 12 dates with at least 12 retained months on each side. Under both pre-AI classifications and the frozen classification, the balanced pseudo-break distributions are centered near zero; none approaches $\beta = -0.1311$. Because the 12 estimates overlap and were designed after the headline was known, the plus-one empirical tail of $\beta/13$ is a resolution floor, not a randomization β -value. Unbalanced early-window extremes are artifacts of having as little as one pre-break month.

+The binary post coefficient is not an event-study design. I estimate quarterly coefficients for all four nonreference quintiles, omitting 2022Q4 (October--November; December is the transition month). The Q5 path has 23 preperiod and 15 postperiod coefficients in pooled and family-month specifications.

-The actual post path remains descriptive rather than a causal event study. A model with separate periods gives $\beta = -0.1356$ in 2023, $\beta = -0.0783$ in 2024, and $\beta = -0.1629$ in 2025--26. The last is distinguishable from 2024 but not from 2023. No detected pretrend and an end-2022 break do not isolate AI from the technology-sector correction, interest rates, return-to-office changes, post-pandemic normalization, or occupational composition.

+Figure-\ref{fig:dynamics} shows the rebuilt Q5 path. Joint tests reject that all 23 preperiod coefficients equal zero in both the pooled ($\beta = -0.0149$) and family-conditioned ($\beta = -0.0053$) models. An insignificant anchored linear slope cannot override those unrestricted tests. January 2023 therefore fails the parallel-evolution diagnostic for causal timing.

-I do not add a national adoption series merely to replace the post indicator. Interacting one national series with fixed occupation exposure would largely reparameterize an already displayed exposure-by-time path. Industry-level BTOS adoption would require an occupation--industry ecological bridge and would still not generate exogenous occupational adoption. Such data could support a separately designed adoption paper, but forcing it into this measurement-centered revision would change both treatment and estimand without resolving the main confound.

+`\begin{figure}[htbp]`
+`\centering`
+`\includegraphics[width=0.94\textwidth]{r3_figure2_dynamics.pdf}`
+`\caption{Quarterly Q5--Q1 young-relative employment coefficients}\label{fig:dynamics}`
+`\tabnote{Coefficients are relative to Q1 and omitted 2022Q4 under rebuilt labels; lines are pointwise 95-percent occupation wild-score intervals. The online appendix reports simultaneous bands and the Q2--Q5 system. The conditioned model gives each SOC2 family its own young-relative monthly path.}`
+`\end{figure}`

-\subsection{Population controls and response composition}

+The calendar-weighted post functional is distinct from the nonlinear static coefficient. It is $\beta = -0.120$ ($[-0.264, 0.024]$) when pooled and $\beta = -0.207$ ($[-0.425, 0.010]$) after family conditioning. Official `\texttt{HonestBID}` 0.2.8 applies smoothness and relative-magnitude restrictions to the full event vector and covariance. Both the full 2017Q1--2022Q3 calibration and a consecutive 2021Q1--2022Q3 post-reopening calibration are reported. Under the latter, the family-conditioned smoothness interval is $\beta = -0.772, -0.007$ when differential-trend bias is restricted to be exactly linear ($\beta = 0$), but expands to $[-1.100, 0.207]$ at the first positive grid value, $\beta = 0.005$. Both conventional intervals already include zero, so no positive zero-exclusion threshold exists. The exercise cannot repair rejected preperiod equality or identify adoption.

-January 2025 introduced unusually large CPS population-control revisions. A literal age-by-occupation counterfactual series ‘‘without’’ those controls cannot be constructed: BLS provides aggregate experimental adjustments, not subgroup weights, and official earlier microdata are not revised. I therefore report what the available data identify rather than rescale cells ad hoc.

+Onset estimates from November 2022 through June 2023 are similar within each structure, but common timing does not distinguish AI from coincident shocks. Ending in December 2024 gives a pooled coefficient of $\beta = -0.111$ ($[-0.201, -0.021]$); its paired movement relative to the July-2026 estimate is $\beta = 0.021$ ($[-0.007, 0.049]$), so the design does not detect a change from extending the endpoint beyond 2024. Ending in late 2025 also gives a pooled coefficient near $\beta = -0.111$, versus $\beta = -0.132$ through July 2026, but those paired endpoint movements are detected. Splitting the postperiod instead gives $\beta = -0.111$ in 2023--24 and $\beta = -0.166$ in 2025--26; their paired difference is $\beta = -0.056$ ($[-0.123, 0.012]$), so the official-weight design does not detect a change between eras. Family-conditioned endpoint changes remain near zero. Survey-gap exclusions are small. Declared post-2020 and saturated seasonal models that did not converge remain recorded failures.

-On the repaired calendar, official weights give $\beta = -0.1346$ and respondent-equivalent cells give $\beta = -0.1348$. Ending in December 2024 gives $\beta = -0.1135$ and $\beta = -0.1056$, respectively. In a joint official-weight model, 2023--24 is $\beta = -0.1108$ and 2025--26 is $\beta = -0.1664$; their paired difference is $\beta = -0.0556$ with interval $[-0.1228, 0.0116]$. Under respondent equivalents the era difference is more negative and detected. The raw official-weight Q5--Q1 log-ratio contrast becomes slightly less negative from December 2024 to January 2025. These checks do not provide a counterfactual weight vintage, but they are inconsistent with revised population weights mechanically creating the full association.

+`\subsection{Linked CPS margins}`

-‘‘Respondent equivalent’’ means that cells are reconstructed from observed employed records after route expansion without their CPS final weights. It changes the empirical cell outcome; dividing all weighted stocks by a common constant would not change the maximizer and is not what is done here.

+Table-\ref{tab:flows} estimates distinct adjacent- and twelve-month margins using validated links and official longitudinal weights. Every primary interval includes zero. Normal-theory MDEs_{80} β values range from 0.115 to 0.336 log point, so these are low-resolution nondetections rather than evidence of absent flow responses.

-\subsection{Age, entry, and mobility}

+`\input{tables/r3_tables5_flows.tex}`

-The pooled older comparison is not the sole source of the pattern. Ages 22--25 versus 26--35 give $\beta = -0.1307$, nearly identical to the frozen pooled result. Comparisons with ages 26--45 and 36--55 are $\beta = -0.1475$ and $\beta = -0.1464$. The contrast against ages 51--65 is smaller, $\beta = -0.0978$, and its paired difference from the pooled comparison is detected. Age 22--25 is an age group, not observed experience; the paper therefore uses ‘‘young’’ rather than treating every respondent as a labor-market entrant.

+Employment exit starts from employed origins; reported occupation change requires employment at both endpoints; entry destination conditions on nonemployment-to-employment links and measures allocation across exposure groups. The last is neither a finding probability nor employer hiring. Twelve-month links are endpoints, not sums of monthly transitions, and retain 50.98 percent of eligible young origins versus 70.55 percent of older origins. Hours ($\beta = -0.119$, $[-0.295, 0.057]$) and conditional weekly earnings ($\beta = -0.0245$, $[-0.1024, 0.0534]$) are also imprecise and condition on employment; the earnings series ends in March 2023 and contains only three postperiod months.

-The main outcome cannot determine whether hiring, separation, or occupation switching generated the stock association. CPS longitudinal links do allow distinct flow estimands, but their intervals are

wide. Entry destination is an allocation conditional on moving from nonemployment to employment, not an employment-finding probability. The employed-to-employed switch frame says still less about hiring. For that reason, the mobility analysis, family-balanced disagreement statistics, rematching benchmark, and flow regressions appear in the online appendix rather than as a main contribution. No immediate-reversal rate is treated as a validated occupation-coding-error rate.

+All flow exposure is defined at the origin where an origin occupation exists. No current occupation is imputed to a nonemployed respondent. Annual endpoints miss intervening jobs and transitions, and positive longitudinal-weight eligibility changes the population; neither horizon is a mechanical decomposition of the stock coefficient.

+ \subsection{Payroll bridge and exposure architecture}

+ BCC's August 2026 19-percent "kept pace" shortfall is a stock-growth comparison, not this regression coefficient or a hiring rate. Complete public membership and tie rules are unavailable, so I use an equal-occupation, tie-preserving approximation and label it a bridge. Its top-two-versus-bottom-three CPS coefficient is $-\$0.0720\$$ ($[-\$0.1218, -\$0.0222\$]$); family-month conditioning gives $-\$0.0168\$$ ($[-\$0.0810, -\$0.0474\$]$), with an undetected paired movement. Differences in universe, ages, outcome, endpoints, employer controls, and worker--firm flows remain.

+ The bridge therefore asks whether a closer public grouping changes the CPS stock pattern; it does not calibrate CPS transitions to proprietary employer flows. The required entry, exit, job-to-job, occupation-switching, aging, and denominator quantities are not jointly observed.

+ Finally, $\$D\lambda$ varies the weight on tasks requiring complementary software. Each row uses rebuilt weights and common draws, and $\$lambda=\5 exactly reproduces beta. Native-tail coefficients range from $-\$0.103\$$ at $\$lambda=\0 to $-\$0.156\$$ at $\$lambda=\1 ; every comparison with beta includes zero, even though 61.6 percent of stock changes quintile at $\$lambda=\0 . This is large classification movement with limited coefficient resolution, not evidence about deployed technology.

+ \input{tables/r3_table6_lambda.tex}

diff --git a/paper/main/sections/08_competing_interpretations.tex b/paper/main/sections/08_competing_interpretations.tex
index 344106a..41b344d 100644
--- a/paper/main/sections/08_competing_interpretations.tex
+++ b/paper/main/sections/08_competing_interpretations.tex
@@ -1,13 +1,7 @@
- \section{Conclusion} \label{sec:competing}
+ \section{Conclusion} \label{sec:conclusion}

-Public CPS data reproduce a negative young-relative employment-stock association under a widely used occupational AI-exposure construction. After repairing the calendar, the top-versus-bottom beta-quintile contrast is $-\$0.1346\$$ log points. Implementing the public top-two-versus-bottom-three grouping used by \code{brynjolfsson2026canaries} gives a smaller but negative $-\$0.0728\$$ contrast. These are descriptive population comparisons, not replications of proprietary hiring results or causal AI estimates.

+Public CPS data contain a sizeable negative association between GPT-4 task exposure and employment for ages 22--25 relative to ages 26--65. The rebuilt Q5--Q1 coefficient is $-\$0.132\$$ log points, but it falls to an imprecise $-\$0.022\$$ when broad occupation families receive their own young-relative monthly paths. Their paired movement is detected; it is a change between conditional estimands, not a causal composition share.

-The decisive fact is where the comparison lies. Broad two-digit occupation-family controls reduce the detailed Q5--Q1 coefficient to about $-\$0.032\$$. The paired movement is detectable, while the residual coefficient is imprecise and draws on only 30 percent of baseline target information. Only four broad families span both exposure tails. Thus, the baseline pattern is principally a comparison across broad occupational families; the design does not precisely isolate a detailed within-family AI gradient.

+Direct within-family tails cover only 29 occupations and are highly imprecise. Preperiod computer use strengthens rather than attenuates the coefficient, showing that computerization is not settled by one control. Unrestricted dynamics reject joint preperiod equality, and linked-CPS flow estimates do not identify a mechanism or replicate employer-match hiring and separation.

-That conclusion should not be embellished into a mechanism the data reject. Low-exposure food-service occupations matter, but removing them attenuates the point estimate by less than one tenth. Broader in-person-service exclusions do not attenuate it. Joint influence exercises show large, oppositely signed occupation contributions rather than one occupation whose removal makes the result vanish. The evidence supports a broad-composition interpretation, not a demonstrated food-service-recovery explanation.

-Nor do alternative exposure estimates permit a horse-race winner. All seven paired beta-minus-alternative Q5--Q1 intervals include zero, with minimum detectable differences as large as 0.169 log points. A confidence interval containing zero means that the design does not detect a difference; it does not establish economic equivalence. The original six implementations are also highly dependent, with two components accounting for 96.11 percent of their score variance. Claims of six independent validations or universal architecture robustness are therefore retired.

-The measurement work remains useful precisely because it distinguishes objects that should not be pooled. Exact-code matching across incompatible occupational vintages is an implementation error. Repairing scores on fixed support is a measurement comparison. Readmitting occupations changes the population. Switching from Q5--Q1 to top-two-versus-bottom-three changes the estimand. Moving from LLM task acceleration to patent overlap changes the construct. Rotating $\$F/\$$ into $\$A/\$$ changes only coordinates. The revised tables report these layers rather than treating every row as interchangeable robustness evidence.

-The paper's claim is consequently narrower and stronger than before. The early-career CPS pattern is real as a descriptive broad-occupation contrast, but this design cannot attribute it specifically to detailed AI exposure. Future causal work needs observed adoption or deployment, a credible source of timing or assignment variation, and outcomes that directly measure hiring and separation. Future measurement work needs occupation-task validation labels collected independently of the constructed indices, repeated ratings or audited use logs, and a taxonomy-stable holdout sample. Until then, occupational AI coefficients should travel with their construct, mapping, support, comparison, and resolution limits.

+The defensible conclusion is therefore comparison-specific. Mapping, support, representation, and conditioning determine what an exposure coefficient measures. A detected movement does not identify why it occurs; nondetection does not establish equivalence. Causal work requires observed adoption or credible assignment and timing variation. This paper instead establishes that the public-CPS young-worker pattern is principally a broad occupational comparison and quantifies what direct within-family and flow evidence cannot resolve.

diff --git a/paper/main/title.tex b/paper/main/title.tex
index 29c293d..3427812 100644
--- a/paper/main/title.tex
+++ b/paper/main/title.tex
@@ -20,10 +20,16 @@
 \vspace{0.15in}
 \noindent\textbf{JEL codes:} J23, J24, O33, C81.

- \noindent\textbf{Keywords:} artificial intelligence, occupational exposure, occupational composition, early-career employment, constructed treatments.
+ \noindent\textbf{Keywords:} artificial intelligence, occupational exposure, occupational composition, young-worker employment, Current Population Survey.

\vspace{0.1in}
 \noindent\textbf{Acknowledgments:} [AUTHOR TO COMPLETE].

+ \noindent\textbf{Funding:} [AUTHOR TO COMPLETE].

+ \noindent\textbf{Conflicts of interest:} [AUTHOR TO COMPLETE].

+ \noindent\textbf{Data and code disclosure:} Replication code and nonconfidential derived outputs will accompany the paper; restricted CPS microdata access and journal-specific disclosure language are [AUTHOR TO COMPLETE].

\end{minipage}
 \vfill
 \end{titlepage}

diff --git a/paper/revision/referee_response.tex b/paper/revision/referee_response.tex
index 2ff8296..2b15cd9 100644
--- a/paper/revision/referee_response.tex
+++ b/paper/revision/referee_response.tex
@@ -3,138 +3,161 @@
 \usepackage{styles/working-paper}
 \usepackage{styles/yax-common}
 \geometry{margin=1in}
 - \hypersetup{pdftitle={Major-Comment Response to Two Referee Reports}, pdfauthor={Lily Luo}}
 + \hypersetup{pdftitle={Response to Two Referee Reports}, pdfauthor={Lily Luo}}
 \setlength{\parindent}{0pt}
 \setlength{\parskip}{0.55em}
 \begin{document}
 \begin{center}
 - {\Large\bfseries Major-Comment Response to Two Referee Reports} \par
 - \vspace{0.4em}
 - {\large AI Exposure or Occupational Composition? \textbackslash(Constructed Measures and Early-Career Employment)} \par
 - \vspace{0.4em}
 + {\Large\bfseries Response to Two Referee Reports} \par
 + \vspace{0.35em}
 + {\large "Occupational AI Exposure and Young-Worker Employment: \textbackslash(Support and Comparisons in the CPS)" } \par
 + \vspace{0.35em}
 Lily Luo \quad September 2026
 \end{center}

- \textbf{(Scope of this revision.)} This document responds to every numbered major comment in the two reports supplied on September 5, 2026. In accordance with the staged revision requested by the author, secondary copyediting and presentation comments are catalogued for the next pass rather than represented as complete. All post-outcome analyses are labeled exploratory; the frozen v1.1 design and results remain intact.

+ \section*{Response to the editor}

- \section*{Revision in one paragraph}

+ We thank both referees for identifying that the earlier manuscript's measurement framing ran ahead of what its comparative evidence could establish. We accepted the central recommendation and rebuilt the article around a narrower question: which occupational comparisons support the public-CPS association between GPT-4 task exposure and the relative employment stock of ages 22--25?

-We accepted the central diagnosis and rewrote the paper around the result the CPS can establish. The corrected Q5--Q1 beta coefficient is $-\$.1346\%$, but it falls to $-\$.0315\%$ after broad-family-specific post changes are absorbed and to $-\$.0317\%$ after family-specific monthly paths are absorbed. The paired movement is detectable, although the conditional estimand changes and retains only 30 percent of baseline target information. Clean service exclusions do not support a food-service mechanism. We now report a primary MDEs_{80} of $.1244$ and paired MDEs for every architecture comparison; all seven paired Q5--Q1 intervals include zero. The title, abstract, introduction, results, and conclusion no longer claim general architecture heterogeneity or AI-specific displacement. The revised contribution is a composition-centered empirical finding joined to a framework that separates implementation error, measurement, construct, support, estimand, representation, and reparameterization changes.

+The revision is not an incremental robustness expansion. We repaired the calendar and reconstructed every preperiod treatment object without postperiod outcomes; re-estimated the entire exposure profile; replaced the permutation argument with explicit broad-family regressions; named the direct within-family support; added occupational-characteristic, industry, education, survey-sampling, timing, and flow analyses; and replaced inconsistent intervals with one machine-readable source of truth. The reconstructed pooled Q5--Q1 estimate is $-\$.0.1321\%$ with occupation wild-score interval $[-.0.2206, -.0.0437]\%$. Allowing each two-digit occupation family its own young-relative calendar-month path gives $-\$.0.0217\%$ with interval $[-.0.1607, 0.1173]\%$. On common draws, the movement is 0.1104 with interval $[0.0107, 0.2102]\%$. This is a detected change between conditional estimands, not a causal decomposition.

-\section*{Response to Referee 1}

+The revised interpretation is deliberately bounded. Direct Q1/Q5 overlap within broad families is sparse and imprecise; unrestricted quarterly coefficients reject joint preperiod equality; and corrected CPS links do not resolve employment exit, reported occupation change, or entry-destination allocation. Computer use does not supply a simple omitted-variable explanation: conditioning on a pre-2022 D*NET computer-use measure makes the coefficient more negative on common support. The article therefore describes the headline as a broad occupational comparison, not an identified AI effect, a computerization effect, or a public replication of proprietary employer hiring results.

+The main paper has been shortened and reorganized around the economic question. Technical derivations, secondary analyses, failures, hashes, and chronology move to a focused online appendix and replication archive. The response below covers every numbered item in the two supplied reports. All new substantive analyses are transparently described as post-outcome exploratory revisions. The repository chronology is internally versioned, not externally preregistered.

+ \section*{Response to Referee 1: major comments}

\subsection*{M1. Precision and comparative language}

-\emph{Response.} Accepted. The main text now reports the realized primary MDEs_{80} of $.1244$ log points next to the $-\$.1311\%$ frozen estimate. A consolidated paired table reports MDEs from $.0609$ to $.1689$ for beta against the three AIOE implementations, alpha, broad, Webb AI, and OECD. Every paired Q5--Q1 interval contains zero. Throughout, we now write “the design does not detect a difference” rather than “the conclusion does not generalize,” and never infer equivalence.

+\emph{Response.} Accepted. Every direct comparison now reports a paired interval built from common draws and a two-sided normal-theory $\mathbf{MDE}_{\{80\}} = (1.96 + 0.84)SE\%$. For the reconstructed pooled model, the realized normal-theory MDE is 0.1266 log point. The family-conditioned coefficient has an MDE of approximately 0.200 , and the paired pooled-to-conditioned movement has an MDE of 0.1454 . We explicitly explain why an observed movement can be significant even when it is smaller than this planning quantity: an MDE is an 80-percent-power summary, not a second rejection threshold. Nondetection is never called equivalence.

-We also diagnosed the prospective-to-realized SE gap. A sharp historical simulation preserves each donor month's full contemporaneous cross-occupation residual vector with a global sign. Its mean SE remains $.01217$, so it closes none of the gap to $.04441$. We report this as a rejection of one proposed mechanism, not a decomposition or validation of the planning simulation. See Sections 1, 4, and 6; Main Table 3 and Figure 4; Appendix D and K.

+The planning-versus-realized gap is no longer used to diagnose a single cause. Instead, the revision reports an 11-design finite-sample audit, a 22-family dependence sensitivity, elapsed-calendar HAC results, and 399 household/sample-unit full refits as separate inferential targets. In the empirical observed-effect simulation, occupation-wild coverage is 0.865 , 0.921 , and 0.889 for the pooled, family-conditioned, and paired targets; family clustering improves the first and third but covers the family-conditioned target only 0.895 of the time. In a clean zero-target design, the corresponding family-conditioned rejection rates are 6.1 percent under occupation weights, 8.9 percent under family weights, and 5.6 percent for the cross-fit oracle. These exercises demonstrate that no single standard-error row exhausts the uncertainty, without pretending to decompose the gap from one simulation.

\subsection*{M2. Effective clustering and CPS rotation}

-\emph{Response.} Addressed within the available survey design. Young-cell respondent equivalents have $p10/\text{median}/p90$ of $0/2/16$; 26.3 percent are zero and 69.9 percent below five. Quarterly aggregation gives $-\$.13103\%$ on the frozen and $-\$.13454\%$ on the repaired calendar, nearly identical to monthly estimates and SEs. A cross-occupation time-score covariance sensitivity increases the SE to $.04909$ at a 12-month Wevey--West lag, with normal interval $[-.2273, -.0349]\%$.

+\emph{Response.} Addressed within the limits of public Basic Monthly files. Young occupation-month cells have $p10/\text{median}/p90$ respondent-equivalent counts of $0/2/16$; 26.25 percent are zero and 69.83 percent have fewer than five. One-sided zeros remain valid likelihood observations; both-age-zero cells contribute no likelihood term. The dynamic analysis is estimated at quarterly frequency, while the static monthly target is retained so its economic object does not change.

-The main text now explains that occupation clustering already permits arbitrary serial dependence within occupation but is not a complete CPS design-based variance. The additional HAC targets common monthly and nearby-month score covariance across occupations. The extract begins in 2017, so literal 2015--19 pseudo-breaks are unavailable. We run all feasible 2017--19 dates and emphasize 12 balanced dates; none approaches $-\$.1311\%$. Their $1/13\%$ empirical tail is explicitly a resolution floor in a dependent, post-outcome diagnostic, not a randomization $\$p$ -value. See Section 4 and Appendix D.

+Occupation clustering permits arbitrary serial dependence within occupation but is not complete CPS design-based inference. We therefore add elapsed-calendar score HAC and a multiplier at the household or sample-unit level. It keeps all observed records of a sampled unit together across months and gives every split-route descendant the same multiplier. Public files lack the confidential design variables or Basic Monthly replicate weights required for a complete survey-design claim. The two uncertainty targets are shown separately rather than mechanically added. The requested 2015--2019 pseudo-break range is infeasible because the extract begins in 2017; feasible historical diagnostics remain descriptive, while the revised paper foregrounds the unrestricted 23-quarter preperiod joint test.

- \subsection*{M3. Regression analogue of the permutation, monotonicity, and stable tails}

+\subsection*{M3. Regression analogue, profile, and direct support}

-\emph{Response.} This result now determines the paper. On the corrected calendar, SOC2-by-young-by-post terms reduce $-\$.1346\%$ to $-\$.0315\%$; SOC2-by-young-by-month paths give $-\$.0317\%$. Paired baseline-to-conditioned movements are approximately $.103$ and their intervals exclude zero. We do not call this a causal decomposition: conditioning changes the estimand, only four of 22 families span Q1 and Q5, and target information falls to 30 percent.

+\emph{Response.} This request now determines the paper. SOC2-by-young-by-calendar-month conditioning moves the reconstructed coefficient from $-\$.0.1321\%$ to $-\$.0.0217\%$; the paired 0.1104 movement has interval $[0.0107, 0.2102]\%$. A less saturated SOC2-by-young-by-post model is also reported. We do not call the movement a percentage “explained,” because family controls can absorb AI-related and non-AI-related evolution.

-We add a three-restriction test of a common Q2--Q5 coefficient ($\$p=.119\%$) and a least-favorable max- $\$t$ monotonicity assessment ($\$p=.393\%$). Neither equality nor monotonicity is established. The 46 always-Q1 and 18 always-Q5 occupations yield $-\$.2120\%$ with interval $[-.4131, -.0109]\%$, but only 15.0 effective information occupations and a 46.7 -percent top-five share. See Section 5, Main Table 2 and Figure 2, and Appendix C/E.

+We report the entire Q2--Q5 profile, a common-draw joint test, and simultaneous intervals rather than infer monotonicity from four point estimates. We also replace the outcome-selected “always-tail” row as the central support check with a transparent direct-tail benchmark: only families containing both reconstructed Q1 and Q5 occupations enter, and the changed population is named. That benchmark has 29 occupations, an inverse-Herfindahl effective count of 6.2 occupations, a 77.8 -percent top-five information share, and a normal-theory MDEs_{80} of 0.4576 log point. The resulting interval is too wide to resolve a direct within-family tail gradient. A continuous demeaned within-family companion states its scale and common-slope restriction and has MDEs_{80} $-.0313\%$ per one within-family standard deviation.

- \subsection*{M4. Influence concentration and service recovery}

+\subsection*{M4. Influence and the low-exposure reference}

-\emph{Response.} Influence is now a headline diagnostic rather than summarized by the absence of sign reversal. Deleting the five occupations with largest absolute prior LOCO movements gives $-\$.1011\%$. Deleting ten or twenty gives $-\$.1522\%$ and $-\$.1555\%$, showing offsetting signed influence. A symmetric signed-tail trim gives $-\$.1281\%$ and continuous Huber down-weighting $-\$.1314\%$. These data-adaptive rows are labeled diagnostics, not preferred estimators.

+\emph{Response.} Influence is promoted as an interpretation diagnostic. The appendix reports leave-one-family-out results, named occupation leave-one-out movements, joint leave- $\$k$ and symmetric-trimming diagnostics from the historical audit, and young/older stock paths. Fast-food and counter workers are influential, but other occupations have offsetting signed influence. Because selecting deletions by observed influence changes the population after seeing outcomes, none is promoted as a preferred estimate.

-We replaced the coarse industry proxy with occupation-major-group exclusions. Dropping Q1 food-service occupations gives $-\$.1196\%$, an 8.8 -percent attenuation. Dropping broader in-person services gives $-\$.1370\%$ or $-\$.1387\%$. We therefore reject the proposed strong food-service mechanism while accepting that the Q1 trajectory and individual occupations matter. See Section 5 and Appendix E/J.

+We no longer claim that a coarse leisure/hospitality code identifies a food-service mechanism. Economically named service exclusions are reported as changed-population diagnostics. The article states that post-pandemic recovery, immigration, wages, and labor supply are possible interpretations requiring separate evidence; it does not treat their plausibility as a regression result.

- \subsection*{M5. CPS population controls}

+\subsection*{M5. CPS population controls and late-2025 production}

-\emph{Response.} The respondent-equivalent result is promoted and defined operationally as cells reconstructed from observed records without final CPS weights. It is not a common rescaling of weighted stocks. On the repaired calendar it is $-\$.1348\%$, versus $-\$.1346\%$ under official weights. Ending in December 2024 gives $-\$.1056\%$ and $-\$.1135\%$, respectively.

+\emph{Response.} We verified the January 2025 and January 2026 control changes, the revised January 2026 file, October 2025 noncollection, and delayed/reweighted November 2025 collection against official documentation. The extract and weight vintages are recorded by hash. The endpoint grid includes the exact requested pre-2025 endpoint through December 2024, late-2025 and pre-2026 endpoints, the full window, and an exclusion of September and November 2025 around the gap. In this post-outcome exploratory grid, the December-2024 pooled coefficient is $-\$.0.1110\%$ ($[-.0208, -.0.0213]\%$); its endpoint-minus-full-window difference is 0.0211 ($[-.0.0073, 0.0495]\%$). The family-conditioned coefficient is $-\$.0.0120\%$ ($[-.0.1657, 0.1417]\%$), with paired difference 0.0097 ($[-.0.0526, 0.0719]\%$). The separately estimated 2023--24 versus 2025--26 postperiod split supplies a different era comparison with paired uncertainty; we do not relabel it as an endpoint regression.

-BLS does not publish age-by-occupation bridge weights for the January 2025 revision; its experimental factors are aggregate. We therefore do not fabricate a counterfactual micro-weight series. The raw official-weight contrast becomes slightly less negative between December 2024 and January 2025, and the official 2025--26 era is not distinguishable from 2023--24. The documentation, bounded inference, and refusal are now explicit. See Section 7 and Appendix F.

+BLS does not publish counterfactual age-by-occupation weights on the prior control basis. We therefore do not manufacture a subgroup adjustment from aggregate factors. Respondent-equivalent cells are retained as a separately defined unweighted outcome, not described as recovered counterfactual weights or a common scalar rescaling.

\subsection*{M6. Selected architecture space}

-\emph{Response.} The revision includes a repository-bounded candidate census with source-integrity, construct, mapping/support, and representation status. It records six selected implementations from

two core families, two operationally admitted external implementations from two other families, one scope failure, and multiple pending or locally unverified candidates. It does not claim a global count.

-The weighted eigenvalue spectrum confirms the referee's concern: PC1 explains 80.04 percent and the first two components 96.11 percent of common-support score variance. The IUT is demoted to a predeclared marginal sign statement over a dependent set; the failed simultaneous band remains visible. Firm- and demand-side measures are not forced into occupation CPS without a defensible unit bridge. See Section 2 and Appendix A.

-\subsection*{M7. Inconsistent confidence intervals}

+ \emph{Response.} Accepted. The technical archive retains the repository-bounded candidate census and pass/fail reasons, while the main article no longer presents six correlated implementations as six independent validations or headlines an intersection-union test. The focused architecture revision reports native and common support separately, coefficient differences rather than significance differences, tail turnover, and characteristic correlations. Firm- and demand-side measures are not forced into an occupation-CPS regression without a defensible unit bridge. The scientific article uses the architecture family as a compact measurement diagnostic rather than its organizing contribution.

- \emph{Response.} Corrected. The only primary frozen interval is the 9,999-draw $[-.217075, -.045073]$. We trace the earlier variants to different finite 999-multiplier sets; the minimum-count row additionally changed support. Pair-specific member intervals are labeled and are not presented as primary. A consistency audit checks the canonical coefficient and interval. See Section 4 and Appendix D.

+ \subsection*{M7. Conflicting intervals}

- \subsection*{M8. Relationship to Brynjolfsson, Chandar, and Chen}

+ \emph{Response.} Corrected. The canonical reconstructed pooled interval is $[-0.2206, -0.0437]$ from the declared 9,999-draw occupation wild-score procedure. Different stochastic targets---22-family dependence, household resampling, and elapsed-time HAC---are labeled rather than printed as interchangeable intervals for one result. A numerical-consistency script checks the load-bearing displayed estimates and refuses a build when declared canonical values differ.

- \emph{Response.} We implement the reproducible public component of BCC: GPT-4 beta with top two employment-weighted quintiles against the bottom three. In CPS this gives $-\$.0728$ with interval $[-.1240, -.0216]$, compared with the corrected YAX Q5-Q1 coefficient $-\$.1346$. Applying that grouping across eight constructions yields native-support estimates from $-\$.0817$ to $-\$.0096$. On literal common support, beta differs detectably from OECD under this grouping but not from the other six alternatives.

+ \subsection*{M8. BCC comparability}

-We label this a bridge, not a replication. The ADP outcome, employer panel, job-title mapping, firm controls, hiring, and separation margins are proprietary or outside CPS. See Sections 2 and 6 and Appendix E.

+ \emph{Response.} We verified the August 12, 2026 paper, public dashboard, grouping description, outcomes, and endpoints. Exact dashboard memberships and tie handling are unavailable, so the CPS exercise is explicitly an equal-occupation, tie-preserving approximation rather than a replication. Its top-two-versus-bottom-three estimate is -0.0720 ; family-by-month conditioning gives $-\$.0168$, and the paired movement interval includes zero. The article compares employment stocks with employment stocks first and keeps the 19-percent ADP "kept pace" statistic separate from regression coefficients and employer-match hiring/separation outcomes.

- \subsection*{M9. January 2023 and adoption}

+ \subsection*{M9. January 2023 and adoption timing}

- \emph{Response.} We supplement the binary coefficient with the existing event-time profile, balanced pre-AI pseudo-breaks, and joint post-era coefficients. We do not introduce a new adoption series in this revision. A national adoption series interacted with fixed exposure would largely reparameterize exposure-by-time dynamics already shown. BTOS would require an occupation--industry ecological bridge and still would not identify exogenous adoption.

+ \emph{Response.} January 2023 is now a chronology split, not an instrument. We report all declared onsets from November 2022 through June 2023, fully interacted quarterly paths for Q2--Q5, joint preperiod tests, seasonality checks, and endpoint comparisons. The post-outcome exploratory December-2024 endpoint gives $-\$.0.1110$ pooled and $-\$.0.0120$ after family-by-month conditioning; neither endpoint-minus-full-window paired interval excludes zero. The official HonestID exercise uses the full quarterly vector and covariance. Its conventional dynamic-functional intervals are $[-\$0.264, 0.024]$ pooled and $[-\$0.425, 0.010]$ after family conditioning. A consecutive 2021Q1--2022Q3 calibration excludes zero for the family-conditioned smoothness interval only under exact linear differential-trend bias ($\$M=0$); the interval includes zero at $\$M=0.005$. A national adoption rate interacted with a fixed occupation score would largely reparameterize time; an ecological industry bridge would change the treatment without creating exogenous adoption. We therefore do not present either as causal identification. The unrestricted preperiod rejection is the binding interpretive fact.

-This is a principled scope decision, not a claim that adoption is irrelevant. The binary term is now called a chronology benchmark, not a causal shock. A separate adoption design remains future work. See Section 7 and Appendix F.

+ \subsection*{M10. F/G parameterization}

- \subsection*{M10. F/G reparameterization}

+ \emph{Response.} Accepted. F/G and the associated rotations are removed from the scientific paper and online appendix. Historical code remains in the replication archive. The concise remaining diagnostic uses the raw task primitives $\$D$ and $\$S$, standardized versions where useful, and the transparent $\$D \rightarrow \lambda \S grid. Changes in coordinates are not reported as economic findings.

- \emph{Response.} Accepted. F/G is removed from the main results and retained in Appendix I as a coordinate transformation. The main text leads with the representative AIOE-plus-beta and direct $\$D, \S models. The task-family centroid's effective raw-unit weights are reported as $\$4.9057D + 1.8104S$, so the embedded $\$S$ -to- $\$D$ weight is .3690. We no longer treat a confidence interval crossing zero after component deletion as evidence of a detected coefficient change.

+ \subsection*{M11. Entry and longitudinal evidence}

- \subsection*{M11. Mobility, entry, and uncertainty}

+ \emph{Response.} Corrected CPS links now distinguish employment exit, reported occupation change among endpoint-employed workers, and Q5-versus-Q1 destination allocation conditional on entry. Adjacent and twelve-month endpoint results use their own risk sets and official longitudinal weights; every primary interval includes zero. Their normal-theory MDEs, $\{80\}$ values range from 0.115 to 0.336 log point and are reported as precision diagnostics, not rejection thresholds. Twelve-month young links retain only 50.98 percent of eligible origins, so the longer horizon does not automatically double information. We never call destination allocation an employment-finding probability or employer hiring rate. The earlier rematching benchmark is removed from the scientific appendix.

- \emph{Response.} The employed-to-employed switch analysis is removed from the paper's main contribution. The appendix now separates the entrant-destination allocation among people observed entering employment from any employment-finding probability. Its beta Q5-Q1 coefficient is $-\$.0888$ with interval $[-.2787, .1011]$ and $\$p=.383$. Immediate reversals are not treated as a validated coding-error rate, so no purported attenuation correction is applied; the appendix also reconciles the prior 9.865-percent statistic with the present sample's 10.336-percent raw conditional rate.

+ \subsection*{M12. Generated-covariate validation}

-The revised mobility audit reports a 20.13-percent family-balanced pairwise disagreement rate and proves that beta adds no new zero-threshold conflict when its task-family endpoints agree. On identical 98.305-percent represented support, realized conflict is 53.2819 percent and the benchmark is 52.3227 percent, a 0.9592-point gap. The declared household-cluster SE is 0.1979 point for the realized rate and 0.0776 point for the gap, versus a 0.00172-point Monte Carlo SE for the benchmark mean. The all-support accounting bound for the gap is $[0.1521, 1.8469]$ points. A second valid repair rule changes the benchmark by only $-\$.0.00157$ point, but neither rule is a canonical uniform draw over feasible derangements. The main text draws no hiring conclusion from switches. See Appendix G/H.

+ \emph{Response.} The appendix now makes the missing validation object explicit. A useful sample would preserve versioned occupation--task units, stratify by occupational family and cross-architecture disagreement, and separately label technical capability, realized time saving with and without complementary software, actual adoption, and task importance. It would require repeated blinded expert ratings and, where possible, audited usage or performance data, with occupations held out from calibration. Sample-size requirements depend on clustered task/rater noise and the target correction; in the absence of such labels we do not invent a universal number or claim that bootstrap inference covers exposure error.

- \subsection*{M12. Generated-covariate validation}

+ \section*{Response to Referee 1: secondary comments}

- \emph{Response.} Appendix A now specifies a validation design: occupation--task units stratified by family and architecture disagreement; separate labels for capability, realized time savings with and without software, adoption, and task importance; blinded independent experts plus audited use logs where feasible; repeated labels; held-out occupations; and versioned task/model/mapping records. We provide a Fisher- $\$2$ power formula and an illustrative independent-unit calculation while emphasizing that clustering, rater noise, sampling weights, and target correction determine the actual sample size. We do not invent one universal number.

+ \begin{enumerate} [leftmargin=1.8em]

+ \item The ten-journal search paragraph is removed from the article; the declared score and date remain only in the evidence ledger.

+ \item The grouped-binomial/conditional-logit representation and log young-to-older stock-ratio estimand are stated explicitly.

+ \item Boundary handling is verified: one-sided zeros remain; both-age-zero cells contribute no likelihood. Minimum-count samples remain outcome-selected sensitivities, not the baseline.

+ \item The common route-share limitation, one-to-many stock, conservation, stable-taxonomy restriction, and changed time/population estimand are stated. A post-2020 fit that fails convergence remains a recorded failure rather than being promoted.

+ \item The OECD capability-gap construction is treated as a different technology boundary and horizon, not a sign-reversed substitute for contemporary LLM exposure.

+ \item Frey--Osborne is not announced as a principal outcome architecture; any retained appearance is labeled an automation characteristic.

+ \item Common-support and native-support architecture rows are separated, as are conditioning and contrast changes.

+ \item The main text uses one uncluttered young/older/ratio figure; additional level paths remain in the appendix.

+ \item Prose and displayed tables use design-appropriate rounding; machine files retain full precision.

+ \item "Respondent-equivalent" is defined on first use as fractional routed-record mass before 2020 and exact record count thereafter, never unique persons.

+ \item The paper calls the retained exposure $\$beta$ and writes its task primitives as $\$D=E1$ and $\$S=E2$. It does not report $\$alpha$ or the upper-bound score as principal outcome architectures. The archive records that the legacy source field `\texttt{dv\_rating\_gamma}` corresponds to the published $\$zeta$ construct; "gamma" and "broad" are not used as competing scientific labels.

+ \item Source and generated text pass a UTF-8/nojibake scan.

+ \item Affiliation, email, acknowledgments, funding, conflicts, and journal disclosures remain visibly marked `\texttt{\{AUTHOR TO COMPLETE\}}`; none is invented.

+ \item The repository chronology is described as internally git-versioned and explicitly not externally registered.

+ \end{enumerate}

- \section*{Response to Referee 2}

+ \section*{Response to Referee 2: major comments}

- \subsection*{M1. Framework separating measurement from the question}

+ \subsection*{M1. Separate measurement from changes in the question}

- \emph{Response.} Accepted and used to reorganize the paper. Main Table 1 distinguishes implementation repair, same-construct measurement, construct change, population/support change, estimand change,

representation change, and pure reparameterization. The paper states positive-affine rank/bin invariance, piecewise-constant bin membership along Δ , and full-rank coordinate invariance. Every empirical row now identifies which layer changed. See Section 2 and Appendix A.

Accepted. The empirical framework distinguishes intended construct, measurement rule, occupational harmonization, retained population, representation, and regression estimand. The crosswalk table separately identifies score correction on fixed support and occupation readmission. A literal vintage mismatch is an implementation error, not a robustness choice. Affine rank invariance and full-rank coordinate invariance are stated briefly; they no longer substitute for the substantive application.

Accepted. Degree of architecture heterogeneity

Accepted. Architecture heterogeneity and direct comparisons

Accepted. The revision separates point-estimate sign, a dependent-set sign test, and paired coefficient differences. Main Table 3 contains direct paired differences, intervals, and MDEs; all Q5–Q1 intervals contain zero. Native-support and BCC-grouping estimates are shown separately rather than pooled into one common-sample experiment. We no longer infer heterogeneity from individual significance patterns.

Accepted. We separate point-estimate signs, evidence about individual coefficients, and paired evidence that two coefficients differ. Every architecture row identifies native or matched support and reports the paired interval and precision. An interval containing zero is described only as failure to detect a difference; it is never evidence of economic equivalence. The broad architecture menu is demoted because the stronger and better-resolved result concerns occupational comparisons.

Accepted. Construct validity of external architectures

Accepted. Construct validity

Accepted. The architecture matrix now records technology scope, primitive, label generation, publication and occupational-information vintages, horizon where verified, mapping, and the downstream hypothesis. OECD is described as a broader capability/demand and horizon construct, not a noisy LLM reliability measure. Webb AI and Webb software are distinguished. The external-admission rule is preserved as an operational screen, while the distinct-cut condition is explicitly representation-dependent rather than construct validity. The paper also reports unconditioned and Webb-conditioned BCC beta results, which are nearly identical for that binary grouping without making a general computerization claim. See Section 2/6 and Appendix A/E.

Accepted. Each retained score is described by its primitive, technology boundary, vintage, horizon, mapping, and support. Webb software is a preperiod computerization control; Webb AI, AIOE, and OECD do not become interchangeable measures merely because each contains “AI” in its label. The focused Δ exercise changes a capability definition and group membership; it does not reveal which technology firms deployed.

Accepted. Reference group and broad occupational variation

Accepted. This is now the substantive center of the paper. We replace “reference dependence” with exact descriptions of changed contrasts, retain the Q1/Q5 stock paths, add profile/equality/monotonicity tests, and estimate both SOC2-by-post and SOC2-by-month conditioned models with information and support diagnostics. The regression results, not the assumption-dependent permutation percentile, determine the framing. See Section 5 and Appendix C/E.

Accepted. The main empirical section now shows Q1 and Q5 young and older stock paths, the full exposure profile, family-by-quintile support, the families spanning both tails, direct-tail and continuous within-family companions, and family/occupation influence. The regression with family-specific monthly paths replaces the assumption-dependent permutation as the central diagnostic. In the post-outcome exploratory December-2024 endpoint check, pooled and family-conditioned coefficients are $\$-0.1110$ and $\$-0.0120$; their respective endpoint-minus-full-window paired intervals are $\$[-0.0073, 0.0495]$ and $\$[-0.0528, 0.0719]$. The conclusion is comparison based: the pooled coefficient relies primarily on broad occupational structure, while direct within-family evidence is sparse and imprecise.

Accepted. Corrected pipeline and bridge uncertainty

Accepted. The repaired 113-month coefficient is now the substantive baseline; the frozen result is a chronology benchmark. We add a fail-closed taxonomy-compatibility gate, one-to-one-target estimates, split-source stock and information shares, age-allocation accounting ranges, and predeclared tilt scenarios. We do not claim the common-proportion bridge resolves age-specific allocation.

Accepted. The fully reconstructed 113-month pipeline is now the sole substantive baseline; historical labels are a labeled checkpoint. All preperiod weights, support, normalizations, cuts, ties, and memberships are regenerated. The March replacement, missing October 2025, transition month, route mass, one-sided zeros, exclusions, and split probabilities are machine audited. The crosswalk expansion conserves weighted stock to machine precision. Age-specific route probabilities remain unavailable, so common shares and bounded allocation sensitivities are disclosed rather than validated by adjacent vintages.

The 490/495 discrepancy is fully reconciled. The raw balanced universe is 539. The 490 support requires beta; the 495 support requires repaired AIOE plus Webb. They overlap in 466, with 24 beta-only and 29 AIOE-plus-Webb-only occupations. See Section 3, the sample-flow table, and Appendix B.

Accepted. Source of randomness

Accepted. Source of randomness and bootstrap mechanics

Accepted. Section 4 now states the target as a stochastic occupation-shock parameter conditional on realized CPS cells and constructed treatment. It lists household survey design, label generation, and mapping allocation as uncovered uncertainty. The nuisance-adjusted score, fitted information, cluster aggregation, finite-cluster factor, Rademacher multipliers, studentizer, critical value, and common-draw paired construction are described explicitly. The failed pseudo-outcome refit is correctly treated as evidence that that construction was inappropriate, neither validation nor invalidation of the score bootstrap. Draws increase from 999 to 9,999 for central revised inference; IUT and simultaneous-band claims remain distinct.

Accepted. The article and Appendix-E state the stochastic target of each interval and the uncertainty it omits. We verify nuisance residualization, score aggregation, fitted information, common-draw pairing, seeds, and covariance rank. Occupation-shock, broad-family, household-sampling, HAC, and simulation results remain distinct. The generated pandemic-shortfall controls were rebuilt on ages 22–65 with logically bounded predictions and a strict one-to-one occupation population. In 399 linked-household refits, we separately regenerate the shortfall and the exposure labels; the shortfall-conditioning movements remain small, but the resulting intervals are labeled sampling sensitivities rather than complete CPS design inference. No procedure is said to cover exposure-label or route-allocation uncertainty beyond the variation it actually regenerates, and no inadmissible pseudo-outcome clipping is used.

Accepted. Primitive units and F/G

Accepted. Direct Δ , AIOE-plus-beta models are presented before the appendix rotation. Effective raw-unit centroid weights are reported, and leave-one-component changes are described as changes in implicit weights. F/G and A/E are explicitly the same fitted column space. See Section 6 and Appendix I.

Accepted. The scientific appendix reports Δ , raw and standardized units, covariance, and the Δ grid. It reconciles $\lambda=5$ exactly with beta under identical mapping, support, weights, and model. F/G is archived but removed from the article and online appendix. Crossing an individual significance threshold is never described as a significant coefficient difference.

Accepted. Mobility scale and benchmark

Accepted. Mobility is demoted and its claims narrowed. Thresholds are decision rules in standardized or rank units, not welfare magnitudes. The family-balanced matrix prevents multiplicity of near-duplicate measures from defining the result. The exact identity $\Delta(\lambda)=\Delta+\lambda\Delta$ shows that beta cannot introduce a new zero-threshold sign conflict when alpha and broad agree. Benchmark and realized conflict are recomputed on identical represented support; omitted support is bounded. Household-linked sampling uncertainty is separated from Monte Carlo error, and at least two valid non-self repair distributions are compared. See Appendix H.

Accepted. We followed the integrated revision priority and removed the rematching/mobility-classification exercise rather than expanding an algorithm-defined benchmark with no welfare scale. The economically defined linked-CPS margins remain: exit, occupation change, and destination allocation conditional on entry. Their risk sets, effect units, link attrition, uncertainty, and comparability limits are explicit. This is a scoped non-adoption, not a claim that mobility is unimportant.

Accepted. Auditable chronology without chronology prose

Accepted. Verifiable chronology without revision-history prose

Accepted. Revision-round and “frozen” language is removed from the scientific narrative. The technical archive retains dated specifications, commit/tag chronology, protected-outcome rules, amendments, seeds, hashes, failures, and result manifests. The article describes new analyses as exploratory and does not imply external registration.

Accepted. Revision-history phrases are removed from the main scientific narrative. The appendix preserves the dated protocol, protected-outcome definition, frozen and confirmatory tags, information set, hypothesis-to-test mapping, scripts, samples, seeds, and output hashes. We do not imply that repository hashes constitute external prospective registration. A result manifest links each headline number to its script and machine-readable output. See Appendix L and the reproducibility package.

Accepted. Response to Referee 2: specific corrections

Accepted. Claims retired

Accepted. The dynamic reference is unambiguous: omitted 2022Q4 contains observed October–November; December is observed but excluded as the transition month. All Q2–Q5 interactions and Webb treatment are documented.

Accepted. Under identical reconstructed support and estimation, $\lambda=5$ reproduces beta exactly. Earlier different coefficients mixed support or treatment contracts and remain only in implementation history.

Accepted. Respondent-equivalent cells remove released final-weight magnitudes record by record; they are not a common scalar rescaling of weighted stocks.

Accepted. The paper consistently says “ages 22–25” or “young workers,” not observed experience. BA/non-BA and exact-age comparisons are reported with simultaneous/paired uncertainty and do not identify a cohort mechanism.

Accepted. Six compact main tables separate data construction, occupational comparison, characteristic conditioning, inference, flows, and the lambda diagnostic. Native, matched-support, and paired quantities are labeled.

Accepted. Longitudinal estimates remain explicitly nondetections on distinct risk sets; no flow coefficient is interpreted as evidence that a mechanism is absent.

The revised paper does not claim a causal AI employment effect, six independent validations, architecture equivalence, a monotone exposure response, a food-service mechanism, a CPS replication of proprietary ADP hiring results, survey-design-complete inference, or a coding-error correction inferred from occupation reversals. It also does not treat an invalid taxonomy merge as a defensible robustness specification.

Accepted. Claims retired and remaining author actions

Accepted. Next revision pass

The paper no longer claims a causal AI effect, a causal composition share, economic equivalence, a monotone dose response, a food-service mechanism, independent validation by correlated scores, complete survey-design inference, or replication of proprietary ADP hiring/separation results. It does not treat an invalid taxonomy merge as a defensible alternative.

The two reports’ secondary comments are preserved in the machine-readable response matrix. The next pass will focus on copyediting, citation alignment, remaining table conventions, and author-supplied affiliation/disclosure fields. None of those items will be marked complete merely because the major empirical revision is complete.

+The only submission items blocked on information outside the empirical record are the author's affiliation, email, acknowledgments, funding, conflicts, and journal-specific disclosures. They remain explicitly marked for author completion. All empirical limitations that cannot be repaired with the authorized data are listed in the revision diagnosis and reproducibility appendix rather than promised as future completed work.

```

\end{document}
diff --git a/paper/revision/revision_diagnosis.tex b/paper/revision/revision_diagnosis.tex
index 3b2771b..e668597 100644
--- a/paper/revision/revision_diagnosis.tex
+++ b/paper/revision/revision_diagnosis.tex
@@ -3,50 +3,66 @@
\usepackage{styles/working-paper}
\usepackage{styles/yax-common}
\geometry{margin=1in}
-\hypersetup{pdftitle={YAX Major-Revision Diagnosis},pdfauthor={Lily Luo}}
+\hypersetup{pdftitle={Revision Diagnosis},pdfauthor={Lily Luo}}
\setlength{\parindent}{0pt}
\setlength{\parskip}{0.6em}
\begin{document}
\begin{center}
-(\Large\bfseries YAX Major-Revision Diagnosis)\par
+(\Large\bfseries Revision Diagnosis)\par
\vspace{0.3em}
-Evidence-led framing decision and remaining work\par
+Evidence-led framing decision, completed work, and unresolved limits\par
September 2026
\end{center}

-\section*{Decision}
+\section*{Framing decision}

-The paper should be a \textbf{composition-centered measurement paper}, titled \emph{AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment}. The strongest defensible substantive result is that the public-CPS young-relative association is largely a comparison across broad occupation families. The strongest methodological result is that implementation repair , same-construct measurement, construct choice, support, estimand, representation, and coordinate changes have to be audited separately.
+The defensible paper is a \textbf{comparison-centered empirical audit}, not an AI-displacement design and not a generic ‘‘measurement choices matter’’ essay. Its question is: what occupational comparisons support the negative association between GPT-4 task exposure and young-relative employment stock in the public CPS?

-The earlier thesis---that robustness is generally statement-specific---was true but too generic. The stronger revised thesis is empirical and falsifiable: after correcting the data, a $-0.1346$ Q5--Q1 association falls to about $-0.032$ when broad-family evolution is absorbed, while the design is too imprecise to resolve most paired architecture differences. The chapter is not a causal AI-displacement paper.
+The answer is specific. The reconstructed pooled Q5--Q1 comparison is $-0.1321$ log point, with 95-percent occupation wild-score interval $[-0.2206, -0.0437]$. Giving every broad occupation family its own young-relative calendar-month path yields $-0.0217$, with interval $[-0.1607, 0.1173]$. Their paired difference is 0.1104, with interval $[0.0107, 0.2102]$. The movement is detected under the declared occupation-shock procedure, but it compares different conditional estimands and is not a causal share.

-\section*{Evidence that forced the decision}
+Direct support is the binding economic qualification. Few broad families contain both exposure tails; a direct-tail sample covers only a small share of employment and cannot precisely resolve the residual gradient. Other families connect the coefficient through intermediate quintiles and common-slope restrictions. The article should therefore say that the pooled pattern is principally a comparison across broad occupational structure, not that ‘‘composition explains’’ a known fraction of an AI effect.

-\begin{enumerate}[leftmargin=1.6em]
-\item \textbf{Broad-family conditioning.} SOC2-by-post and SOC2-by-month specifications yield $-0.0315$ and $-0.0317$, versus $-0.1346$ at baseline. Paired movements of approximately .103 have intervals excluding zero.
-\item \textbf{Support cost.} Only four of 22 SOC2 families contain both Q1 and Q5, and conditioned target information is 30 percent of baseline. The residual detailed-exposure coefficient is weakly measured and is not a causal remainder.
-\item \textbf{Food-service mechanism rejected.} Removing Q1 food-service occupations attenuates the coefficient only 8.8 percent. Broader in-person-service exclusions make it slightly more negative.
-\item \textbf{Architecture heterogeneity unresolved.} The primary MDEs, {80}$ is .1244. All seven paired beta-minus-alternative Q5--Q1 intervals include zero; paired MDEs reach .1689. Point-estimate differences cannot carry claims of detected heterogeneity or equivalence.
-\item \textbf{Direct public BOC bridge.} Their top-two-versus-bottom-three beta grouping produces a smaller $-0.0728$ CPS coefficient. This anchors the audit to the motivating classification while preserving the boundary around proprietary hiring evidence.
-\item \textbf{Construction dependence.} Two components explain 96.11 percent of the six core score implementations. The prior six-measure sign exercise is a dependent-set statement, not six independent validations.
-\item \textbf{Data corrections do not explain the negative association.} March repair moves $-0.1311$ to $-0.1346$. One-to-one-target and stable-taxonomy estimates remain negative. Population-control and respondent-equivalent diagnostics do not reveal a mechanical January-2025 explanation.
-\item \textbf{Mobility is diagnostic rather than mechanistic.} A family-balanced pairwise statistic is 20.13 percent, while 53.28 percent is a six-way ‘‘any conflict’’ event. The aligned-support realized-minus-rematching gap is 0.959 point with a declared household-cluster sampling SE of 0.0776 point, but the algorithm-defined benchmark and switch-only sample do not identify hiring or coding error.
+\section*{Evidence that determines the interpretation}
+
+\begin{enumerate}[leftmargin=1.6em]
+\item \textbf{Corrected baseline.} Five wrongly selected March ASEC samples are replaced by Basic Monthly files; October 2025 is absent; December 2022 is the transition month. All 71-month preperiod weights, normalizations, cuts, ties, and memberships are rebuilt. Nine occupations change quintile, but reconstructed-minus-historical treatment moves the coefficient only 0.00245, with interval spanning zero.
+\item \textbf{Broad-family sensitivity.} The pooled-to-SOC2-month movement is 0.1104 and statistically distinguishable from zero on common draws. The family-conditioned coefficient itself remains imprecise. A direct-tail and continuous within-family analysis expose the support and functional-form cost.
+\item \textbf{Computerization and pandemic shortfalls are not one-control explanations.} A May 2020 O*NET computer-use measure is highly correlated with beta exposure. On common support it acts as a suppressor, making the coefficient more negative. The repaired shortfall exercise bounds impossible trend predictions, uses a strict 363-occupation population, and regenerates the controls in linked-household draws; its pooled conditioning movements are 0.0004 for total stock and 0.0011 for the young-relative measure, with sampling intervals containing zero. Remotability and other characteristics likewise do not yield detected movements. These are correlated diagnostics, not causal horse races, and the shortfall result does not establish that pandemic dynamics are absent.
+\item \textbf{Industry and education.} Conditioning occupation--industry microcells on 13 broad-industry post shifts moves $-0.1376$ to $-0.0987$; the paired 0.0389 interval excludes zero while the residual remains negative. BA and non-BA coefficients differ by only $-0.0154$, but their paired interval $[-0.1557, 0.1250]$ is too wide for equivalence. Exact-age evidence is nonmonotone and does not isolate experience or cohort mechanisms.
+\item \textbf{Preperiod dynamics.} The fully interacted quarterly Q5 path rejects joint equality over 23 preperiod coefficients in both pooled and family-conditioned models. A linear-slope nondetection cannot override that result. Official HonestDiD intervals for the dynamic post functional conventionally include zero in both structures. Under the consecutive 2021Q1--2022Q3 calibration, the family-conditioned smoothness interval excludes zero only at the exact-linear restriction $\$M=0$ and includes zero at $\$M=0.005$. January 2023 consequently remains descriptive chronology, not causal timing.
+\item \textbf{Later observations matter.} Extending from 2025 endpoints through July 2026 makes the pooled coefficient roughly 0.02 log point more negative on paired draws. Family-conditioned endpoint movements remain near zero. This is an informative data-extension result, not proof that AI adoption intensified.
+\item \textbf{Flows do not resolve the mechanism.} Corrected adjacent and twelve-month CPS links give intervals containing zero for employment exit, reported occupation change, and exposure allocation conditional on entry. Annual links have substantially lower young-worker retention. Hours and conditional earnings are also imprecise. None is an employer hiring rate.
+\item \textbf{External comparability is bounded.} An equal-occupation approximation to BOC's top-two-versus-bottom-three grouping gives $-0.0720$ in CPS and $-0.0168$ after family-month conditioning; their paired movement is not detected at five percent. Exact dashboard membership, employer controls, and proprietary payroll flows cannot be reproduced.
+\item \textbf{Mapping matters for population, not just score values.} A literal SOC-vintage mismatch retains 3.33 percent of computer-and-mathematical employment; documented routing restores 97.7 percent. Fixed-support score repair changes little, while readmission changes the population. The invalid merge is an implementation failure, not a robustness specification attributed to another paper.
+\item \textbf{Inference remains conditional.} Occupation wild scores, 22-family dependence weights, household/sample-unit full refits, elapsed-calendar HAC, and the completed 11-design finite-sample audit answer different stochastic questions. In the empirical observed-effect design, neither occupation nor family clustering attains nominal coverage for every target; the clean zero-target family-month rejection rates are 6.1 percent, 8.9 percent, and 5.6 percent for occupation weights, family weights, and the cross-fit oracle. Public Basic Monthly files do not permit complete CPS design-based inference, and these variances cannot be added without a valid decomposition.
\end{enumerate}

-\section*{What the revised paper may say}
+\section*{What the paper may and may not claim}
+
+The paper may say that public CPS data contain a sizeable negative young-relative employment-stock association under GPT-4 beta exposure; that broad-family conditioning produces a detected movement to an imprecise near-zero coefficient; that direct within-family tail support is thin; that a simple computer-use control does not attenuate the result; that unrestricted preperiod differences undermine causal timing; and that available flows do not identify a mechanism.
+
+It may not say that AI caused an individual employment decline, that occupational composition causally explains a percentage of an AI effect, that the residual coefficient is an AI effect, that computerization or reopening is the cause, that nondetection means economic equivalence, or that CPS reproduces ADP employer hiring and separation results.

-It may say that public CPS data reproduce a negative young-relative association under beta exposure and the public BOC grouping; that broad-family conditioning absorbs most of the point-estimate magnitude; that the movement is detectable but changes the estimand; that food-service exclusions do not explain the result; that mapping errors can remove nearly all computer/math coverage; and that available precision does not resolve most architecture differences.
+\section*{Completed revision package}

-It may not say that AI caused employment loss, that 76 percent is a causal share, that the remaining $-0.032$ is an AI effect, that food-service recovery is the mechanism, that intervals containing zero establish equivalence, that six measures independently validate the finding, or that CPS reproduces ADP hiring and separation evidence.
+The editable package contains a rewritten main manuscript, a focused online appendix, a point-by-point response, this diagnosis, a source diff, a machine-readable comment-response matrix, the

```


specification and results registries, module-level code and tests, source and environment manifests, covariance/influence outputs, figure/table generators, failure logs, and a numerical-consistency audit. A deterministic source-token audit records a 39.2-percent contraction of the included abstract and Sections 1--8 relative to commit \texttt{6b8d85e}. Restricted microdata, credentials, and direct identifiers are excluded.

-\section*{Major-comment completion}

+The scientific appendix no longer contains the mobility-rematching benchmark or F/G coordinate machinery. Historical artifacts remain available in the replication archive. Revision chronology is likewise moved out of the article; it is described honestly as internal git ordering rather than external prospective registration.

-Both reports' numbered major comments have been translated into an exact response matrix. The major empirical program now covers composition, profile and monotonicity, stable tails, joint influence, service exclusions, canonical intervals, realized MDEs, quarterly aggregation, time-score covariance, pre-AI pseudo-breaks, population controls, the BCC grouping, architecture dependence, corrected mapping, age-allocation uncertainty, and mobility/rematching uncertainty. The manuscript and appendix have been reorganized around the economic question rather than revision chronology.

+\section*{Unresolved items}

-Two requests receive principled non-adoptions. No counterfactual January-2025 age-by-occupation weights are created because official subgroup bridge factors do not exist. No national adoption or industry ETOS series is appended merely to reparameterize event time or introduce an unvalidated occupation--industry bridge. Both limits are explicit.

+The following limits remain after feasible execution:

-\section*{Remaining work before submission}

+\begin{itemize}[leftmargin=1.8em]

+ \item observed firm adoption, employer identities, and employer-match hiring/separation are unavailable in the CPS;

+ \item public files do not supply all confidential survey-design variables or counterfactual age-by-occupation weights for population-control revisions;

+ \item exact BCC dashboard occupation memberships and tie rules are unavailable;

+ \item adjacent occupational coding vintages do not validate age-specific route shares;

+ \item no external gold-standard label identifies a latent true occupational AI exposure;

+ \item several direct-tail, flow, subgroup, and conditional estimates remain imprecise;

+ \item affiliation, email, acknowledgments, funding, conflicts, and required journal disclosures require author input and remain \texttt{[AUTHOR TO COMPLETE]}.

+\end{itemize}

-This is the \textbf{major-advice revision}, not the final copyedit. The next pass should reconcile all secondary comments line by line, finalize citations and table formatting, and ask the author to complete affiliation, email, acknowledgments, funding, conflicts, and data-use disclosures. The paper should also receive an independent econometric review of the maintained occupation-shock inference before being sent to a top journal. These open items do not alter the framing decision.

+These are not reasons to add unrestricted specification searches. They define the boundary of the paper's contribution. Before submission, the author should complete the metadata/disclosure fields and obtain independent econometric review of the maintained shock-cluster interpretation.

\end{document}

```
diff --git a/paper/revision/source_diff.tex b/paper/revision/source_diff.tex
new file mode 100644
index 0000000..2f4e28b
--- /dev/null
+++ b/paper/revision/source_diff.tex
@@ -0,0 +1,27 @@
\documentclass[9pt,letterpaper]{article}
\usepackage[margin=0.55in]{geometry}
\usepackage[T1]{fontenc}
\usepackage{listings}
\usepackage[hideinks]{hyperref}
\hypersetup{pdftitle={Source Diff for the Substantive Revision},pdfauthor={Lily Luo}}
\setlength{\parindent}{0pt}
\begin{document}
\begin{center}
+{\Large\bfseries Source Diff for the Substantive Revision}\par
\vspace{0.3em}
+Baseline commit \texttt{6b8d85e} to the final revision working tree\par
\vspace{0.2em}
+September 2026
\end{center}
+
+This change-marked artifact is a unified source diff. It covers the main manuscript, online appendix, generated table sources, and revision documents. Binary artifacts are not embedded in the patch.
+
+The canonical editable files and generated PDFs accompany this document.
+
+\lstinputlisting[
+ basicstyle=\ttfamily\tiny,
+ breaklines=true,
+ breakatwhitespace=false,
+ columns=fullflexible,
+ keepspaces=true,
+ showstringspaces=false
+]{revision/source_diff.txt}
\end{document}
diff --git a/paper/tables/r3_appendix_architecture_pairs.tex b/paper/tables/r3_appendix_architecture_pairs.tex
new file mode 100644
index 0000000..a76423f
--- /dev/null
+++ b/paper/tables/r3_appendix_architecture_pairs.tex
@@ -0,0 +1,18 @@
\begin{table}[\htbp]
\centering
\caption{Post-outcome exploratory task-weight architectures on common support}\label{tab:appendix_architecture_pairs}
\small
\setlength{\tabcolsep}{4pt}
\resizebox{\textwidth}{!}{%
\begin{tabular}{llrrrr}
\toprule
+Architecture pair & Alternative coefficient & Beta coefficient & Alternative &  $\beta$  & Paired  $\mathbf{MDE}_{80}$  \\
\midrule
+$D$ versus $D+0.5$ &  $-0.103$  &  $-0.132$  &  $0.029$  &  $[-0.044, 0.103]$  &  $0.106$  \\
+$D+0.25$ versus $D+0.5$ &  $-0.101$  &  $-0.132$  &  $0.031$  &  $[-0.005, 0.068]$  &  $0.055$  \\
+$D+0.75$ versus $D+0.5$ &  $-0.143$  &  $-0.132$  &  $-0.011$  &  $[-0.039, 0.017]$  &  $0.041$  \\
+$D+S$ versus $D+0.5$ &  $-0.156$  &  $-0.132$  &  $-0.024$  &  $[-0.064, 0.017]$  &  $0.058$  \\
\bottomrule
\end{tabular}}
\end{table}
+\tabnote{All entries are post-outcome exploratory Q5--Q1 coefficients estimated on the identical 468-occupation support. Beta is $D+0.5$. At each alternative weight, normalization, cutoffs, and tie-preserving quintile memberships are recomputed, so a paired movement combines score and membership changes rather than holding the treatment groups fixed. The intervals use 9,999 common occupation multiplier draws, preserving covariance within each pair. Every paired interval contains zero; this means that the design does not detect a difference, not that the architectures are economically equivalent.}
\end{table}
diff --git a/paper/tables/r3_appendix_bcc_bridge.tex b/paper/tables/r3_appendix_bcc_bridge.tex
new file mode 100644
index 0000000..bcaf34e
--- /dev/null
+++ b/paper/tables/r3_appendix_bcc_bridge.tex
@@ -0,0 +1,23 @@
\begin{table}[\htbp]
\centering
\caption{Post-outcome exploratory BCC-style grouping bridge}\label{tab:appendix_bcc_bridge}
\scriptsize
\setlength{\tabcolsep}{3pt}
\resizebox{\textwidth}{!}{%
\begin{tabular}{llrrrr}
\toprule
+Grouping rule & Conditioning structure & Coefficient &  $\beta$  & Change from baseline &  $\beta$  & Paired  $\mathbf{MDE}_{80}$  \\
\midrule
+Multicolumn{5}{l}{\textit{Panel A. Public-dashboard equal-occupation approximation}} \\
+Equal occupation & Occupation and calendar-month FE &  $-0.072$  &  $[-0.122, -0.022]$  & --- & --- \\
+Equal occupation & SOC2 by young by post &  $-0.015$  &  $[-0.080, 0.050]$  &  $0.057$  &  $[-0.001, 0.115]$  &  $0.083$  \\
+Equal occupation & SOC2 by young by calendar month &  $-0.017$  &  $[-0.081, 0.047]$  &  $0.055$  &  $[-0.002, 0.113]$  &  $0.082$  \\
\addlinespace
\end{tabular}}
\end{table}
```

```

\multicolumn{5}{l}{\textit{Panel B. Historical employment-weighted approximation}} \\\
+Employment weighted & Occupation and calendar-month FE &  $\$-0.075$  &  $[-0.126, -0.025]$  & --- & --- \\\
+Employment weighted & SOC2 by young by post &  $\$-0.027$  &  $[-0.095, 0.041]$  &  $\$0.048$  &  $[-0.011, 0.108]$  &  $\$0.086$  \\\
+Employment weighted & SOC2 by young by calendar month &  $\$-0.027$  &  $[-0.093, 0.040]$  &  $\$0.048$  &  $[-0.011, 0.107]$  &  $\$0.086$  \\\
+bottomrule
+end{tabular}}
+tabnote{All rows are post-outcome exploratory CPS bridges, not replications of BCC's proprietary ADP analysis. Each row compares GPT-4-beta Q4--Q5 with Q1--Q3 on the same 468 occupations, includes
the standardized historical Webb-software control, and uses 9,999 common occupation-level Rademacher multipliers. Paired changes compare each SOC2-conditioned row with its own unconditioned
baseline. Panel A follows the public dashboard's equal-occupation description as closely as public information permits. Panel B uses earlier employment-weighted cuts formed from 108 months of
young-plus-older CPS stock, including postperiod months, and is only a grouping-rule diagnostic. Exact BCC occupation membership, universe, cutoff and tie algorithm are not public, so
concordance is unverified. The BCC and CPS objects also differ in ages, denominator, endpoint, employer controls, and observed worker--firm flows. Null-containing paired intervals are
nondetections, not equivalence.}
+end{table}
diff --git a/paper/tables/r3_appendix_characteristic_estimates.tex b/paper/tables/r3_appendix_characteristic_estimates.tex
new file mode 100644
index 0000000..173ef2c
--- /dev/null
+++ b/paper/tables/r3_appendix_characteristic_estimates.tex
@@ -0,0 +1,27 @@
+begin{table}[htbp]
+centering
+caption{Post-outcome exploratory characteristic-conditioning estimates}\label{tab:appendix_characteristic_estimates}
+scriptsize
+setlength{\tabcolsep}{3pt}
+resizebox{\textwidth}{!}{%
+begin{tabular}{lrrrrr}
+toprule
+Conditioning block & Occupations & Baseline coefficient & Augmented coefficient & [95\% CI] & Paired change & [95\% CI] & Paired  $\mathbf{MDE}_{\{80\}}$  \\\
+midrule
+multicolumn{6}{l}{\textit{Panel A. One-at-a-time maximal-support comparisons}} \\\
+Computer use & 455 &  $\$-0.096$  &  $\$-0.197$  &  $[-0.316, -0.077]$  &  $\$-0.101$  &  $[-0.176, -0.025]$  &  $0.108$  \\\
+Remotability & 408 &  $\$-0.115$  &  $\$-0.122$  &  $[-0.222, -0.022]$  &  $\$-0.007$  &  $[-0.048, 0.033]$  &  $0.058$  \\\
+Preperiod wage & 457 &  $\$-0.144$  &  $\$-0.161$  &  $[-0.248, -0.075]$  &  $\$-0.017$  &  $[-0.044, 0.009]$  &  $0.038$  \\\
+Education requirement & 404 &  $\$-0.100$  &  $\$-0.105$  &  $[-0.203, -0.008]$  &  $\$-0.005$  &  $[-0.027, 0.016]$  &  $0.031$  \\\
+Routine-task intensity & 449 &  $\$-0.137$  &  $\$-0.129$  &  $[-0.221, -0.036]$  &  $\$0.008$  &  $[-0.011, 0.027]$  &  $0.027$  \\\
+Manual/physical content & 455 &  $\$-0.140$  &  $\$-0.142$  &  $[-0.289, 0.004]$  &  $\$-0.002$  &  $[-0.092, 0.089]$  &  $0.130$  \\\
+SOC2 by young by post & 468 &  $\$-0.132$  &  $\$-0.022$  &  $[-0.162, 0.119]$  &  $\$0.111$  &  $[0.008, 0.213]$  &  $0.146$  \\\
+addlinespace
+multicolumn{6}{l}{\textit{Panel B. Cumulative comparisons on the 341-occupation common support}} \\\
+Computer use & 341 &  $\$-0.124$  &  $\$-0.248$  &  $[-0.391, -0.106]$  &  $\$-0.125$  &  $[-0.211, -0.038]$  &  $0.124$  \\\
+remotability & 341 &  $\$-0.124$  &  $\$-0.250$  &  $[-0.388, -0.111]$  &  $\$-0.126$  &  $[-0.210, -0.042]$  &  $0.121$  \\\
+wage, education, routine, and manual & 341 &  $\$-0.124$  &  $\$-0.233$  &  $[-0.390, -0.077]$  &  $\$-0.110$  &  $[-0.221, 0.002]$  &  $0.164$  \\\
+bottomrule
+end{tabular}}
+tabnote{All entries are post-outcome exploratory and were designed after the initial outcome pattern was known. Each baseline--augmented pair uses identical occupations, the canonical rebuilt
corrected-preperiod beta groups and Webb normalization, and 9,999 common occupation-level Rademacher multipliers. Panel A uses each characteristic's maximal finite subset; Panel B uses the
lateral 341-occupation common support, which covers 76.77 percent of primary-support January 2017--November 2022 treatment-construction stock. For comparison, the unaugmented coefficient on
all 468 occupations is  $\$-0.132$ , while the unaugmented coefficient after imposing the 341-occupation common support is  $\$-0.124$ ; that support-only movement is kept separate from every paired
conditioning change. Rows at and after the superseded unbounded pandemic-shortfall construction are omitted; the corrected shortfall analysis uses a separate 363-occupation population and
appears in Table--\ref{tab:appendix_shortfall}. A confidence interval containing zero is a nondetection, not evidence of equivalence; conditioning does not identify a causal AI component.}
+end{table}
diff --git a/paper/tables/r3_appendix_characteristic_information.tex b/paper/tables/r3_appendix_characteristic_information.tex
new file mode 100644
index 0000000..515f99d
--- /dev/null
+++ b/paper/tables/r3_appendix_characteristic_information.tex
@@ -0,0 +1,27 @@
+begin{table}[htbp]
+centering
+caption{Post-outcome exploratory support and target-information diagnostics}\label{tab:appendix_characteristic_information}
+scriptsize
+setlength{\tabcolsep}{3pt}
+resizebox{\textwidth}{!}{%
+begin{tabular}{lrrrrr}
+toprule
+Conditioning block & Employment coverage (\%) & Information retained (\%) & Target VIF-like ratio & Effective occupation count & Top-five information share (\%) \\\
+midrule
+multicolumn{6}{l}{\textit{Panel A. One-at-a-time maximal-support comparisons}} \\\
+Computer use & 96.75 & 29.02 & 3.45 & 52.58 & 22.34 \\\
+Remotability & 88.58 & 52.66 & 1.90 & 29.25 & 30.65 \\\
+Preperiod wage & 97.93 & 59.45 & 1.68 & 37.86 & 26.32 \\\
+Education requirement & 84.54 & 63.60 & 1.57 & 27.48 & 31.24 \\\
+Routine-task intensity & 98.45 & 65.19 & 1.53 & 43.26 & 24.59 \\\
+Manual/physical content & 96.85 & 30.61 & 3.27 & 51.05 & 22.34 \\\
+SOC2 by young by post & 100.00 & 20.81 & 4.81 & 53.06 & 21.87 \\\
+addlinespace
+multicolumn{6}{l}{\textit{Panel B. Cumulative comparisons on the 341-occupation common support}} \\\
+Computer use & 76.77 & 29.33 & 3.41 & 48.88 & 21.98 \\\
+remotability & 76.77 & 28.46 & 3.51 & 44.97 & 24.72 \\\
+wage, education, routine, and manual & 76.77 & 21.08 & 4.74 & 58.00 & 20.28 \\\
+bottomrule
+end{tabular}}
+tabnote{All diagnostics are post-outcome exploratory. Employment coverage is the share of primary-support January 2017--November 2022 corrected treatment-construction stock retained. "Information
retained" is nuisance-adjusted target information divided by fixed-effect-adjusted raw target information; its reciprocal is the reported target VIF-like ratio. These are design-information
diagnostics, not an estimate of latent exposure reliability. The effective occupation count is the inverse concentration of occupation-level conditional target-information shares, and the
final column is the share supplied by the five largest occupations. In the 341-occupation common-support sample, the 2017--2019-employment-weighted correlations of beta exposure with computer
use, remotability, wage, education requirement, routine-task intensity, and manual/physical content are 0.807, 0.562, 0.429, 0.421, 0.197, and  $\$-0.748$ , respectively. Rows based on the
superseded unbounded pandemic-shortfall construction are omitted. Substantial information loss makes residual coefficients less precise and does not purify an AI treatment.}
+end{table}
diff --git a/paper/tables/r3_appendix_endpoint_sensitivity.tex b/paper/tables/r3_appendix_endpoint_sensitivity.tex
new file mode 100644
index 0000000..ccc3380
--- /dev/null
+++ b/paper/tables/r3_appendix_endpoint_sensitivity.tex
@@ -0,0 +1,27 @@
+begin{table}[htbp]
+centering
+caption{Post-outcome exploratory endpoint sensitivity}\label{tab:app_endpoints}
+scriptsize
+setlength{\tabcolsep}{3pt}
+resizebox{\textwidth}{!}{%
+begin{tabular}{lrrrrr}
+toprule
+Endpoint & Months & Coefficient & [95\% CI] & Endpoint  $\$-\$$  full & [paired 95\% CI] & Paired  $\mathbf{MDE}_{\{80\}}$  \\\
+midrule
+multicolumn{6}{l}{\textit{Pooled}} \\\
+Through December 2024 & 95 &  $\$-0.1110$  &  $[-0.2008, -0.0213]$  &  $\$0.0211$  &  $[-0.0073, 0.0495]$  &  $0.0407$  \\\
+Through September 2025 & 104 &  $\$-0.1096$  &  $[-0.1978, -0.0214]$  &  $\$0.0225$  &  $[0.0053, 0.0397]$  &  $0.0245$  \\\
+Through December 2025 (actual gap) & 106 &  $\$-0.1109$  &  $[-0.1979, -0.0239]$  &  $\$0.0212$  &  $[0.0057, 0.0367]$  &  $0.0221$  \\\
+Through July 2026, excluding September and November 2025 & 111 &  $\$-0.1337$  &  $[-0.2236, -0.0438]$  &  $\$-0.0016$  &  $[-0.0075, 0.0043]$  &  $0.0084$  \\\
+Through July 2026 (reference) & 113 &  $\$-0.1321$  &  $[-0.2206, -0.0437]$  & Reference & --- & \\\
+addlinespace
+multicolumn{6}{l}{\textit{SOC2 by calendar month}} \\\
+Through December 2024 & 95 &  $\$-0.0120$  &  $[-0.1657, 0.1417]$  &  $\$0.0097$  &  $[-0.0526, 0.0719]$  &  $0.0897$  \\\
+Through September 2025 & 104 &  $\$-0.0219$  &  $[-0.1616, 0.1178]$  &  $\$-0.0002$  &  $[-0.0345, 0.0341]$  &  $0.0491$  \\\
+Through December 2025 (actual gap) & 106 &  $\$-0.0170$  &  $[-0.1555, 0.1215]$  &  $\$0.0047$  &  $[-0.0248, 0.0342]$  &  $0.0422$  \\\
+Through July 2026, excluding September and November 2025 & 111 &  $\$-0.0252$  &  $[-0.1660, 0.1155]$  &  $\$-0.0035$  &  $[-0.0131, 0.0060]$  &  $0.0137$  \\\
+Through July 2026 (reference) & 113 &  $\$-0.0217$  &  $[-0.1607, 0.1173]$  & Reference & --- & \\\

```

+\\bottomrule
+\\end{tabular}]

+\\tabnote{All rows use the rebuilt corrected-preperiod treatment and the same occupation support. The December 2024 row is the requested pre-2025 endpoint; the December 2025 row retains the actual missing October survey month. The shutdown-month row excludes September and November 2025 and does not interpolate October. Nonreference endpoint intervals and paired differences use the common endpoint-grid draws. To avoid printing a second seed-dependent interval for an identical target, each full-window reference uses its declared canonical single-target interval. The MDE is the two-sided five-percent, 80-percent normal-theory precision diagnostic, not an economic-equivalence threshold.}

+\\end{table}
diff --git a/paper/tables/r3_appendix_exit_components.tex b/paper/tables/r3_appendix_exit_components.tex
new file mode 100644
index 0000000..16803fb
--- /dev/null
+++ b/paper/tables/r3_appendix_exit_components.tex
@@ -0,0 +1,16 @@
+\\begin{table}[\\htbp]
+\\centering
+\\caption{Employment-exit components in linked CPS records}\\label{tab:app_exit_components}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{llrrr}
+\\toprule
+Endpoint & Component & Raw events & Coefficient & 95\\% interval \\\br/>
+\\midrule
+Adjacent month & Unemployment entry & 32,488 & $\$0.055$ & $[-0.272, 0.381]$ \\\br/>
+Adjacent month & Labor-force exit & 74,480 & $\$0.135$ & $[-0.064, 0.333]$ \\\br/>
+Twelve month & Unemployment at endpoint & 27,330 & $\$0.246$ & $[-0.295, 0.787]$ \\\br/>
+Twelve month & Out of labor force at endpoint & 78,773 & $\$0.078$ & $[-0.181, 0.337]$ \\\br/>
+\\bottomrule
+\\end{tabular}]}

+\\tabnote{Rows use official longitudinal weights and the same young-relative Q5--Q1 log-rate-ratio contrast as the core flow table. Counts are repeated link events rather than unique persons. Unemployment and out-of-labor-force events exhaust observed employment exits, but their nonlinear coefficients do not add. Every interval includes zero.}

+\\end{table}
diff --git a/paper/tables/r3_appendix_family_information_precision.tex b/paper/tables/r3_appendix_family_information_precision.tex
new file mode 100644
index 0000000..e80665d
--- /dev/null
+++ b/paper/tables/r3_appendix_family_information_precision.tex
@@ -0,0 +1,16 @@
+\\begin{table}[\\htbp]
+\\centering
+\\caption{Family-support information and precision diagnostics}\\label{tab:app_family_information}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{llrrrrrr}
+\\toprule
+Target and structure & Occ. & $\$I$ (millions) & $\$I/I_{\\mathbf{raw}}$ & $\$I/I_{\\mathbf{parent}}$ & Effective occ. & Top-five share & Normal MDE $_{\\$80}$ \\\br/>
+\\midrule
+Q5--Q1, pooled & 468 & 26.289 & 0.378 & 1.000 & 43.2 & 0.245 & 0.1266 \\\br/>
+Q5--Q1, SOC2 $\$\\times$ month & 468 & 7.803 & 0.112 & 0.297 & 53.3 & 0.218 & 0.1998 \\\br/>
+Direct-tail Q5--Q1, SOC2 $\$\\times$ month & 29 & 0.669 & 0.080 & 0.308 & 6.2 & 0.778 & 0.4576 \\\br/>
+Continuous within-SOC2 slope, SOC2 $\$\\times$ month & 468 & 261.278 & 0.583 & 0.965 & 48.0 & 0.249 & 0.0313 \\\br/>
+\\bottomrule
+\\end{tabular}]}

+\\tabnote{ $\$I$ is nuisance-adjusted Fisher information after projecting the target score on the model's nuisance score; $\$I_{\\mathbf{raw}}$ is the raw weighted treatment sum of squares. "Parent" is the corresponding pooled model on the same target and population, so the direct-tail and continuous rows each use their own pooled parent. Because target scales and populations differ, absolute $\$I$ is not comparable across all rows. Effective occupations are the inverse Herfindahl count of occupation-level information contributions; the top-five share is the sum of the five largest contributions. Fitted information and its effective count are design-concentration diagnostics: neither replaces the nominal occupation-cluster count nor measures first-stage CPS sampling precision. MDE $_{\\$80}$ =2.80\\widehat{SE} uses the occupation-cluster standard error and is a normal-theory precision diagnostic, not a rejection result. The direct-tail MDE is 0.4576 log point; the continuous MDE is 0.0313 per one within-family standard deviation (0.1084 raw beta units).}

+\\end{table}
diff --git a/paper/tables/r3_appendix_family_support.tex b/paper/tables/r3_appendix_family_support.tex
new file mode 100644
index 0000000..f70f0af
--- /dev/null
+++ b/paper/tables/r3_appendix_family_support.tex
@@ -0,0 +1,42 @@
+\\begin{center}
+\\footnotesize
+\\begin{longtable}{@{}r p{0.25\\textwidth} r r r l c r@{}}
+\\caption{Broad-family exposure support}\\label{tab:app_family_support}\\\\
+\\toprule
+SOC2 & Family & Occs. & Stock (\\%) & Beta range & Quintiles & Q1/Q5 & Q5 info. (\\%) \\\br/>
+\\midrule
+\\endfirstthead
+\\multicolumn{8}{l}{\\footnotesize \\tshape Table \\thetable{} continued}\\\\
+\\toprule
+SOC2 & Family & Occs. & Stock (\\%) & Beta range & Quintiles & Q1/Q5 & Q5 info. (\\%) \\\br/>
+\\midrule
+\\endthead
+\\midrule\\multicolumn{8}{l}{\\footnotesize Continued on next page}\\\\
+\\endfoot
+\\bottomrule
+\\endlastfoot
+11 & Management & 29 & 13.56 & 0.367--0.586 & 3|4|5 & No & 3.04 \\\br/>
+13 & Business and Financial Operations & 28 & 6.22 & 0.405--0.636 & 3|4|5 & No & 5.99 \\\br/>
+15 & Computer and Mathematical & 15 & 4.31 & 0.500--0.950 & 4|5 & No & 0.28 \\\br/>
+17 & Architecture and Engineering & 19 & 2.28 & 0.406--0.727 & 3|4|5 & No & 1.87 \\\br/>
+19 & Life, Physical, and Social Science & 21 & 1.25 & 0.344--0.723 & 3|4|5 & No & 0.54 \\\br/>
+21 & Community and Social Service & 12 & 1.31 & 0.111--0.439 & 1|2|3 & No & 0.20 \\\br/>
+23 & Legal & 4 & 1.34 & 0.425--0.571 & 3|4|5 & No & 0.28 \\\br/>
+25 & Educational Instruction and Library & 7 & 4.53 & 0.229--0.567 & 2|3|5 & No & 3.20 \\\br/>
+27 & Arts, Design, Entertainment, Sports, and Media & 25 & 1.80 & 0.000--0.958 & 1|2|3|4|5 & Yes & 4.86 \\\br/>
+29 & Healthcare Practitioners and Technical & 37 & 6.38 & 0.067--0.618 & 1|2|3|4|5 & Yes & 3.61 \\\br/>
+31 & Healthcare Support & 14 & 3.66 & 0.074--0.875 & 1|2|3|5 & Yes & 7.26 \\\br/>
+33 & Protective Service & 17 & 2.28 & 0.036--0.415 & 1|2|3 & No & 1.42 \\\br/>
+35 & Food Preparation and Serving Related & 10 & 3.27 & 0.000--0.511 & 1|2|4 & No & 10.84 \\\br/>
+37 & Building and Grounds Cleaning and Maintenance & 6 & 2.51 & 0.000--0.355 & 1|2|3 & No & 1.29 \\\br/>
+39 & Personal Care and Service & 17 & 2.51 & 0.015--0.400 & 1|2|3 & No & 2.97 \\\br/>
+41 & Sales and Related & 17 & 10.06 & 0.133--0.806 & 1|2|3|4|5 & Yes & 17.40 \\\br/>
+43 & Office and Administrative Support & 47 & 11.21 & 0.216--0.975 & 2|3|4|5 & No & 18.92 \\\br/>
+45 & Farming, Fishing, and Forestry & 5 & 0.15 & 0.043--0.292 & 1|2 & No & 0.14 \\\br/>
+47 & Construction and Extraction & 30 & 6.14 & 0.000--0.417 & 1|2|3 & No & 1.97 \\\br/>
+49 & Installation, Maintenance, and Repair & 31 & 3.56 & 0.000--0.410 & 1|2|3 & No & 2.64 \\\br/>
+51 & Production & 52 & 4.14 & 0.000--0.438 & 1|2|3 & No & 5.67 \\\br/>
+53 & Transportation and Material Moving & 25 & 7.52 & 0.031--0.413 & 1|2|3 & No & 5.59 \\\br/>
+\\end{longtable}

+\\tabnote{Stock is each family's share of preperiod employment on the 468-occupation rebuilt support. Q5 information is the family's share of nuisance-adjusted fitted information for the family-conditioned Q5 target; it is neither a sampling-cluster share nor pre-outcome residual-exposure support. Beta ranges and memberships use the corrected preperiod treatment contract. Only Q1/Q5 rows enter the changed-population direct-tail exercise.}

+\\end{center}
diff --git a/paper/tables/r3_appendix_heterogeneity.tex b/paper/tables/r3_appendix_heterogeneity.tex
new file mode 100644
index 0000000..7691df8
--- /dev/null
+++ b/paper/tables/r3_appendix_heterogeneity.tex
@@ -0,0 +1,20 @@
+\\begin{table}[\\htbp]
+\\centering
+\\caption{Industry, education, and exact-age comparisons}\\label{tab:app_heterogeneity}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{llrrrr}
+\\bottomrule
+\\end{tabular}]}

```

\toprule
+Model & Occupations & Coefficient & Pointwise 95\% interval & Simultaneous 95\% interval \\
+midrule
+Occupation--industry-cell baseline & 468 & \$-0.138\$ & \$[-0.228,-0.048]\$ & --- \\
+Industry-conditioned & 468 & \$-0.099\$ & \$[-0.188,-0.009]\$ & --- \\
+BA or higher & 429 & \$-0.138\$ & \$[-0.275,-0.001]\$ & \$[-0.293,0.017]\$ \\
+No BA & 429 & \$-0.123\$ & \$[-0.231,-0.014]\$ & \$[-0.243,-0.002]\$ \\
+Age 22 & 439 & \$-0.093\$ & \$[-0.212,0.026]\$ & \$[-0.239,0.053]\$ \\
+Age 23 & 439 & \$-0.140\$ & \$[-0.252,-0.028]\$ & \$[-0.280,-0.0003]\$ \\
+Age 24 & 439 & \$-0.222\$ & \$[-0.358,-0.086]\$ & \$[-0.394,-0.050]\$ \\
+Age 25 & 439 & \$-0.082\$ & \$[-0.188,0.024]\$ & \$[-0.212,0.047]\$ \\
+bottomrule
+end{tabular}}
+{tabnote[Both industry rows use occupation--broad-industry--age--month weighted-stock cells. The baseline includes common Q2--Q5 and Webb interactions; the conditioned row additionally includes 13
broad-industry-specific young-by-post shifts. Education and age collections use their own common occupational support and 9,999 common occupation multipliers. The four-age common-score
equality test gives  $\chi^2(3)=8.21$ ,  $p=.0419$ , but every simultaneous age-minus-pooled interval includes zero; no individual age departure is promoted.]}
+end{table}
diff --git a/paper/tables/r3_appendix_honestdid.tex b/paper/tables/r3_appendix_honestdid.tex
new file mode 100644
index 0000000..bc06ef8
--- /dev/null
+++ b/paper/tables/r3_appendix_honestdid.tex
@@ -0,0 +1,33 @@
+{begin{table}[htbp]
+{centering
+{caption[Official HonestDiD sensitivity intervals]\label{tab:app_honestdid}
+{small
+{begin{tabular}{llrrr}
+{toprule
+{Structure & Calibration & Restriction & Lower & Upper \\
+{midrule
+{Pooled & 2017Q1--2022Q3 & Conventional & \$-0.264\$ & \$0.024\$ \\
+ & & Smoothness,  $\$M=0\$$  & \$-0.140\$ & \$0.016\$ \\
+ & & Smoothness,  $\$M=0.005\$$  & \$-0.482\$ & \$0.516\$ \\
+ & & Relative magnitude,  $\$|bar M=0\$$  & \$-0.260\$ & \$0.019\$ \\
+ & & Relative magnitude,  $\$|bar M=0.5\$$  & \$-1.023\$ & \$0.789\$ \\
+{Pooled & 2021Q1--2022Q3 & Conventional & \$-0.264\$ & \$0.024\$ \\
+ & & Smoothness,  $\$M=0\$$  & \$-0.230\$ & \$0.204\$ \\
+ & & Smoothness,  $\$M=0.005\$$  & \$-0.540\$ & \$0.495\$ \\
+ & & Relative magnitude,  $\$|bar M=0\$$  & \$-0.260\$ & \$0.019\$ \\
+ & & Relative magnitude,  $\$|bar M=0.5\$$  & \$-0.985\$ & \$0.742\$ \\
+{addlinespace
+{SOC2 monthly paths & 2017Q1--2022Q3 & Conventional & \$-0.425\$ & \$0.010\$ \\
+ & & Smoothness,  $\$M=0\$$  & \$-0.351\$ & \$0.007\$ \\
+ & & Smoothness,  $\$M=0.005\$$  & \$-0.869\$ & \$0.352\$ \\
+ & & Relative magnitude,  $\$|bar M=0\$$  & \$-0.420\$ & \$0.007\$ \\
+ & & Relative magnitude,  $\$|bar M=0.5\$$  & \$-2.222\$ & \$1.884\$ \\
+{SOC2 monthly paths & 2021Q1--2022Q3 & Conventional & \$-0.425\$ & \$0.010\$ \\
+ & & Smoothness,  $\$M=0\$$  & \$-0.772\$ & \$-0.007\$ \\
+ & & Smoothness,  $\$M=0.005\$$  & \$-1.100\$ & \$0.207\$ \\
+ & & Relative magnitude,  $\$|bar M=0\$$  & \$-0.420\$ & \$0.007\$ \\
+ & & Relative magnitude,  $\$|bar M=0.5\$$  & \$-2.222\$ & \$1.884\$ \\
+{bottomrule
+end{tabular}}
+{tabnote[The target is the observed-month-weighted average of the 15 post-quarter Q5--Q1 coefficients relative to 2022Q4. Intervals use the full occupation-cluster covariance and official HonestDiD
0.2.8. Smoothness  $\$M\$$  is in log points per quarter squared;  $\$|bar M\$$  bounds post changes relative to the largest signed preperiod change. Both the full 2017Q1--2022Q3 calibration and the
consecutive post-reopening 2021Q1--2022Q3 calibration are fixed and reported. The recent-window family-path interval excludes zero only at the exact-linear-bias restriction  $\$M=0\$$  and includes
zero at  $\$M=0.005\$$ . Because each conventional interval includes zero, positive zero-exclusion robustness is undefined, not an omitted threshold.]}
+end{table}
diff --git a/paper/tables/r3_appendix_lofo.tex b/paper/tables/r3_appendix_lofo.tex
new file mode 100644
index 0000000..7e277e1
--- /dev/null
+++ b/paper/tables/r3_appendix_lofo.tex
@@ -0,0 +1,20 @@
+{begin{table}[htbp]
+{centering
+{caption[Largest leave-one-family-out movements for the conditioned Q5 target]\label{tab:app_lofo}
+{resizebox{\textwidth}{\height}{%
+{begin{tabular}{rllrrr}
+{toprule
+{Omitted SOC2 & Family & Preperiod stock (\%) & Re-estimated coefficient & Movement [95\% paired CI] \\
+{midrule
+{43 & Office and Administrative Support & 11.21 & -0.0614 & -0.0397 [-0.1131,0.0337] \\
+{35 & Food Preparation and Serving Related & 3.27 & 0.0174 & 0.0391 [-0.0292,0.1074] \\
+{31 & Healthcare Support & 3.66 & -0.0534 & -0.0317 [-0.0772,0.0139] \\
+{13 & Business and Financial Operations & 6.22 & 0.0099 & 0.0316 [-0.0006,0.0638] \\
+{39 & Personal Care and Service & 2.51 & -0.0457 & -0.0240 [-0.0496,0.0015] \\
+{25 & Educational Instruction and Library & 4.53 & -0.0351 & -0.0134 [-0.0383,0.0115] \\
+{17 & Architecture and Engineering & 2.28 & -0.0339 & -0.0123 [-0.0330,0.0085] \\
+{33 & Protective Service & 2.28 & -0.0096 & 0.0121 [-0.0050,0.0291] \\
+{bottomrule
+end{tabular}}
+{tabnote[The parent SOC2-by-calendar-month coefficient is  $\$-0.0217\$$ . Families are ranked by the absolute coefficient movement, so this is an outcome-informed influence diagnostic rather than a
preferred deletion rule. Quintiles and the continuous centering scale remain fixed; all 22 deletions are available in the machine-readable file.]}
+end{table}
diff --git a/paper/tables/r3_appendix_population_eras.tex b/paper/tables/r3_appendix_population_eras.tex
new file mode 100644
index 0000000..00a63c2
--- /dev/null
+++ b/paper/tables/r3_appendix_population_eras.tex
@@ -0,0 +1,16 @@
+{begin{table}[htbp]
+{centering
+{caption[Post-outcome exploratory comparison of early and later postperiods]\label{tab:appendix_population_eras}
+{small
+{setlength{\tabcolsep}{4pt}
+{resizebox{\textwidth}{\height}{%
+{begin{tabular}{rllrrrr}
+{toprule
+{Cell construction & 2023--2024 coefficient [95\% CI] & 2025--2026 coefficient [95\% CI] & Later minus early [95\% CI] & Paired  $\$|mathrm{MDE}|_{.80}\$ \\
+{midrule
+{Official final-weight stocks & \$-0.111\%; [-0.197,-0.024]\$ & \$-0.166\%; [-0.270,-0.062]\$ & \$-0.056\%; [-0.123,0.012]\$ & 0.097 \\
+{Respondent-equivalent routed counts & \$-0.103\%; [-0.193,-0.012]\$ & \$-0.183\%; [-0.277,-0.089]\$ & \$-0.080\%; [-0.139,-0.021]\$ & 0.085 \\
+{bottomrule
+end{tabular}}
+{tabnote[This timing comparison was produced after outcomes were opened and is post-outcome exploratory. Both rows use the corrected 113-month calendar, the same 468 occupations, a joint two-era
specification, and 9,999 common occupation multipliers within each paired later-minus-early contrast. Official final-weight stocks are the paper's employment-stock object. Respondent-
equivalent values are fractional routed-record counts after occupational crosswalking; they are neither distinct survey respondents nor a recovered counterfactual CPS weight vintage. The
official-weight interval does not detect an era difference, whereas the respondent-equivalent interval excludes zero. That divergence does not identify response composition or show that
population-control revisions caused the later estimate.]}
+end{table}
diff --git a/paper/tables/r3_appendix_precision_summary.tex b/paper/tables/r3_appendix_precision_summary.tex
new file mode 100644
index 0000000..fb0883c
--- /dev/null
+++ b/paper/tables/r3_appendix_precision_summary.tex
@@ -0,0 +1,20 @@
+{begin{table}[htbp]
+{centering
+{caption[Post-outcome exploratory comparison of early and later postperiods]\label{tab:appendix_population_eras}
+{small
+{setlength{\tabcolsep}{4pt}
+{resizebox{\textwidth}{\height}{%
+{begin{tabular}{rllrrrr}
+{toprule
+{Cell construction & 2023--2024 coefficient [95\% CI] & 2025--2026 coefficient [95\% CI] & Later minus early [95\% CI] & Paired  $\$|mathrm{MDE}|_{.80}\$ \\
+{midrule
+{Official final-weight stocks & \$-0.111\%; [-0.197,-0.024]\$ & \$-0.166\%; [-0.270,-0.062]\$ & \$-0.056\%; [-0.123,0.012]\$ & 0.097 \\
+{Respondent-equivalent routed counts & \$-0.103\%; [-0.193,-0.012]\$ & \$-0.183\%; [-0.277,-0.089]\$ & \$-0.080\%; [-0.139,-0.021]\$ & 0.085 \\
+{bottomrule
+end{tabular}}
+{tabnote[This timing comparison was produced after outcomes were opened and is post-outcome exploratory. Both rows use the corrected 113-month calendar, the same 468 occupations, a joint two-era
specification, and 9,999 common occupation multipliers within each paired later-minus-early contrast. Official final-weight stocks are the paper's employment-stock object. Respondent-
equivalent values are fractional routed-record counts after occupational crosswalking; they are neither distinct survey respondents nor a recovered counterfactual CPS weight vintage. The
official-weight interval does not detect an era difference, whereas the respondent-equivalent interval excludes zero. That divergence does not identify response composition or show that
population-control revisions caused the later estimate.]}
+end{table}
diff --git a/paper/tables/r3_appendix_precision_summary.tex b/paper/tables/r3_appendix_precision_summary.tex
new file mode 100644
index 0000000..fb0883c
--- /dev/null
+++ b/paper/tables/r3_appendix_precision_summary.tex
@@ -0,0 +1,20 @@
+{begin{table}[htbp]
+{centering
+{caption[Post-outcome exploratory comparison of early and later postperiods]\label{tab:appendix_population_eras}
+{small
+{setlength{\tabcolsep}{4pt}
+{resizebox{\textwidth}{\height}{%
+{begin{tabular}{rllrrrr}
+{toprule
+{Cell construction & 2023--2024 coefficient [95\% CI] & 2025--2026 coefficient [95\% CI] & Later minus early [95\% CI] & Paired  $\$|mathrm{MDE}|_{.80}\$ \\
+{midrule
+{Official final-weight stocks & \$-0.111\%; [-0.197,-0.024]\$ & \$-0.166\%; [-0.270,-0.062]\$ & \$-0.056\%; [-0.123,0.012]\$ & 0.097 \\
+{Respondent-equivalent routed counts & \$-0.103\%; [-0.193,-0.012]\$ & \$-0.183\%; [-0.277,-0.089]\$ & \$-0.080\%; [-0.139,-0.021]\$ & 0.085 \\
+{bottomrule
+end{tabular}}
+{tabnote[This timing comparison was produced after outcomes were opened and is post-outcome exploratory. Both rows use the corrected 113-month calendar, the same 468 occupations, a joint two-era
specification, and 9,999 common occupation multipliers within each paired later-minus-early contrast. Official final-weight stocks are the paper's employment-stock object. Respondent-
equivalent values are fractional routed-record counts after occupational crosswalking; they are neither distinct survey respondents nor a recovered counterfactual CPS weight vintage. The
official-weight interval does not detect an era difference, whereas the respondent-equivalent interval excludes zero. That divergence does not identify response composition or show that
population-control revisions caused the later estimate.]}
+end{table}$$$ 
```

```

\begin{table}[htbp]
\centering
\caption{Consolidated normal-theory precision diagnostics}\label{tab:appendix_precision_summary}
\scriptsize
\resizebox{\textwidth}{!}{%
\begin{tabular}{lll r}
\toprule
Analysis & Quantity & Scale &  $\mathbf{MDE}_{.80}$  \\
\midrule
Core pooled & Q5--Q1 coefficient & Log young-to-older stock ratio & 0.1266 \\
Family-month & Q5--Q1 coefficient & Log young-to-older stock ratio & 0.1998 \\
Family-month vs. pooled & Paired coefficient difference & Log young-to-older stock ratio & 0.1454 \\
Direct tall within family & Q5--Q1 coefficient & Log young-to-older stock ratio & 0.4576 \\
Continuous within family & Exposure slope & One within-family exposure SD & 0.0313 \\
Computer-use conditioning & Paired coefficient difference & Log young-to-older stock ratio & 0.1131 \\
Remotability conditioning & Paired coefficient difference & Log young-to-older stock ratio & 0.0510 \\
Industry conditioning & Paired coefficient difference & Log young-to-older stock ratio & 0.0444 \\
BA vs. non-BA & Paired coefficient difference & Log young-to-older stock ratio & 0.2014 \\
Architecture comparisons & Paired coefficient differences & Log young-to-older stock ratio & 0.0609--0.1689 \\
Adjacent exit & Q5--Q1 coefficient & Log conditional transition-rate ratio & 0.254 \\
Adjacent occupation change & Q5--Q1 coefficient & Log conditional transition-rate ratio & 0.174 \\
Adjacent entry destination & Q5--Q1 coefficient & Log conditional allocation ratio & 0.260 \\
Twelve-month exit & Q5--Q1 coefficient & Log conditional transition-rate ratio & 0.336 \\
Twelve-month occupation change & Q5--Q1 coefficient & Log conditional transition-rate ratio & 0.115 \\
Twelve-month entry destination & Q5--Q1 coefficient & Log conditional allocation ratio & 0.282 \\
\bottomrule
\end{tabular}}
\tabnote{$\mathbf{MDE}_{.80}=(1.96+0.84)\widehat{\text{SE}}$ is the two-sided five-percent normal-theory minimum detectable quantity at 80-percent power. It summarizes resolution on the displayed scale; it is not a wild-score critical value, equivalence margin, or acceptance threshold. Architecture entries report the observed range across seven beta-minus-alternative comparisons. Flow rows use their own risk sets and should not be compared as if they estimated one common probability.}
\end{table}

diff --git a/paper/tables/r3_appendix_primitive_ds.tex b/paper/tables/r3_appendix_primitive_ds.tex
new file mode 100644
index 0000000..3ed72ad
--- /dev/null
+++ b/paper/tables/r3_appendix_primitive_ds.tex
\begin{table}[htbp]
\centering
\caption{Post-outcome exploratory direct and software-complement primitives}\label{tab:appendix_primitive_ds}
\small
\setlength{\tabcolsep}{4pt}
\resizebox{\textwidth}{!}{%
\begin{tabular}{llrrrr}
\toprule
Units or contrast & Term & Estimate & [95\% CI] & Bootstrap SE &  $\mathbf{MDE}_{.80}$  \\
\midrule
Raw share &  $\mathbf{D} \times \mathbf{E}$  young &  $\mathbf{D} \times \mathbf{E}$  post &  $-\$-0.189$ ; [-0.354,-0.024] &  $\$ 0.085$  & 0.239 \\
Raw share &  $\mathbf{S} \times \mathbf{E}$  young &  $\mathbf{S} \times \mathbf{E}$  post &  $-\$-0.089$ ; [-0.172,-0.005] &  $\$ 0.043$  & 0.121 \\
Preperiod SD &  $\mathbf{D} \times \mathbf{E}$  young &  $\mathbf{D} \times \mathbf{E}$  post &  $-\$-0.031$ ; [-0.058,-0.004] &  $\$ 0.014$  & 0.039 \\
Preperiod SD &  $\mathbf{S} \times \mathbf{E}$  young &  $\mathbf{S} \times \mathbf{E}$  post &  $-\$-0.028$ ; [-0.054,-0.002] &  $\$ 0.014$  & 0.038 \\
One SD in each primitive &  $\mathbf{D} \times \mathbf{S}$  joint change & &  $-\$-0.059$ ; [-0.098,-0.020] &  $\$ 0.020$  & 0.056 \\
\bottomrule
\end{tabular}}
\tabnote{All entries are post-outcome exploratory and use the same 468 occupations and 9,999 common occupation multipliers. The joint model includes  $\mathbf{D}\mathbf{E}$ ,  $\mathbf{S}\mathbf{E}$ , and standardized Webb software, each interacted with young  $\mathbf{D} \times \mathbf{E}$  post. Here  $\mathbf{D} \times \mathbf{E}$  is the share of tasks classified as directly accelerable and  $\mathbf{S} \times \mathbf{E}$  is the share requiring complementary software; beta satisfies  $\mathbf{D} \times \mathbf{E} \leq 0.5$ . The preperiod employment-weighted means (standard deviations) are 0.144 (0.164) for  $\mathbf{D}$  and 0.463 (0.315) for  $\mathbf{S}$ . The common-draw correlation between the two standardized coefficient draws is 0.062. Component-row MDEs use the analytic occupation-clustered standard errors; the joint-change MDE uses its common-draw standard error. The final row is the declared one-standard-deviation increase in both primitives, not a paired difference between two technologies. The parameterization does not measure adoption and is not a causal decomposition.}
\end{table}

diff --git a/paper/tables/r3_appendix_profile.tex b/paper/tables/r3_appendix_profile.tex
new file mode 100644
index 0000000..29fe495
--- /dev/null
+++ b/paper/tables/r3_appendix_profile.tex
\begin{table}[htbp]
\centering
\caption{Exposure profile before and after broad-family monthly paths}\label{tab:app_profile}
\scriptsize
\resizebox{\textwidth}{!}{%
\begin{tabular}{llrrrrrrrr}
\toprule
& \multicolumn{3}{c}{Pooled} & \multicolumn{3}{c}{SOC2  $\mathbf{D} \times \mathbf{E}$  month} & Paired movement \\
\cmidrule{1r}{2-4}\cmidrule{1r}{5-7}
Group & Coef. & Pointwise 95\% CI & Simultaneous 95\% CI & Coef. & Pointwise 95\% CI & Simultaneous 95\% CI & Coef. & [95\% CI] \\
\midrule
Q2 & -0.0784 & [-0.1546,-0.0023] & [-0.1716,0.0147] & -0.0247 & [-0.1228,0.0734] & [-0.1431,0.0937] & 0.0537 & [-0.0051,0.1126] \\
Q3 & -0.0500 & [-0.1224,0.0225] & [-0.1383,0.0383] & 0.0169 & [-0.0859,0.1197] & [-0.1043,0.1381] & 0.0669 & [-0.0047,0.1386] \\
Q4 & -0.0896 & [-0.1578,-0.0214] & [-0.1737,-0.0055] & -0.0126 & [-0.1302,0.1051] & [-0.1510,0.1259] & 0.0770 & [-0.0138,0.1679] \\
Q5 & -0.1321 & [-0.2206,-0.0437] & [-0.2393,-0.0249] & -0.0217 & [-0.1607,0.1173] & [-0.1836,0.1403] & 0.1104 & [0.0107,0.2102] \\
\bottomrule
\end{tabular}}
\tabnote{Q1 is omitted. Simultaneous intervals use each model's common four-target maximum- $|t|$  critical value. To prevent the same Q5 target carrying seed-dependent pointwise intervals, the Q5 pointwise and paired cells reproduce the declared single-target canonical intervals; its simultaneous cells remain the four-target family-profile result. All rows use rebuilt labels and 9,999 common occupation draws within each declared collection.}
\end{table}

diff --git a/paper/tables/r3_appendix_shortfall.tex b/paper/tables/r3_appendix_shortfall.tex
new file mode 100644
index 0000000..f1a45fd
--- /dev/null
+++ b/paper/tables/r3_appendix_shortfall.tex
\begin{table}[htbp]
\centering
\caption{Corrected bounded pandemic-shortfall diagnostics}\label{tab:appendix_shortfall}
\scriptsize
\begin{tabular}{llrrrrrrrrrr}
\toprule
Shortfall & Model & Baseline Q5--Q1 & Conditioned Q5--Q1 & Movement & [sampling interval] \\
\midrule
Total stock & Pooled &  $-\$-0.1641$  &  $-\$-0.1637$  &  $\$ 0.00040$ ; [-0.00655,0.00661] \\
Total stock & SOC2 &  $\mathbf{D} \times \mathbf{E}$  month &  $-\$-0.0870$  &  $\$-0.0875$  &  $\$-0.00050$ ; [-0.01377,0.00970] \\
Young relative & Pooled &  $-\$-0.1641$  &  $-\$-0.1630$  &  $\$ 0.00114$ ; [-0.00509,0.00573] \\
Young relative & SOC2 &  $\mathbf{D} \times \mathbf{E}$  month &  $-\$-0.0870$  &  $\$-0.0809$  &  $\$ 0.00613$ ; [-0.00889,0.01835] \\
\bottomrule
\end{tabular}}
\tabnote{The common population contains 363 one-to-one-mapped occupations with sufficient preperiod histories. Total-employment predictions are bounded below at zero and young-share predictions to  $[0,1]$ . Movement is the conditioned minus baseline exposure coefficient. Brackets are basic intervals from 399 linked-household draws that regenerate the shortfall but hold exposure labels fixed. They are released-weight sampling sensitivities, not occupation-wild or complete CPS design-based intervals, and are not mechanically combined with the article's shock-cluster variance. Joint label regeneration gives the same displayed point estimates and very similar uncertainty.}
\end{table}

diff --git a/paper/tables/r3_appendix_webb_support.tex b/paper/tables/r3_appendix_webb_support.tex
new file mode 100644
index 0000000..3b46fc1
--- /dev/null
+++ b/paper/tables/r3_appendix_webb_support.tex
\begin{table}[htbp]
\centering
\caption{Corrected bounded pandemic-shortfall diagnostics}\label{tab:appendix_shortfall}
\scriptsize
\begin{tabular}{llrrrrrrrrrr}
\toprule
Shortfall & Model & Baseline Q5--Q1 & Conditioned Q5--Q1 & Movement & [sampling interval] \\
\midrule
Total stock & Pooled &  $-\$-0.1641$  &  $-\$-0.1637$  &  $\$ 0.00040$ ; [-0.00655,0.00661] \\
Total stock & SOC2 &  $\mathbf{D} \times \mathbf{E}$  month &  $-\$-0.0870$  &  $\$-0.0875$  &  $\$-0.00050$ ; [-0.01377,0.00970] \\
Young relative & Pooled &  $-\$-0.1641$  &  $-\$-0.1630$  &  $\$ 0.00114$ ; [-0.00509,0.00573] \\
Young relative & SOC2 &  $\mathbf{D} \times \mathbf{E}$  month &  $-\$-0.0870$  &  $\$-0.0809$  &  $\$ 0.00613$ ; [-0.00889,0.01835] \\
\bottomrule
\end{tabular}}
\tabnote{The common population contains 363 one-to-one-mapped occupations with sufficient preperiod histories. Total-employment predictions are bounded below at zero and young-share predictions to  $[0,1]$ . Movement is the conditioned minus baseline exposure coefficient. Brackets are basic intervals from 399 linked-household draws that regenerate the shortfall but hold exposure labels fixed. They are released-weight sampling sensitivities, not occupation-wild or complete CPS design-based intervals, and are not mechanically combined with the article's shock-cluster variance. Joint label regeneration gives the same displayed point estimates and very similar uncertainty.}
\end{table}

```

```

+\\begin{table}[\\htbp]
+\\centering
+\\caption{Post-outcome exploratory Webb conditioning and support}\\label{tab:appendix_webb_support}
+\\small
+\\setlength{\\tabcolsep}{4pt}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{lrrr}
+\\toprule
+\\specification & Occupations & Beta coefficient [95\\% CI] & Same-support paired change [95\\% CI] \\\\
+\\midrule
+Without Webb, fixed primary support & 468 &  $-\$0.134$ ;  $[-0.219, -0.049]$  &  $-\$0.002$ ;  $[-0.010, 0.007]$  \\\\
+With Webb, fixed primary support & 468 &  $-\$0.132$ ;  $[-0.220, -0.044]$  & Reference \\\\
+Without Webb, broader beta-valid support & 490 &  $-\$0.135$ ;  $[-0.218, -0.051]$  & Not paired \\\\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{All entries are post-outcome exploratory. The first two rows hold occupations, beta cutoffs, and the outcome specification fixed; the displayed paired change is without-Webb minus with-Webb,
+uses common draws, and has  $\\mathrm{MDE}_{\\{80\\}}=0.013$ . Its interval containing zero is a nondetection, not equivalence. The final row relaxes Webb availability and therefore changes support
+and beta cutoffs; it has no paired interpretation. Webb conditioning is an occupational-software proxy, not realized software or AI adoption.}
+\\end{table}
diff --git a/paper/tables/r3_table1_data.tex b/paper/tables/r3_table1_data.tex
new file mode 100644
index 0000000..183bf15
--- /dev/null
+++ b/paper/tables/r3_table1_data.tex
@@ -0,0 +1,20 @@
+\\begin{table}[\\htbp]
+\\centering
+\\caption{Corrected CPS reconstruction and analysis support}\\label{tab:data}
+\\begin{tabular}{lrr}
+\\toprule
+ & Count & Definition \\\\
+\\midrule
+Expected calendar months & 115 & Jan. 2017--July 2026 \\\\
+Observed source months & 114 & October 2025 absent \\\\
+Static analysis months & 113 & December 2022 excluded \\\\
+Eligible employed source records & 5,322,047 & ages 18--65, positive final weight \\\\
+Route-expanded rows & 6,188,956 & fractional descendants allowed \\\\
+Primary occupations & 468 & finite beta and Webb; positive pre-stock \\\\
+Occupation--month grid & 52,884 & 468 occupations  $\\times$  113 months \\\\
+Positive-total fitted cells & 51,891 & one-sided zeros retained \\\\
+Both-age-zero cells & 993 & no likelihood contribution \\\\
+\\bottomrule
+\\end{tabular}
+\\tabnote{Source records are not independent observations for regression inference. Before 2020, one source record can generate several fractional crosswalk descendants. Counts, routing checks, named
+support, exclusions, and restricted-input hashes are generated by the corrected-calendar data audit.}
+\\end{table}
diff --git a/paper/tables/r3_table2_occupation.tex b/paper/tables/r3_table2_occupation.tex
new file mode 100644
index 0000000..8caef01
--- /dev/null
+++ b/paper/tables/r3_table2_occupation.tex
@@ -0,0 +1,13 @@
+\\begin{table}[\\htbp]
+\\centering
+\\caption{Pooled and broad-family exposure comparisons}\\label{tab:occupation}
+\\begin{tabular}{lrrrr}
+\\toprule
+\\specification & Coefficient & 95\\% interval & Paired movement &  $\\mathrm{MDE}_{\\{80\\}}$  \\\\
+\\midrule
+Pooled Q5--Q1 &  $-\$0.132$  &  $[-0.2206, -0.0437]$  & --- & 0.1266 \\\\
+SOC2  $\\times$  young &  $\\times$  month &  $-\$0.0217$  &  $[-0.1607, 0.1173]$  &  $\\times$  0.1104 & 0.1998 \\\\
+\\bottomrule
+\\end{tabular}
+\\tabnote{The weighted-stock criterion is the grouped-binomial conditional likelihood for the two age-specific employment-stock counts. Coefficients parameterize log ratios of conditional means; no
+observed log ratio is constructed, and one-sided zero-stock cells remain in the criterion. The second row changes the conditioning estimand. Intervals use 9,999 occupation wild-score draws.
+The paired movement is the second-row coefficient minus the first on common draws; its interval is  $\\times$   $[0.0107, 0.2102]$  & its  $\\mathrm{MDE}_{\\{80\\}}$  is 0.1454. MDEs are normal-theory precision
+diagnostics, not rejection thresholds.}
+\\end{table}
diff --git a/paper/tables/r3_table3_characteristics.tex b/paper/tables/r3_table3_characteristics.tex
new file mode 100644
index 0000000..84bf02a
--- /dev/null
+++ b/paper/tables/r3_table3_characteristics.tex
@@ -0,0 +1,15 @@
+\\begin{table}[\\htbp]
+\\centering
+\\caption{Exposure conditioning on selected occupational dimensions}\\label{tab:characteristics}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{lrrrr}
+\\toprule
+Conditioning dimension & Occupations & Baseline & Augmented & Paired change [95\\% CI] \\\\
+\\midrule
+D*NET computer use & 408 &  $-\$0.107$  &  $-\$0.212$  &  $-\$0.105$ ;  $[-0.183, -0.027]$  \\\\
+Dingel--Neiman remotability & 408 &  $-\$0.107$  &  $-\$0.116$  &  $-\$0.009$ ;  $[-0.044, 0.027]$  \\\\
+Broad industry in microcells & 468 &  $-\$0.138$  &  $-\$0.099$  &  $\\times$  0.039;  $[0.008, 0.070]$  \\\\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{The first two rows use the same 408-occupation support and reconstructed treatment; each characteristic is interacted with young  $\\times$  post. The industry row changes the cell objective to
+occupation  $\\times$  broad-industry  $\\times$  age  $\\times$  month and gives 13 industries distinct young-relative post changes. Common bootstrap draws form paired intervals. Coefficient
+survival or movement is descriptive and does not identify a causal mechanism.}
+\\end{table}
diff --git a/paper/tables/r3_table4_inference.tex b/paper/tables/r3_table4_inference.tex
new file mode 100644
index 0000000..862ccb9
--- /dev/null
+++ b/paper/tables/r3_table4_inference.tex
@@ -0,0 +1,18 @@
+\\begin{table}[\\htbp]
+\\centering
+\\caption{Inference sensitivities for the core occupational comparison}\\label{tab:inference}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{lrrr}
+\\toprule
+Procedure and target & Pooled 95\\% CI & Conditioned 95\\% CI & Paired-movement 95\\% CI \\\\
+\\midrule
+\\multicolumn{4}{l}{\\textit{Same pooled/SOC2-by-calendar-month targets}} \\\\
+Occupation wild score, SOC2-by-month &  $[-0.221, -0.044]$  &  $[-0.161, 0.117]$  &  $\\times$   $[0.011, 0.210]$  \\\\
+Twenty-two-family Webb wild score &  $[-0.211, -0.053]$  &  $[-0.153, 0.110]$  &  $\\times$   $[0.006, 0.215]$  \\\\
+\\addlinespace
+\\multicolumn{4}{l}{\\textit{Sampling-oriented companion with regenerated preperiod labels}} \\\\
+Household full refit, 399 draws &  $[-0.186, -0.076]$  &  $[-0.124, 0.066]$  &  $\\times$   $[0.035, 0.177]$  \\\\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{All rows report the reconstructed pooled, SOC2-by-calendar-month, and paired-movement coefficients. The 22-family row permits common shocks within broad occupational families. The household
+row uses 399 mean-one linked-household full refits and regenerates preperiod exposure labels; it is conditional on released CPS final weights and is not complete CPS design inference. The
+procedures target different perturbations and are not mechanically combined.}
+\\end{table}
diff --git a/paper/tables/r3_table5_flows.tex b/paper/tables/r3_table5_flows.tex

```

```

new file mode 100644
index 0000000..d83d017
--- /dev/null
+++ b/paper/tables/r3_table5_flows.tex
@@ -0,0 +1,18 @@
+\begin{table}[htbp]
+\centering
+\caption{Corrected CPS flow margins}\label{tab:flows}
+\resizebox{\textwidth}{!}{%
+\begin{tabular}{lrrrr}
+\toprule
+Endpoint and margin & Raw link records at risk / entries & Coefficient & 95\% interval & Normal MDE$_{80}$ \\
+\midrule
+Adjacent employment exit & 3,346,227 & $0.132$ & $[-0.042,0.307]$ & 0.254 \\
+Adjacent reported-occupation change & 3,207,598 & $0.003$ & $[-0.118,0.123]$ & 0.174 \\
+Adjacent entry-destination allocation & 96,981 & $-0.070$ & $[-0.250,0.110]$ & 0.260 \\
+Twelve-month employment exit & 1,399,376 & $0.123$ & $[-0.111,0.357]$ & 0.336 \\
+Twelve-month reported-occupation endpoint change & 1,110,024 & $-0.018$ & $[-0.097,0.061]$ & 0.115 \\
+Twelve-month entry-destination allocation & 82,573 & $-0.052$ & $[-0.251,0.147]$ & 0.282 \\
+\bottomrule
+\end{tabular}}
+\end{table}
+\tabnote{Counts are repeated endpoint-link records, not unique persons; an individual may contribute more than one eligible link. Coefficients are post-January-2023 changes in the ages-22--25 versus ages-26--65, beta-Q5 versus beta-Q1 log rate ratio. Entry destination is a log allocation ratio conditional on an observed entry, not an employment-finding or employer-hiring rate. Primary intervals use 9,999 occupation wild-score draws and official longitudinal weights. MDE$_{80}$=2.80$\widehat{\text{SE}}$ uses the analytic occupation-cluster standard error and is a normal-theory precision diagnostic, not the wild-score rejection threshold and not evidence that a smaller effect is absent.}
+\end{table}
diff --git a/paper/tables/r3_table6_lambda.tex b/paper/tables/r3_table6_lambda.tex
new file mode 100644
index 0000000..2b3fb0c
--- /dev/null
+++ b/paper/tables/r3_table6_lambda.tex
@@ -0,0 +1,17 @@
+\begin{table}[htbp]
+\centering
+\caption{Weighting direct and software-complement tasks}\label{tab:lambda}
+\resizebox{\textwidth}{!}{%
+\begin{tabular}{lrrrr}
+\toprule
+$_{\lambda}$ in $D*\lambda$ S$ & Q5--Q1 coefficient & Difference from $_{\lambda}=5$ [95\% CI] & Stock changing group (\%) \\
+\midrule
+0.00 & $-0.103$ & $0.029$; $[-0.044,0.103]$ & 61.6 \\
+0.25 & $-0.101$ & $0.031$; $[-0.005,0.068]$ & 25.5 \\
+0.50 & $-0.132$ & --- & 0.0 \\
+0.75 & $-0.143$ & $-0.011$; $[-0.039,0.017]$ & 10.2 \\
+1.00 & $-0.156$ & $-0.024$; $[-0.064,0.017]$ & 25.5 \\
+\bottomrule
+\end{tabular}}
+\end{table}
+\tabnote{All rows use 468 occupations, reconstructed preperiod weights, tie-preserving native groups, Webb software conditioning, and 9,999 common occupation draws. ‘‘Stock changing group’’ is preperiod employment in occupations whose quintile differs from beta ($\lambda=5$). The coefficient path is descriptive of the capability boundary; no paired beta comparison detects a difference at five percent.}
+\end{table}

```